

# Classical Machine Learning approaches to detect ADHD from eye movements

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# **Classical Machine Learning approaches to detect ADHD from eye movements**

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# Contents

|                                                            |           |
|------------------------------------------------------------|-----------|
| <b>Acknowledgement</b>                                     | <b>ix</b> |
| <b>Abstract</b>                                            | <b>xi</b> |
| <b>1 Introduction</b>                                      | <b>1</b>  |
| <b>2 Background and State of the Art</b>                   | <b>5</b>  |
| 2.1 Attention Deficit and Hyperactivity Disorder . . . . . | 5         |
| 2.2 Symptoms of ADHD in Children and Adults . . . . .      | 6         |
| 2.3 How ADHD is Diagnosed . . . . .                        | 7         |
| 2.4 Eye-tracking technologies and data . . . . .           | 8         |
| 2.5 Machine Learning methods: a brief overview . . . . .   | 11        |
| 2.5.1 Supervised Learning . . . . .                        | 11        |
| 2.5.2 Unsupervised Learning . . . . .                      | 11        |
| 2.5.3 Regression/Classifications . . . . .                 | 12        |
| 2.5.4 Tree-based methods . . . . .                         | 13        |
| 2.5.5 Support Vector Machine(SVM) . . . . .                | 15        |
| 2.6 State of the Art . . . . .                             | 16        |
| <b>3 Data Description and Experimental Methodology</b>     | <b>21</b> |
| 3.1 Data Collection . . . . .                              | 21        |
| 3.2 Data Preprocessing and Visualization . . . . .         | 23        |
| 3.3 Methodology . . . . .                                  | 25        |
| 3.3.1 Feature Extraction . . . . .                         | 26        |
| 3.3.2 Variance Inflation Factor (VIF) . . . . .            | 34        |
| <b>4 Results</b>                                           | <b>35</b> |
| 4.1 Visualization of features . . . . .                    | 35        |

---

|                                           |                                             |           |
|-------------------------------------------|---------------------------------------------|-----------|
| 4.2                                       | VIF on Features . . . . .                   | 36        |
| 4.3                                       | Logistic Regression . . . . .               | 37        |
| 4.4                                       | Decision Trees . . . . .                    | 42        |
| 4.5                                       | Random Forests . . . . .                    | 45        |
| 4.6                                       | Gradient Boosting . . . . .                 | 47        |
| 4.7                                       | Naive Bayes . . . . .                       | 48        |
| 4.8                                       | Support Vector Machine . . . . .            | 50        |
| 4.9                                       | Comparison of Model Performance . . . . .   | 51        |
| 4.10                                      | AIC and BIC in Model Selection . . . . .    | 52        |
| 4.11                                      | Model Parameters . . . . .                  | 53        |
| 4.12                                      | Results Excluding Metafeatures . . . . .    | 54        |
| <b>5</b>                                  | <b>Discussion and Conclusions</b>           | <b>61</b> |
| 5.1                                       | Summary of Key Findings . . . . .           | 61        |
| 5.2                                       | Interpretation and Implications . . . . .   | 63        |
| 5.3                                       | Limitations and Future Directions . . . . . | 64        |
| <b>Appendix A Feature Lists and Plots</b> |                                             | <b>65</b> |
| <b>Bibliography</b>                       |                                             | <b>93</b> |

# List of Figures

|      |                                                                                                                                             |    |
|------|---------------------------------------------------------------------------------------------------------------------------------------------|----|
| 2.1  | An illustration of a basic eye-tracking system [22]. . . . .                                                                                | 8  |
| 2.2  | A regression example [33]. . . . .                                                                                                          | 13 |
| 2.3  | A classification example [34]. . . . .                                                                                                      | 14 |
| 3.1  | Gaze trajectories (left) and fixation-saccade patterns (right) of an ADHD participant (top) and TD participant (bottom). . . . .            | 24 |
| 3.2  | Gaze trajectories after outlier removal (left) and blink removal (right) for an ADHD participant (top) and TD participant (bottom). . . . . | 25 |
| 3.3  | Flowchart of the ADHD detection methodology. . . . .                                                                                        | 27 |
| 3.4  | Angle between gaze points. . . . .                                                                                                          | 28 |
| 3.5  | Angle histogram of a sample ADHD (top) and TD (bottom) participant. . . . .                                                                 | 29 |
| 3.6  | Log velocity histogram of a sample ADHD (top) and TD (bottom) participant. . . . .                                                          | 30 |
| 3.7  | Pupil histogram of a sample ADHD (top) and TD (bottom) participant. . . . .                                                                 | 31 |
| 3.8  | Cross-correlation between log velocity and pupil size of a sample ADHD (top) and TD (bottom) participant. . . . .                           | 32 |
| 3.9  | Horizontal or vertical movement dominance using using $\cos(2\theta)$ . . . . .                                                             | 33 |
| 4.1  | The feature set for ADHD and TD participants. . . . .                                                                                       | 36 |
| 4.2  | Scatter and box plots of Diff_Y_Mean and Diff_Y_Kurtosis. . . . .                                                                           | 37 |
| 4.3  | Scatter and box plots of PartA_score and z.ASRS_score. . . . .                                                                              | 38 |
| 4.4  | Scatter and box plots of SymSpan.PartialScore and z.sspan_score. . . . .                                                                    | 39 |
| 4.5  | Features vs different VIF thresholds. . . . .                                                                                               | 40 |
| 4.6  | The feature set of 49 features after applying the VIF threshold of 20. . . . .                                                              | 41 |
| 4.7  | Confusion matrix applying logistic regression. . . . .                                                                                      | 42 |
| 4.8  | Feature coefficients after applying logistic regression. . . . .                                                                            | 43 |
| 4.9  | Confusion matrix after applying decision trees. . . . .                                                                                     | 44 |
| 4.10 | Feature importance after applying decision trees. . . . .                                                                                   | 45 |
| 4.11 | Confusion matrix using random forest. . . . .                                                                                               | 46 |

|      |                                                                                                                                 |    |
|------|---------------------------------------------------------------------------------------------------------------------------------|----|
| 4.12 | Feature importance after applying random forests. . . . .                                                                       | 47 |
| 4.13 | Feature importance after applying gradient boosting. . . . .                                                                    | 48 |
| 4.14 | Feature importance after applying gradient boosting. . . . .                                                                    | 49 |
| 4.15 | Confusion matrix applying Naive Bayes. . . . .                                                                                  | 50 |
| 4.16 | Confusion matrix applying Support Vector Machine. . . . .                                                                       | 50 |
| 4.17 | Cross-validation performances of different classifiers, including standard deviation. . . . .                                   | 52 |
| 4.18 | VIF versus accuracies for 5-fold cross-validation of RF experiments. . . . .                                                    | 53 |
| 4.19 | Feature importance after applying logistic regression without metafeatures. . . . .                                             | 55 |
| 4.20 | Feature importance after applying decision trees without metafeatures. . . . .                                                  | 56 |
| 4.21 | Feature importance after applying random forests without metafeatures. . . . .                                                  | 57 |
| 4.22 | Feature importance after applying gradient boosting without metafeatures. . . . .                                               | 58 |
| 4.23 | VIF versus number of features (left) and VIF versus accuracies using random forests (right), both without metafeatures. . . . . | 59 |
| 4.24 | Cross-validation performances of different classifiers, including standard deviation without metafeatures. . . . .              | 60 |
| A.1  | The set of all 66 features to classify ADHD and TD participants . . . . .                                                       | 66 |
| A.2  | Scatter and box plots of X_Mean . . . . .                                                                                       | 66 |
| A.3  | Scatter and box plots of X_Median . . . . .                                                                                     | 67 |
| A.4  | Scatter and box plots of X_Std . . . . .                                                                                        | 67 |
| A.5  | Scatter and box plots of X_Min . . . . .                                                                                        | 67 |
| A.6  | Scatter and box plots of X_Max . . . . .                                                                                        | 68 |
| A.7  | Scatter and box plots of X_Skewness . . . . .                                                                                   | 68 |
| A.8  | Scatter and box plots of X_Kurtosis . . . . .                                                                                   | 68 |
| A.9  | Scatter and box plots of Y_Mean . . . . .                                                                                       | 69 |
| A.10 | Scatter and box plots of Y_Median . . . . .                                                                                     | 69 |
| A.11 | Scatter and box plots of Y_Std . . . . .                                                                                        | 69 |
| A.12 | Scatter and box plots of Y_Min . . . . .                                                                                        | 70 |
| A.13 | Scatter and box plots of Y_Max . . . . .                                                                                        | 70 |
| A.14 | Scatter and box plots of Y_Skewness . . . . .                                                                                   | 70 |
| A.15 | Scatter and box plots of Y_Kurtosis . . . . .                                                                                   | 71 |
| A.16 | Scatter and box plots of Pupil_Mean . . . . .                                                                                   | 71 |
| A.17 | Scatter and box plots of Pupil_Median . . . . .                                                                                 | 71 |
| A.18 | Scatter and box plots of Pupil_Std . . . . .                                                                                    | 72 |
| A.19 | Scatter and box plots of Pupil_Min . . . . .                                                                                    | 72 |



|                                                                   |    |
|-------------------------------------------------------------------|----|
| A.20 Scatter and box plots of Pupil_Max . . . . .                 | 72 |
| A.21 Scatter and box plots of Pupil_Skewness . . . . .            | 73 |
| A.22 Scatter and box plots of Pupil_Kurtosis . . . . .            | 73 |
| A.23 Scatter and box plots of Diff_X_Mean . . . . .               | 73 |
| A.24 Scatter and box plots of Diff_X_Median . . . . .             | 74 |
| A.25 Scatter and box plots of Diff_X_Std . . . . .                | 74 |
| A.26 Scatter and box plots of Diff_X_Min . . . . .                | 74 |
| A.27 Scatter and box plots of Diff_X_Max . . . . .                | 75 |
| A.28 Scatter and box plots of Diff_X_Skewness . . . . .           | 75 |
| A.29 Scatter and box plots of Diff_X_Kurtosis . . . . .           | 75 |
| A.30 Scatter and box plots of Diff_Y_Median . . . . .             | 76 |
| A.31 Scatter and box plots of Diff_Y_Std . . . . .                | 76 |
| A.32 Scatter and box plots of Diff_Y_Min . . . . .                | 76 |
| A.33 Scatter and box plots of Diff_Y_Max . . . . .                | 77 |
| A.34 Scatter and box plots of Diff_Y_Skewness . . . . .           | 77 |
| A.35 Scatter and box plots of Log_Velocity_Mean . . . . .         | 77 |
| A.36 Scatter and box plots of Log_Velocity_Median . . . . .       | 78 |
| A.37 Scatter and box plots of Log_Velocity_Std . . . . .          | 78 |
| A.38 Scatter and box plots of Log_Velocity_Min . . . . .          | 78 |
| A.39 Scatter and box plots of Log_Velocity_Max . . . . .          | 79 |
| A.40 Scatter and box plots of Log_Velocity_Skewness . . . . .     | 79 |
| A.41 Scatter and box plots of Log_Velocity_Kurtosis . . . . .     | 79 |
| A.42 Scatter and box plots of Angle_Mean . . . . .                | 80 |
| A.43 Scatter and box plots of Angle_Median . . . . .              | 80 |
| A.44 Scatter and box plots of Angle_Std . . . . .                 | 80 |
| A.45 Scatter and box plots of Angle_Min . . . . .                 | 81 |
| A.46 Scatter and box plots of Angle_Max . . . . .                 | 81 |
| A.47 Scatter and box plots of Angle_Skewness . . . . .            | 81 |
| A.48 Scatter and box plots of Angle_Kurtosis . . . . .            | 82 |
| A.49 Scatter and box plots of $\cos(2\theta)$ _Mean . . . . .     | 82 |
| A.50 Scatter and box plots of $\cos(2\theta)$ _Median . . . . .   | 82 |
| A.51 Scatter and box plots of $\cos(2\theta)$ _Std . . . . .      | 83 |
| A.52 Scatter and box plots of $\cos(2\theta)$ _Min . . . . .      | 83 |
| A.53 Scatter and box plots of $\cos(2\theta)$ _Max . . . . .      | 83 |
| A.54 Scatter and box plots of $\cos(2\theta)$ _Skewness . . . . . | 84 |
| A.55 Scatter and box plots of $\cos(2\theta)$ _Kurtosis . . . . . | 84 |

|                                                                                                             |    |
|-------------------------------------------------------------------------------------------------------------|----|
| A.56 Scatter and box plots of maximum values of cross-correlation between pupil size and velocity . . . . . | 84 |
| A.57 Scatter and box plots of minimum values of cross-correlation between pupil size and velocity . . . . . | 85 |
| A.58 Scatter and box plots of fixation length . . . . .                                                     | 85 |
| A.59 Scatter and box plots of succade length . . . . .                                                      | 85 |
| A.60 Scatter and box plots of ravens_score . . . . .                                                        | 86 |
| A.61 Scatter and box plots of z.ravens_score . . . . .                                                      | 86 |

# List of Tables

|     |                                                                                                           |    |
|-----|-----------------------------------------------------------------------------------------------------------|----|
| 4.1 | Model accuracy comparison using 70-30 train-test split . . . . .                                          | 51 |
| 4.2 | Mean accuracy and standard deviation of classifiers with different cross-validation folds . . . . .       | 51 |
| 4.3 | Model accuracy comparison using 70-30 train-test split without considering metafeatures . . . . .         | 54 |
| 4.4 | Mean accuracy and standard deviation of classifiers using cross-validation without metaFeatures . . . . . | 59 |



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# Abstract

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder that affects cognitive and behavioral functions, often leading to difficulties in daily life. Traditional diagnostic methods, which are primarily based on subjective behavioral assessments, can be time-consuming and prone to inaccuracies. Eye movement analysis has emerged as a promising tool for identifying objective biomarkers associated with ADHD, as individuals with the disorder often exhibit distinct patterns in saccades, fixations, and gaze behavior. Recent advancements in machine learning have enabled the development of algorithms capable of processing these complex eye movement patterns to detect ADHD with greater accuracy and efficiency. This research explores the potential of combining classic machine learning techniques with eye-tracking data to improve ADHD diagnosis, offering a faster and more scalable alternative to conventional methods. Such innovations have the potential to improve early detection and support personalized intervention strategies, significantly advancing the field of neurodevelopmental diagnostics. Thus, classical machine learning models were evaluated for classifying ADHD and TD participants using eye-tracking data. The dataset included 60 features derived from eye movement behavior—such as variability in pupil size and gaze angle—and 6 metafeatures obtained from psychological assessments, including symptom scores and cognitive test results. When using a 70-30 train-test split, the Random Forest model achieved the highest accuracy of 86% with metafeatures and 78% without. Cross-validation confirmed this trend, with Random Forest maintaining the best performance, reaching 78% accuracy with metafeatures and 67% without. These findings show that although self-reported symptoms are strong predictors, eye-tracking measures such as standard deviation of pupil size and gaze angle also contribute significantly to classification. This supports the use of gaze-based features as objective markers for ADHD, complementing traditional diagnostic approaches.





# Chapter 1

## Introduction

Attention Deficit Hyperactivity Disorder (ADHD) is a developmental condition that involves challenges with inattention, hyperactivity, and impulsive behavior. These symptoms can disrupt day to day life and reduce overall quality of life. ADHD is prevalent in both developed and developing countries and is diagnosed more frequently in males than females [1]. While the prevalence of the disorder has remained consistent over the past 30 years, increased awareness and recognition among clinicians have led to a higher rate of diagnoses in recent years compared to previous decades. ADHD is commonly the result of an interplay between genetic and environmental factors that build up over time [2]. Environmental influences often take effect early, either during pregnancy or shortly after birth. In rare circumstances, symptoms resembling ADHD can emerge due to extreme early-life deprivation [3] a single genetic mutation [2], or early-life traumatic brain injuries [4].

Although these insights provide valuable information about potential contributors to ADHD, they are not directly useful for diagnostic purposes. Evidence strongly supports connections between environmental factors and the development of ADHD, with some demonstrating a clear causal relationship. However, many of these associations could stem from interactions between genetic traits and environmental influences. For this reason, such factors are more accurately described as correlates of ADHD rather than direct causes. Additionally, the genetic and environmental risks associated with ADHD are not unique to the condition, highlighting its multifaceted nature.

Traditional methods for diagnosing ADHD rely heavily on behavioral assessments and subjective evaluations, which can lead to delayed or inaccurate diagnoses. However, recent advancements in machine learning (ML) have opened new possibilities for improving the precision and efficiency of ADHD detection by using objective physiological data, such as

eye movement patterns. Specific abnormalities in saccadic eye movements, fixation duration, and gaze behavior have been linked to the disorder, making them valuable biomarkers. Machine learning algorithms can analyze these patterns to identify subtle differences between individuals with and without ADHD, enabling automated and unbiased diagnosis. By integrating advanced ML models with eye-tracking technology, clinicians may gain access to a powerful tool that complements traditional methods, ultimately leading to earlier detection and personalized treatment strategies. One innovative area of research focuses on studying eye movement patterns, which are closely tied to attention and cognitive functions.

Tracking eye movements offers valuable insights into how people process visual information, stay focused, and react to their surroundings. For individuals with ADHD, these movements often display unique patterns, such as difficulty maintaining focus, frequent shifts in gaze, or slower response times. By using eye-tracking tools, researchers and medical professionals can identify subtle signs of ADHD, providing a non-invasive, effective, and data-driven method for diagnosis and ongoing monitoring. This research aims to evaluate the potential of eye-tracking technology as both a diagnostic tool for ADHD. By examining gaze behaviors, fixation times, and saccadic eye movements, the study aims to deepen our understanding of how ADHD affects attention and visual processing, making the way for more accurate and tailored treatment approaches.

The adoption of machine learning (ML) and deep learning (DL) models has significantly advanced ADHD research. These models have been trained using diverse datasets, such as brain imaging (MRI), physiological signals (EEG, ECG), and eye-tracking data, to classify ADHD with high accuracy. Among these approaches, eye-tracking stands out for its noninvasive nature and its ability to provide real-time information on attentional and cognitive dysfunctions in individuals with ADHD. Eye-tracking systems have shown effectiveness in detecting impairments in attention regulation, inhibitory control, and emotional processing, which are commonly associated with ADHD. Despite progress, challenges remain. Research has largely focused on specific data modalities, such as magnetic resonance imaging(MRI), while fewer studies have explored the integration of multimodal datasets, including wearable devices, to improve diagnostic capabilities and accessibility. Moreover, there is a shortage of publicly available datasets for eye-tracking and other physiological measures, limiting reproducibility and broader application. However, AI-powered approaches, combined with eye-tracking, offer promising potential to create scalable, cost-effective, and objective tools to support ADHD diagnosis and treatment.

Research questions guide the focus and scope of this study by identifying key aspects to be

explored and evaluated. In the context of this thesis, which investigates the use of classical machine learning techniques for detecting ADHD using eye movement data, the research questions aim to assess the effectiveness, relevance of features, and methodological impacts on classification performance. These questions are intended to provide a structured basis for analysis and experimentation throughout the study. The research questions addressed in this thesis are as follows:

- **RQ1:** Can classical machine learning models be used to distinguish between ADHD and typically developed (TD) individuals based on gaze data?
- **RQ2:** Which classical machine learning model provides the highest performance for classifying ADHD and TD?
- **RQ3:** How does model performance change when meta-features are included compared to when they are not?



# Chapter 2

## Background and State of the Art

This chapter provides a foundational overview of the key concepts relevant to this study. It begins by outlining the characteristics, symptoms, and diagnosis of Attention Deficit Hyperactivity Disorder (ADHD), followed by a brief introduction to eye-tracking technologies and their role in behavioral research. Core machine learning techniques used for classification tasks are also introduced to support the methodological framework of this work. Finally, the chapter presents a summary of existing research efforts that combine eye-tracking data and machine learning for ADHD detection, highlighting recent progress and identifying current trends.

### 2.1 Attention Deficit and Hyperactivity Disorder

Attention-Deficit/Hyperactivity Disorder (ADHD) is a widespread neurodevelopmental condition characterized by persistent patterns of inattention, impulsivity, and hyperactivity. ADHD is recognized as a lifelong condition that can significantly impact academic performance, workplace productivity, and social relationships. While it is commonly diagnosed in childhood, many individuals continue to experience its effects well into adulthood, often requiring ongoing support and management strategies. Globally, ADHD affects around 5.9 percent of children and adolescents and approximately 2.5 percent of adults, spanning all cultural and socioeconomic groups.

The disorder has been extensively researched across multiple disciplines, including psychology, neuroscience, and education, leading to a growing understanding of its complexities. Advancements in diagnostic criteria and awareness have contributed to an increase in ADHD diagnoses, highlighting the importance of early identification and intervention. Despite this,

many individuals remain undiagnosed or misdiagnosed, particularly in populations where awareness of the condition is limited. Moreover, ADHD presents differently across individuals, with varying degrees of severity and a broad spectrum of challenges that can influence daily functioning.

Additionally, ongoing research continues to explore innovative therapeutic strategies, including neurofeedback, cognitive training, and lifestyle modifications. Given the widespread impact of ADHD, interdisciplinary research remains essential for refining diagnostic methods, optimizing treatment approaches, and enhancing the quality of life for those affected.

## 2.2 Symptoms of ADHD in Children and Adults

Attention-Deficit/Hyperactivity Disorder (ADHD) presents distinct yet overlapping symptoms in children and adults, as documented by recent research. The condition, marked by patterns of inattention, hyperactivity, and impulsivity, manifests differently across age groups due to variations in developmental and life circumstances.

In children, ADHD is characterized by persistent inattention, hyperactivity, and impulsivity that are inappropriate for their developmental level. These symptoms must be present for at least six months and significantly impair daily functioning. Common signs include difficulty focusing on tasks, frequent careless mistakes, forgetfulness, and an inability to follow through with instructions. Hyperactivity is often expressed through constant fidgeting, an inability to stay seated, running or climbing in inappropriate situations, and excessive talking [5]. Recent research highlights additional challenges faced by children with ADHD. A study found that children with ADHD often experience a reduced quality of life and report lower happiness levels compared to their peers. They may also face higher instances of bullying, including sibling bullying, which can further exacerbate emotional and behavioral difficulties [6]. Moreover, inattention in children is strongly associated with academic underperformance, low self-esteem, and difficulty forming and maintaining social relationships [7].

While many children with ADHD retain symptoms into adulthood, the presentation often changes. Adults with ADHD continue to experience inattention, which may manifest as poor time management, procrastination, difficulty prioritizing tasks, and frequent forgetfulness. However, hyperactivity is often less pronounced and may be replaced by an internal sense of restlessness or irritability [8]. A unique aspect of adult ADHD is emotional dysregulation, which includes heightened irritability, mood swings, and difficulty managing frustration or stress. Emotional dysregulation is increasingly recognized as a core component of adult ADHD,

as reflected in the Wender Utah criteria, which emphasize emotional instability alongside traditional symptoms [9]. Adults with ADHD are also more prone to risky behaviors, including impulsive decision-making and difficulty maintaining stable relationships or employment [10].

Additionally, research indicates that ADHD symptoms may diminish with age, particularly hyperactivity and impulsivity, but inattention often remains persistent, requiring ongoing management [8]. Furthermore, gender differences in ADHD presentation are increasingly recognized. Girls and women with ADHD are more likely to exhibit inattention and emotional symptoms rather than hyperactivity, leading to underdiagnosis and misdiagnosis in these populations [11].

In conclusion, ADHD symptoms evolve across the lifespan, requiring age-specific diagnostic and therapeutic approaches. While hyperactivity may be more visible in children, emotional and executive functioning challenges dominate adults. Understanding these developmental transitions is critical for effective intervention and support.

## 2.3 How ADHD is Diagnosed

ADHD is diagnosed by licensed clinicians through a structured evaluation process that includes interviews with patients, parents, or caregivers to document whether the individual meets diagnostic criteria [5, 12]. The diagnosis is not based on rating scales, neuropsychological tests, or brain imaging but relies on clinical judgment supported by standardized guidelines [13, 14]. Key diagnostic criteria include developmentally inappropriate levels of hyperactivity, impulsivity, and/or inattention that persist for at least six months, occur across multiple settings (e.g., home and school), and cause significant impairment in functioning. Additionally, symptoms must have been present since childhood and cannot be better explained by another disorder [5, 12, 15].

Despite criticisms of subjectivity, ADHD meets the Robins and Guze criteria for mental disorder validity, as diagnoses are consistent across cultures and settings, predict future difficulties (e.g., academic or occupational impairments, risk for substance abuse), and align with genetic and neurological findings [8]. The disorder is classified into three subtypes: primarily inattentive, primarily hyperactive-impulsive, or combined [5]. Inattention is associated with academic challenges and low self-esteem, while hyperactivity-impulsivity often leads to behaviors like peer rejection, risky activities, and accidental injuries [7].

ADHD frequently co-occurs with other conditions such as anxiety, depression, and autism spectrum disorders, which do not rule out an ADHD diagnosis [10, 16]. Studies show that

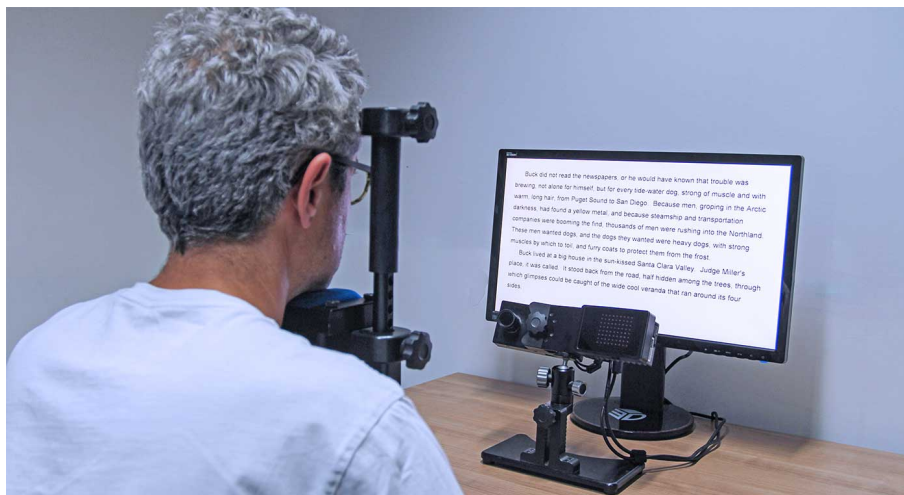


Figure 2.1. An illustration of a basic eye-tracking system [22].

ADHD affects individuals across all IQ levels [17] and often persists into adolescence and adulthood, with symptoms of inattention remaining prominent [8]. Epidemiological data also reveal that younger children in a class are more likely to be diagnosed [18].

## 2.4 Eye-tracking technologies and data

Eye-tracking technology is a groundbreaking tool that monitors and analyzes eye movements, gaze patterns, and pupil responses to understand cognitive and behavioral processes. By capturing where and how long a person looks at specific objects or areas, it provides insights into attention, perception, and decision-making. This technology has found applications in diverse fields, including healthcare, marketing, education, and human-computer interaction, driving innovation in both research and industry.

Eye-tracking systems operate using various mechanisms. A common method is infrared light reflection, where infrared light is projected onto the eyes and cameras capture reflections from the cornea and retina to determine the direction of gaze. Another widely used approach is pupil-corneal reflection, which measures the relative positions of the pupil and corneal reflections for precise tracking. These technologies are further enhanced by machine learning algorithms that interpret gaze data and generate visualizations such as heat maps and gaze paths, providing a clear representation of attention patterns [19–21]. Figure 2.1 shows an example of a basic eye tracking system where a person is looking at the screen wearing an eye tracking device.

There are several types of eye-tracking system. Remote eye trackers, which use stationary cameras, are ideal for usability testing and consumer research. Wearable eye-trackers, like



glasses, are suited for real-world environments, such as sports or driving simulations. Screen-based systems are commonly integrated into laptops or virtual reality (VR) headsets for controlled studies, while mobile devices and embedded systems make eye tracking more accessible through smartphones and augmented reality (AR) applications [23, 24].

The applications of eye-tracking technology are vast. In healthcare, it helps diagnose neurological disorders such as Parkinson's disease and autism spectrum disorder (ASD) by analyzing irregular gaze patterns [24, 25]. In human-computer interaction, it powers assistive devices for people with disabilities, enabling gaze-based navigation and robotic control. Eye-tracking is also a key tool in marketing, where it measures advertising effectiveness and analyzes customer behavior in retail environments. Furthermore, in gaming and VR, eye-tracking creates immersive experiences by enabling gaze-controlled interactions and responsive content [19, 23].

In education, eye-tracking helps identify learning difficulties such as dyslexia by analyzing reading behaviors. It also informs instructional design by revealing how learners allocate attention between multimedia elements. The integration of machine learning and AI further improves the capabilities of the technology, enabling more accurate predictions of cognitive states and emotions [21]. Despite its promise, eye-tracking technology faces challenges. Data privacy concerns arise from the sensitive biometric information collected, while the lack of standardization in data collection and analysis complicates cross-study comparisons. However, ongoing advancements, including the miniaturization of devices and the incorporation of eye tracking into everyday technology, are driving their broader adoption [20, 23].

Eye-tracking technology continues to revolutionize the way we study human behavior and interaction. Its ability to decode visual attention and cognitive processes holds immense potential for applications in research, healthcare, education, and beyond, making it a cornerstone of innovation in the digital age.

Variables play a central role in research, programming, and data science, representing data elements that can take on different values. Choosing the correct type of variable and understanding its properties is crucial for accurate data collection, analysis, and interpretation. Their classification depends on their properties, roles, and the context in which they are used, ranging from categorical and numerical classifications to their roles as independent or dependent variables. Understanding the nature of variables ensures robust study design, reliable outcomes, and effective data organization and interpretation in both academic and practical applications.

Variables are essential in research, programming, and data science, representing elements that can assume different values. Choosing the right type of variable and understanding its properties is critical for accurate data collection, analysis, and interpretation. Variables are classified

based on their properties, roles, and contexts, ranging from numerical and categorical types to independent or dependent roles. Proper understanding ensures robust study designs, reliable outcomes, and effective data organization.

Variables are broadly categorized into numerical, categorical, Boolean, text, and date/time types. Numerical variables represent measurable quantities and are further divided into discrete or continuous types. Discrete variables take specific whole-number values, such as the number of children in a family [26]. Continuous variables, on the other hand, can assume any value within a range, like height or temperature. Categorical variables represent groups or labels and include nominal and ordinal types. Nominal variables, such as blood types, lack inherent order, while ordinal variables, like education levels, have a meaningful order [27].

Boolean variables represent binary states like true/false and are widely used in programming and decision-making. Text variables store character sequences, such as names or descriptions, and date/time variables capture temporal data, essential for time-series analysis and event tracking.

The roles of variables in studies are crucial for understanding causality. Independent variables are manipulated to observe their effects on dependent variables. For example, in studying sleep duration's impact on cognitive performance, sleep duration is the independent variable, and performance scores are the dependent variable [28]. Controlled variables, like age or diet, are kept constant to isolate relationships, while confounding variables, such as stress, may influence outcomes alongside the independent variable [29].

Measurement scales classify variables as nominal, ordinal, interval, or ratio. Nominal variables, like marital status, lack order, while ordinal variables, such as satisfaction ratings, have an order but unequal intervals. Interval variables, like temperature in Celsius, have equal intervals but no absolute zero. Ratio variables, like weight, include equal intervals and an absolute zero, enabling meaningful ratios [27].

Advanced classifications include latent variables, used to infer constructs like intelligence [30], and derived variables, calculated from others, such as BMI. In programming, static variables retain values across calls, while dynamic variables exist temporarily, optimizing flexibility [31].

Understanding variables is foundational for effective research, data analysis, and programming, driving decision-making and discovery in diverse fields.

## 2.5 Machine Learning methods: a brief overview

Machine Learning (ML) is a subfield of artificial intelligence (AI) focused on enabling systems to learn and improve from experience without explicit programming. ML relies on algorithms and statistical models to identify patterns in data and make predictions or decisions. These methods can be categorized into four main types: supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning, each tailored to address specific problem types and data structures.

### 2.5.1 Supervised Learning

Supervised learning is one of the most widely used ML approaches, involving training models on labeled data. Labeled data means that for each input, there is a corresponding output, enabling the model to learn a mapping between the two. The primary goal is to generalize from the training data and make accurate predictions on unseen data. Supervised learning is commonly applied in tasks like classification, which predicts categorical labels (e.g., determining whether an email is spam or not), and regression, which predicts continuous values (e.g., forecasting house prices or stock trends).

Popular algorithms in supervised learning include Linear Regression, Logistic Regression, Decision Trees, Random Forests, and Support Vector Machines (SVM). Neural networks, particularly deep learning models, have also gained prominence in supervised learning, achieving state-of-the-art results in tasks like image recognition and text classification [28].

### 2.5.2 Unsupervised Learning

Unsupervised learning, unlike supervised learning, deals with data that does not have labeled outputs. The goal is to uncover hidden patterns or structures within the data, making it ideal for exploratory data analysis. Clustering and dimensionality reduction are two common tasks in this domain. Clustering involves grouping data points into categories based on similarity, such as segmenting customers into distinct demographic groups. Dimensionality reduction techniques, such as Principal Component Analysis (PCA) and t-SNE, reduce the number of features while retaining essential information, enabling better data visualization and preprocessing [26].

Popular algorithms in unsupervised learning include K-Means Clustering, Hierarchical Clustering, and Gaussian Mixture Models (GMM). Deep learning techniques like autoencoders are also employed for unsupervised feature learning [32].

### 2.5.3 Regression/Classifications

Regression and classification are two fundamental types of supervised learning techniques in machine learning. Both aim to establish relationships between input data (features) and output data (labels), enabling predictive analysis. However, they differ in the nature of the outputs they predict and the problems they address.

Regression deals with predicting continuous numerical values based on input features. Its primary goal is to identify patterns in data to estimate a dependent variable, often referred to as the target, as a function of one or more independent variables [26]. For instance, linear regression, one of the simplest and widely used methods, assumes a linear relationship between variables and fits a line that minimizes the error between predicted and actual values. This technique is well-suited for tasks such as predicting housing prices, stock market trends, or temperature changes. Polynomial regression extends this concept by fitting non-linear relationships, enabling it to handle more complex datasets. Regularized regression techniques, such as Ridge and Lasso regression, help prevent overfitting by adding penalty terms to the loss function, making them ideal for high-dimensional data scenarios [28]. These techniques highlight the versatility of regression models in solving prediction tasks involving numerical data.

The Figure 2.2 illustrates a simple linear regression model, where a red regression line is fitted to a scatter plot of data points. The blue dots represent individual data observations, showing a clear positive correlation between the independent variable (x-axis) and the dependent variable (y-axis). The red line represents the line of best fit, which minimizes the squared differences between actual and predicted values. This regression model captures the linear trend in the data and is useful for predicting the dependent variable from new input values. The dispersion of points around the line reflects the residual error in the model.

On the other hand, classification focuses on categorizing data points into discrete classes or categories. Unlike regression, the output of classification is not a numerical value but a class label, such as “spam” or “not spam” in an email filtering system. Binary classification tasks involve two classes, whereas multi-class classification handles problems with three or more categories. For example, classifying handwritten digits (0-9) or identifying types of animals in an image are common multi-class classification problems [32]. Popular algorithms include decision trees, random forests, support vector machines (SVMs), and neural networks. Decision trees split data based on feature conditions, creating a hierarchy that makes predictions. Random forests, an ensemble method, aggregate multiple decision trees to enhance accuracy and reduce overfitting. SVMs excel in tasks requiring high-dimensional data separation, while neural

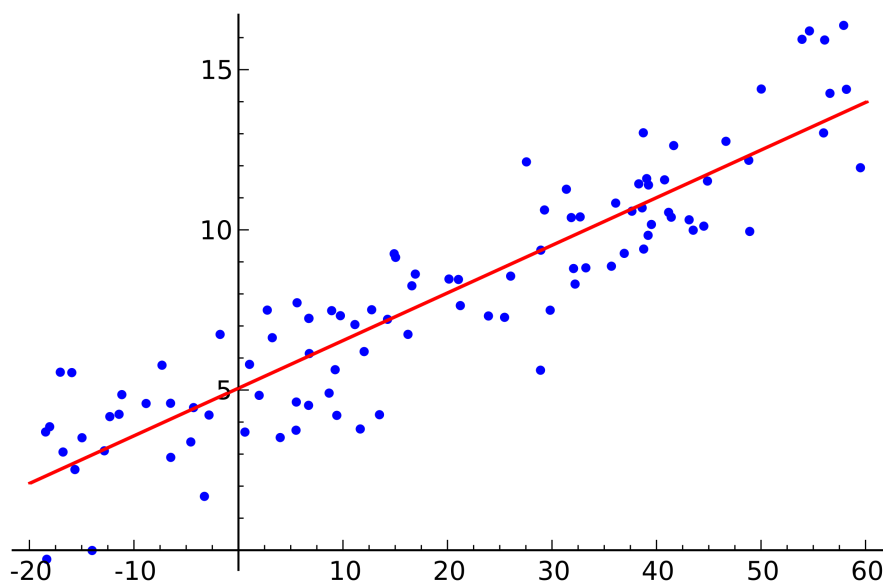


Figure 2.2. A regression example [33].

networks and deep learning methods dominate complex applications like image recognition and natural language processing [31]. Classification is extensively used in fields such as fraud detection, medical diagnosis, and sentiment analysis. The distinction between regression and classification lies in their output and purpose. Regression aims to predict continuous variables and quantify relationships, while classification assigns categorical labels to data points. Both methods are fundamental in machine learning and have transformed decision-making in various domains, from finance to healthcare, enabling more accurate and data-driven insights [30].

The Figure 2.3 shows a basic example of binary classification in two dimensions. It displays two groups of data points: blue circles labeled as Class A and green triangles labeled as Class B. These classes are separated by a straight black line, which represents the decision boundary learned by a classifier. The X and Y axes represent two features used to distinguish the classes. Any new data point falling on one side of the line will be classified as Class A, while points on the other side will be classified as Class B. This is a clear example of linearly separable data. Thus, the figure highlights the fundamental concept of supervised classification in machine learning.

#### 2.5.4 Tree-based methods

Tree-based methods have emerged as powerful tools in machine learning, particularly in healthcare applications where interpretability and robust performance are crucial. These methods create decision rules based on features that hierarchically partition the data into

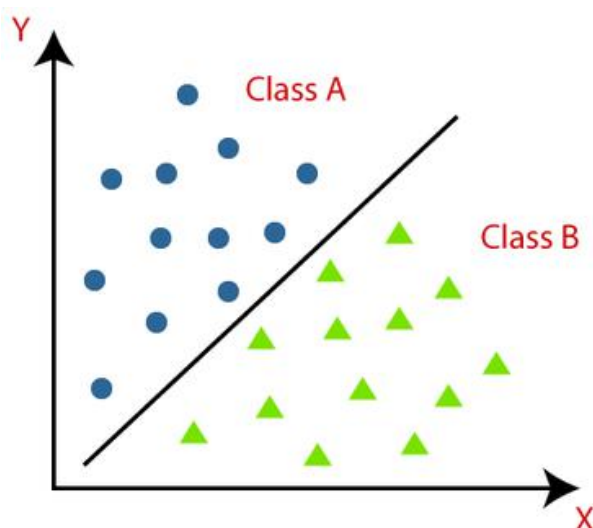


Figure 2.3. A classification example [34].

increasingly homogeneous subsets [35]. In the context of ADHD detection through eye movements, tree-based methods offer several advantages, including their ability to handle non-linear relationships and capture complex interactions between various eye-tracking metrics.

The simplest form, decision trees, offers interpretable models but often suffers from overfitting. To address this limitation, ensemble methods like Random Forests were developed, combining multiple trees through bootstrap aggregation and random feature selection [36]. These ensemble approaches significantly improve predictive accuracy while maintaining interpretability, making them particularly valuable for medical applications. Random Forests have demonstrated success in analyzing eye movement patterns for ADHD detection, achieving classification accuracies between 85-90% [37].

Further advancing the field, Gradient Boosting Machines (GBM), particularly XGBoost [38] and LightGBM [39], build trees sequentially to correct previous errors, offering enhanced predictive accuracy and efficient handling of large datasets. In ADHD detection through eye movements, these methods excel at identifying important features such as saccadic movement patterns, fixation durations, and spatial attention distribution. They naturally capture complex interactions between different eye movement metrics while providing interpretable decision rules that can be validated by clinical experts.

While tree-based methods offer numerous advantages, including robust handling of missing values and built-in feature selection, they require careful consideration of hyperparameter tuning and validation procedures to prevent overfitting. Cross-validation and independent test set evaluation are essential practices, along with clinical expert review of decision rules to

ensure medical relevance [40]. Recent developments in the field focus on integrating tree-based methods with deep learning approaches and developing more efficient algorithms for handling temporal dependencies in eye movement data.

### 2.5.5 Support Vector Machine(SVM)

Support Vector Machines (SVMs) have established themselves as powerful and versatile machine learning algorithms since their introduction by Vapnik and colleagues [41]. The fundamental principle of SVMs lies in finding an optimal hyperplane that maximizes the margin between different classes in the feature space while minimizing classification errors [42]. In their basic form, SVMs construct a hyperplane or set of hyperplanes in a high-dimensional space, which can be used for classification, regression, or other tasks. The algorithm achieves optimal separation by maximizing the margin between classes in the training data, where the margin is defined as the minimal distance from the separating hyperplane to the nearest training data points [43]. These closest points are termed support vectors, as they solely determine the position of the separating hyperplane.

One of the most powerful aspects of SVMs is the kernel trick, which allows the algorithm to operate in a high-dimensional feature space without explicitly computing the coordinates of the data in that space [44]. Common kernel functions include linear, polynomial, and radial basis function (RBF), each offering different approaches to non-linear classification. The choice of kernel and its parameters significantly influences the model's performance and must be carefully considered during the design process [45]. In practical applications, SVMs have demonstrated remarkable success across various domains. In medical diagnosis, they have shown promising results in disease classification and patient monitoring [46]. Their ability to handle high-dimensional data while maintaining good generalization properties makes them particularly suitable for biological data analysis [47]. Recent advances have focused on improving computational efficiency and adapting SVMs for large-scale applications through techniques like sequential minimal optimization [48]. Despite their advantages, SVMs face certain challenges, including sensitivity to feature scaling and parameter selection. The choice of kernel parameters and regularization constants often requires careful tuning through cross-validation. Modern implementations address these challenges through automated parameter optimization and efficient training algorithms [49].



## 2.6 State of the Art

Recent advancements in eye-tracking and machine learning have presented novel methods for identifying patterns associated with ADHD through visual behavior analysis. By examining fixation, gaze shifts, and responses to visual stimuli, researchers have created models that reliably distinguish between individuals with and without ADHD. These methods not only improve diagnostic accuracy but also provide more accessible and objective tools. This work delves into state-of-the-art studies that utilize eye-tracking data and machine learning techniques to enhance our understanding of ADHD and lay the foundation for more effective assessment strategies.

The Lévy flight model has gained significant traction as a robust framework for analyzing gaze trajectories, particularly in the context of neurodevelopmental disorders such as Attention-Deficit/Hyperactivity Disorder (ADHD). The study by Papanikolaou et al. (2024) underscores the model's effectiveness, demonstrating that children with ADHD exhibit lower Lévy exponents ( $\alpha$ ), indicative of more erratic gaze shifts compared to their typically developing peers [50]. This deviation in gaze patterns highlights the potential of Lévy flight models to assist in ADHD diagnosis by offering a scale-invariant characterization of gaze dynamics. Brockmann and Geisel (2000) observed that visual scan-paths exhibit power-law dependencies, a hallmark of Lévy flight behavior. Their findings emphasize that such dynamics are present even in controlled tasks with pre-planned visual targets [51]. Similarly, Stephen et al. (2009) extended this understanding by identifying Lévy-like diffusion in eye movements during spoken-language comprehension. Their work reinforced the hypothesis that gaze trajectories follow a Lévy flight pattern, particularly in cognitive tasks involving linguistic processing [52]. Boccignone and Ferraro (2004) modeled gaze movements as constrained random walks. They demonstrated that these models effectively replicate human visual foraging behaviors on saliency maps. This alignment with Lévy flight dynamics underscores their utility in simulating naturalistic gaze patterns [53]. Moreover, Viswanathan et al. (1996) provided a biological perspective by revealing Lévy flight patterns in the foraging behaviors of albatrosses. This groundbreaking work established that Lévy flights are optimal foraging strategies, a principle that extends to human visual information search processes [54]. Wang et al. (2023) contributed to this body of literature by examining nonlinear features of gaze behavior in children with autism spectrum disorder (ASD). Their study revealed that children with ASD exhibit distinct gaze patterns, characterized by larger Lévy exponents, compared to typically developing peers [55]. This finding underscores the broader applicability of Lévy flight models in identifying attention-related anomalies across neurodevelopmental conditions, including ADHD.



Deng and Smith [56] studied the application of natural viewing tasks, which replicate everyday visual activities, to differentiate ADHD from typical behavior. By collecting eye movement data as participants viewed various real-world images and videos, they analyzed features like fixation duration, saccade amplitude, and gaze distribution using support vector machines (SVMs). Their model achieved 85% accuracy, demonstrating that naturalistic viewing conditions can yield meaningful diagnostic markers. This research highlights the potential of non-invasive and ecologically valid methods in ADHD diagnosis.

Jayawardena and Lee [57] focused on memory-intensive tasks, asking participants to recall visual stimuli after a short delay. The study showed that these tasks placed considerable cognitive demands on working memory, leading to distinct gaze patterns among individuals with ADHD. Using decision tree classifiers to analyze fixation stability, saccadic latency, and re-fixation rates, they achieved a classification accuracy of 78%. This work illustrates the utility of combining memory load with machine learning for better understanding and identifying ADHD.

Brown and Martinez [58] examined the relationship between reading behavior and ADHD-related eye movement abnormalities. Participants read text passages while their eye movements were recorded, providing data on fixation duration, regression frequency, and saccade length. Applying SVMs to this data, they reached a classification accuracy of 83%. Their results demonstrated that ADHD individuals displayed longer fixations, more regressions, and less efficient reading patterns, showcasing the diagnostic value of reading tasks paired with machine learning.

Zhang and Lin [21] used visual search tasks as a diagnostic approach for ADHD. Participants were required to find target objects among distractors, and logistic regression models were applied to analyze gaze patterns, fixation counts, search times, and spatial gaze distributions. The study found that ADHD individuals exhibited more erratic search patterns and took longer to locate targets. With an accuracy of 80%, the research highlighted the effectiveness of visual search tasks combined with logistic regression in distinguishing ADHD from non-ADHD participants.

Lee and Chen [59] introduced a video-based attention task approach, where participants viewed dynamic scenes and responded to target stimuli. By employing k-Nearest Neighbors (k-NN) classifiers on gaze data, they identified ADHD-specific behaviors such as increased gaze shifts and longer fixation latencies. Their method achieved 82% accuracy, demonstrating that dynamic stimuli could more effectively capture attentional lapses related to ADHD. This study suggested that k-NN models could be effectively used for video-based ADHD assessments.

Patel and Kumar [60] explored cognitive interference tasks that challenged participants to ignore distracting stimuli while identifying relevant targets. Using Naïve Bayes classifiers, they analyzed eye movement metrics like distractor fixation duration, saccadic velocity, and pupil dilation. The method achieved a classification accuracy of 75%, indicating that Naïve Bayes can reliably detect ADHD-related gaze disruptions caused by cognitive conflict. This research demonstrated the importance of analyzing eye movements under challenging cognitive conditions to refine ADHD diagnostic models.

Kim and Park [61] employed visual exploration tasks, which allowed participants to examine complex scenes without specific instructions. The study utilized Linear Discriminant Analysis (LDA) to classify ADHD and non-ADHD participants based on metrics like spatial fixation dispersion and revisit frequency. With an accuracy of 79%, the findings indicated that ADHD individuals displayed more scattered and less systematic gaze patterns. This approach demonstrated that unstructured visual tasks could yield diagnostic insights into ADHD-related visual behavior.

Wang and Zhang [62] designed multimedia learning tasks that incorporated both visual and auditory components. By analyzing gaze stability, fixation duration, and attentional shifts with Random Forest classifiers, they achieved an accuracy of 86%. The study showed that realistic, multimedia-rich tasks were effective in identifying attentional difficulties associated with ADHD. The use of Random Forest models proved to be highly robust in handling diverse gaze data, further emphasizing the value of incorporating multimedia stimuli in ADHD diagnostics.

Martinez and Lee [63] focused on smooth pursuit tasks, in which participants tracked moving targets on a screen. The analysis, which examined pursuit gain, smoothness, and catch-up saccade frequency, revealed notable differences between ADHD and control groups. SVMs achieved an accuracy of 81%, underscoring the diagnostic potential of smooth pursuit eye movements and highlighting how SVMs can effectively capture nuanced tracking performance differences in ADHD populations.

Roberts and Davis [64] explored virtual reality (VR) environments to diagnose ADHD. Participants interacted with VR scenarios that simulated real-world attention challenges, such as tracking objects and responding to stimuli in a three-dimensional space. Eye movement data analyzed with decision tree classifiers achieved an accuracy of 84%. The study demonstrated that VR tasks provide a unique and engaging diagnostic environment that can reveal ADHD-related attentional deficits more naturally and effectively than traditional setups.

Nguyen and Tran [65] investigated multisensory processing tasks that required participants to

respond to simultaneous visual and auditory stimuli. Logistic regression models applied to gaze metrics such as fixation duration on auditory cues and cross-modal saccadic latencies achieved an accuracy of 77%. The research highlighted that analyzing sensory integration challenges provided valuable diagnostic information and that logistic regression could effectively capture these ADHD-related patterns.

Fernandez and Lopez [66] used Rapid Serial Visual Presentation (RSVP) tasks, where participants identified specific targets within a quickly displayed image sequence. By applying k-NN classifiers, they achieved an accuracy of 80%. ADHD individuals exhibited slower response times and reduced target identification accuracy, reflecting attentional lapses and slower visual processing. This study underscored the diagnostic value of RSVP tasks and demonstrated the efficacy of k-NN in detecting ADHD-related gaze behaviors.

Olsen and Smith [67] studied attentional blink tasks, which measure the ability to detect a second target presented shortly after the first. They found that ADHD participants experienced longer and more pronounced attentional blinks, indicating impaired temporal attention. Naïve Bayes classifiers reached an accuracy of 76%, confirming that attentional blink metrics serve as reliable indicators of ADHD-related attentional deficits. This research provided deeper insights into the temporal dynamics of attention in ADHD diagnosis.

Schmidt and Taylor [68] analyzed free-viewing tasks where participants explored visual scenes without a specific goal. Using Random Forest classifiers, they examined gaze metrics like fixation frequency, dispersion, and saccadic amplitudes, achieving an accuracy of 82%. Their findings demonstrated that even unstructured viewing conditions could provide meaningful diagnostic markers, further validating the usefulness of free-viewing tasks and robust machine learning models in ADHD detection.

Chen and Wang [69] examined decision-making tasks where participants made choices among multiple options while their gaze was tracked. Logistic regression models revealed that ADHD individuals exhibited more erratic gaze patterns and longer decision times. The model reached an accuracy of 81%, highlighting that analyzing gaze behaviors during decision-making processes can provide valuable insights into attentional control deficits in ADHD populations.

Park and Kim [70] investigated auditory-visual tasks that required participants to process inputs from both modalities simultaneously. SVMs were applied to classify ADHD and non-ADHD participants, achieving an accuracy of 80%. The study found that ADHD individuals displayed increased gaze shifts and longer fixations when integrating auditory and visual information, demonstrating the benefits of multimodal tasks in improving ADHD detection sensitivity.

Jones and Evans [71] explored memory recall tasks, requiring participants to retrieve previously shown information while their eye movements were monitored. Decision tree classifiers identified differences in gaze stability and fixation transitions, achieving a classification accuracy of 79%. This study highlighted that memory-related gaze patterns are a valuable diagnostic feature for ADHD, showing that combining memory recall tasks with machine learning could enhance diagnostic accuracy.

Yadav and Singh [72] used visual scanning tasks where participants searched for specific targets within complex arrays. With k-NN classifiers, the study achieved an accuracy of 82%, identifying that ADHD individuals displayed inefficient scanning strategies and increased revisits to distractor areas. This work demonstrated that visual scanning patterns are robust indicators of ADHD and that k-NN models can effectively differentiate these behaviors.

Liu and Zhang [73] examined smooth pursuit features, including pursuit gain and corrective saccade frequency. By using Naïve Bayes classifiers, they reached an accuracy of 77%, demonstrating that smooth pursuit metrics are reliable diagnostic markers for ADHD. This study reinforced the value of analyzing pursuit eye movements in ADHD diagnosis and showed that even simple probabilistic models can yield meaningful results.

Smith and Taylor [74] investigated task-switching activities that required participants to shift attention between different visual stimuli. Random Forest models were used to identify differences in gaze shifts, fixation durations, and response times, achieving an accuracy of 83%. Their findings highlighted that task-switching metrics provide clear markers of ADHD-related attentional challenges, showcasing the effectiveness of Random Forests in handling dynamic and context-dependent eye movement data.

These studies demonstrate that integrating eye-tracking data with machine learning algorithms can significantly improve ADHD detection. By using well-designed tasks that isolate particular attentional and cognitive deficits, researchers have achieved high classification accuracy through various models, including SVMs, decision trees, and Random Forests. This body of work emphasizes the value of nuanced, data-driven approaches in the diagnostic process, moving beyond traditional subjective methods. As research in this area advances, it holds promise for developing more reliable, timely, and accessible ADHD detection methods, ultimately improving outcomes for individuals with this condition.

# Chapter 3

## Data Description and Experimental Methodology

This chapter provides an overview of the data used in the study and the approach taken to prepare and analyze it. It begins with a description of how the dataset was collected through an experimental setup involving eye-tracking and cognitive tasks. Basic details about the participants and the recording process are also included. The chapter then outlines the steps followed to clean and preprocess the raw data, such as removing outliers and blinks. Thus, it introduces the overall methodology used to build and evaluate machine learning models, offering a foundation for the results presented in later chapters.

### 3.1 Data Collection

The experimental procedure was designed by researchers at the University of Oslo to investigate irony processing in adults with ADHD, using a combination of eye-tracking, cognitive tasks, and self-report measures [75]. Initially, 130 participants were recruited, all of whom were native Norwegian speakers with no known history of dyslexia or other learning disorders. After exclusions due to technical failures and comorbid diagnoses such as Autism Spectrum Disorder (ASD), the final dataset included 119 participants, divided into two groups: 60 adults with ADHD and 59 neurotypical controls.

Participants silently read short stories presented on a computer screen while their eye movements were recorded using an SR EyeLink 1000+ eye-tracker. Each participant completed 37 trials, resulting in a total of 4,403 trials across the entire dataset. The eye movements were recorded using different sampling rates: 46 participants were recorded at 1000 Hz, while the

remaining 73 participants were recorded at 500 Hz. This means the eye-tracker collected either 1000 or 500 data points per second, depending on the participant.

On average, each trial lasted about 32.01 seconds, with a standard deviation of 13.28 seconds, indicating some variability in trial length. The shortest trial was obtained from only one trial of a participant, which had only 0.10 seconds duration. The longest trial lasted 160.99 seconds. Altogether, the total recording time for all trials in the dataset amounts to approximately 39.15 hours of eye-tracking data.

The tasks comprised 24 stories containing a target phrase presented in a literal or ironic condition, as well as 12 filler stories used as controls. The order of the stories was randomized for each participant, and the exposure of paired stories was balanced between literal and ironic conditions.

Eye-tracking data were analyzed in relation to three regions of interest: the context, the target phrase, and the spillover region following the target phrase. These measures helped assess reading times, fixation durations, and regression likelihood to determine differences in processing between the ADHD and control groups. Following the reading task, participants answered two comprehension questions per story: one assessed general understanding of the content, while the other aimed at their interpretation of the ironic or literal phrase. This comprehension task helped capture an explicit understanding of irony. In addition, participants completed two cognitive tasks to assess working memory and fluid intelligence. The Shortened Symmetry Span Task (SSPAN) measured working memory capacity by requiring participants to recall spatial sequences while performing a symmetry judgment task. Fluid intelligence was assessed using a timed version of Raven's advanced progressive matrices, which tested participants' reasoning and problem-solving skills. Self-reported ADHD symptoms were evaluated using the Adult ADHD Self-Report Scale (ASRS), which provided sub-scores for inattention and hyperactivity/impulsivity. The medication status of the participants was recorded and some participants underwent a 24-hour medication withdrawal to assess the effects of unmedicated ADHD. The session lasted approximately 60 to 90 minutes and ethical approval was obtained from the Norwegian research regulatory authorities.

The collected data included eye-tracking metrics such as first-pass reading time, total reading time, fixation durations, and regression probability, in addition to comprehension accuracy and response times. Cognitive measures were analyzed as independent predictors to examine their role in irony processing. This multi-faceted methodology provided a comprehensive approach to understanding ADHD-related differences in irony processing, linking eye-tracking data with behavioral and cognitive assessments.

## 3.2 Data Preprocessing and Visualization

Data preprocessing is a crucial step in building an effective machine learning model for classification. The raw dataset often contains noise, missing values, and inconsistencies that must be addressed before training the model. The first step is data cleaning, where irrelevant or redundant data is removed. The dataset consists of 119 participants, where there are 60 with ADHD and 59 with TD. The average number of rows in the dataset for a participant is around a million (i.e., 1006806). The participants were left eye focused, and hence the dataset contains mainly horizontal coordinate  $x$  position, vertical coordinate  $y$  position, pupil size, and classification (i.e., fixation or saccade).

Since the majority of the frame rates were 500 Hz (i.e., 73 participants at 500 Hz and 46 at 1000 Hz), downsampling was done on the data of those 46 participants to make them 500 Hz. This was done by skipping one data point from two consecutive data points. While data processing, only one trial of a participant had a short trial length of 0.10 seconds. Hence, only that trial was ignored for further processing.

Figure 3.1 (left) shows an example of gaze trajectory from the first 20000 points, where the top figure is for an ADHD participant (1142) and the bottom one is for a TD participant (1212). In this case, the class is 0 if the point belongs to fixation and 1 for saccade. The Figure 3.1 (right) shows an example of fixation and saccade from the first 20000 points for the ADHD and TD participants.

Outlier cleaning is an essential step in preprocessing eye-tracking data to ensure accurate gaze analysis. A common approach involves filtering gaze points that fall outside the screen boundaries. Since eye-tracking data records the  $x$  and  $y$  coordinates of gaze positions, values beyond the screen's defined range (e.g., a resolution of 1920x1080 pixels) are considered invalid. These outliers may result from tracking errors, calibration drift, or rapid eye movements that momentarily disrupt tracking accuracy. Removing such data points ensures the reliability of fixation and saccade analysis. To implement this, a valid range for gaze coordinates is defined as:  $0 \leq x_{\text{left}} \leq W$  and  $0 \leq y_{\text{left}} \leq H$ , where  $W$  and  $H$  represent the screen width and height, respectively. Any data points outside these boundaries are excluded from further processing. This method effectively removes extreme values while preserving natural eye movement variations.

Furthermore, filtering outliers enhances visualization clarity, facilitating the interpretation of fixation clusters and saccade trajectories. By ensuring that only valid gaze points are analyzed, the accuracy and reliability of eye-tracking studies is improved, benefiting research and clinical

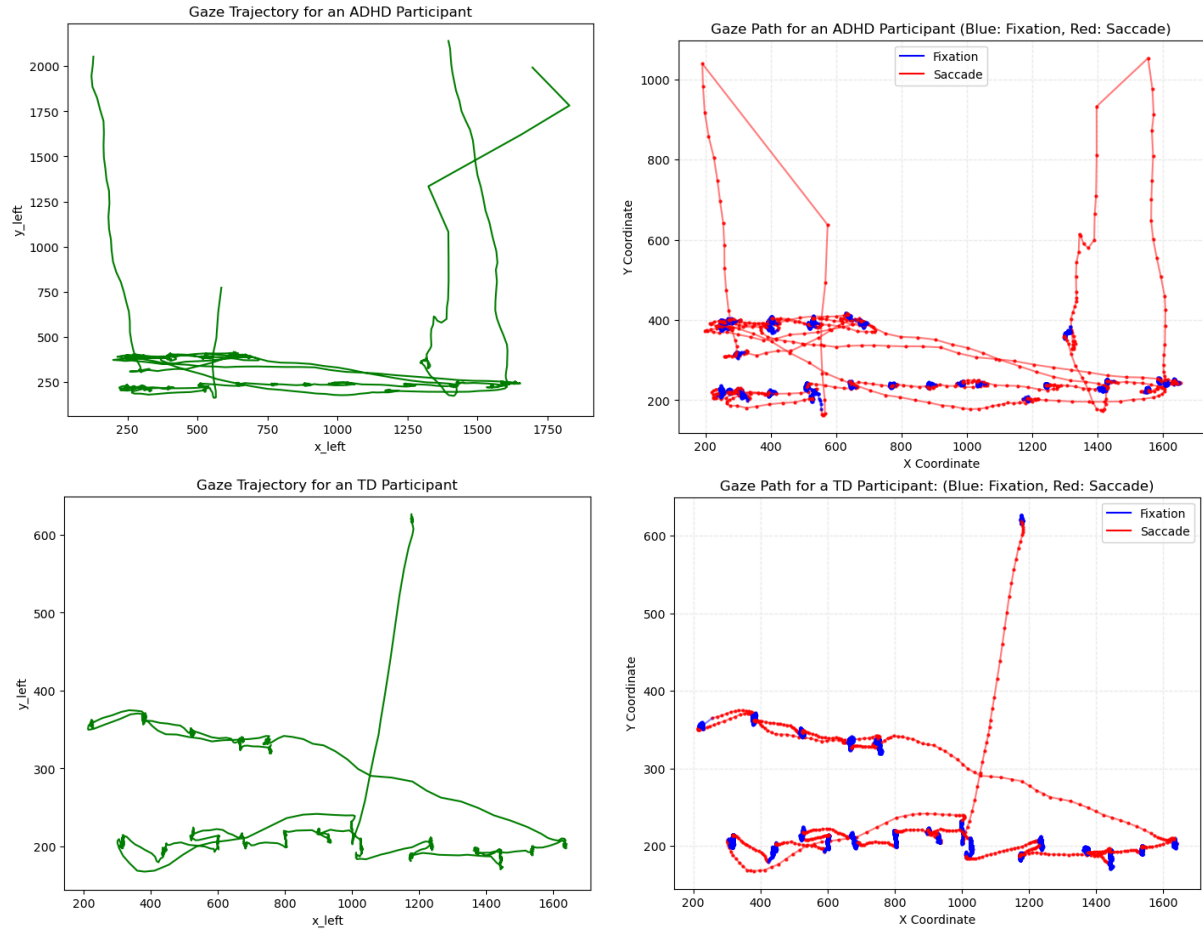


Figure 3.1. Gaze trajectories (left) and fixation-saccade patterns (right) of an ADHD participant (top) and TD participant (bottom).

applications. Figure 3.2 (left) shows an example of outline-removed gaze trajectories from the first 20000 points of the same ADHD and TD participant mentioned.

Blink removal is a critical preprocessing step in eye-tracking studies, ensuring that only valid gaze points are analyzed. Blinks introduce missing or invalid data points due to the temporary occlusion of the eye-tracking system. When a blink occurs, the gaze coordinates are either absent or recorded as extreme values, often falling outside the screen range. If not handled properly, these missing values can disrupt the analysis of fixations and saccades, leading to inaccurate conclusions about visual attention and oculomotor behavior.

A common approach to blink detection is based on pupil data. If the pupil size is recorded as zero or significantly lower than the normal range, it is indicative of a blink. Removing blinks improves the accuracy of gaze trajectory visualization and classification tasks, particularly in studies involving clinical populations such as individuals with ADHD. The presence of frequent and prolonged blinks can distort gaze metrics, making it essential to apply systematic blink-



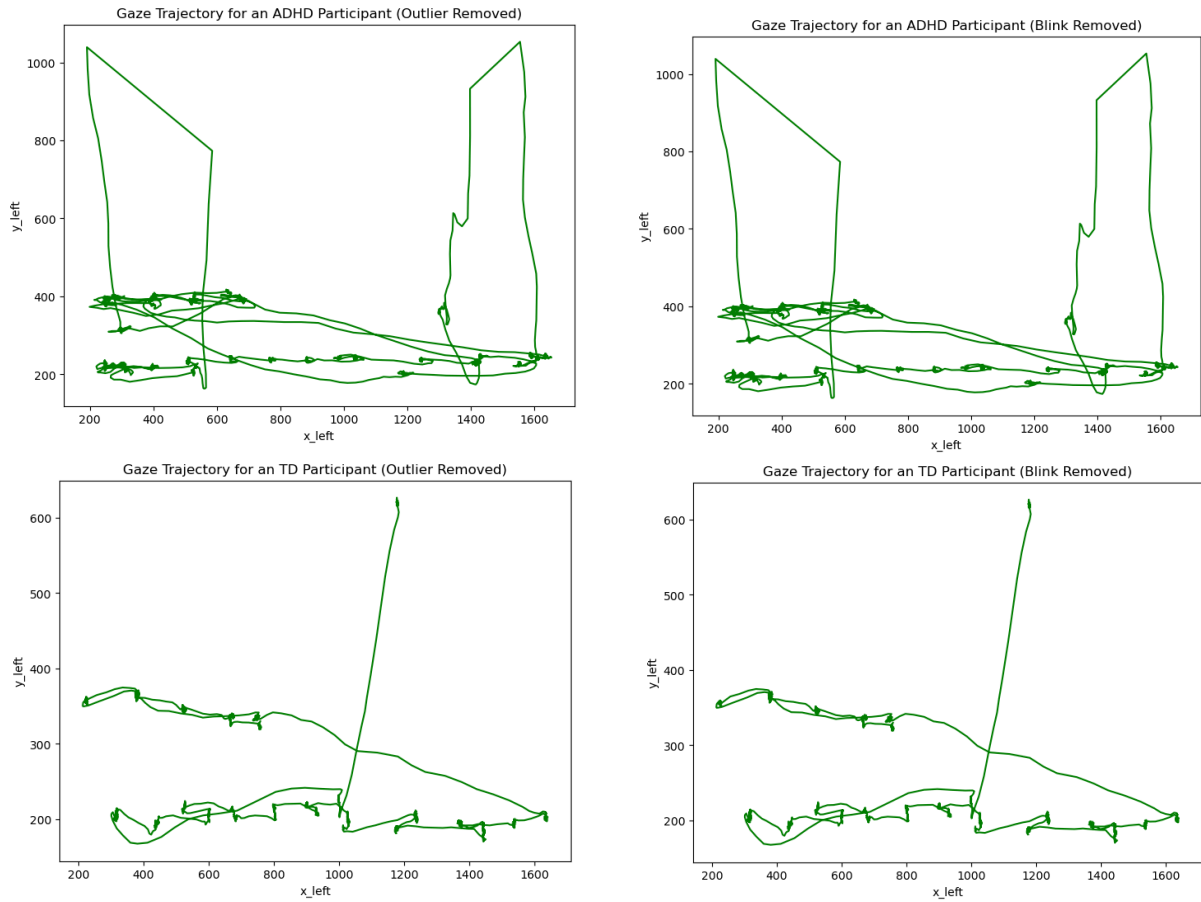


Figure 3.2. Gaze trajectories after outlier removal (left) and blink removal (right) for an ADHD participant (top) and TD participant (bottom).

filtering techniques. Figure 3.2 (right) shows an example of blink-removed gaze trajectories from the first 20000 points of the same ADHD and TD participant mentioned above. The threshold for removing the blink points is the first percentile of all pupil data. By eliminating blink artifacts, a clearer representation of natural eye movement behavior is achieved, facilitating a more accurate comparison between group participants. This preprocessing step is crucial to ensure that gaze data reflect genuine visual attention patterns rather than artifacts introduced by involuntary blinks, enhancing the reliability of findings in both experimental and applied eye-tracking research.

### 3.3 Methodology

The methodology for ADHD detection using classical machine learning follows a structured workflow that involves data collection, preprocessing, feature extraction, model training, and evaluation.

Eye tracking data were first collected from both ADHD and typically developing participants while performing the reading tasks. The raw data includes gaze coordinates, pupil dilation, fixation duration, saccade characteristics, and blink rates, recorded at high temporal resolution. Given the sensitivity of eye-tracking data, preprocessing is essential to remove noise, outliers, and artifacts. This step includes blink removal, filtering gaze points outside screen boundaries, and interpolating missing values to ensure data consistency.

Feature extraction is performed to derive meaningful metrics from the preprocessed gaze data. Commonly extracted features include fixation-related measures (e.g. duration), saccadic features (e.g. velocity and frequency), and statistical measures such as gaze entropy. These features serve as quantitative representations of visual attention and oculomotor control, highlighting the behavioral differences between individuals with ADHD and TD.

The extracted features are then used to train classical machine learning models such as Support Vector Machines (SVM), Random Forest (RF), k-Nearest Neighbors (k-NN), and Logistic Regression (LR). A stratified cross-validation strategy is used to prevent overfitting and ensure robustness. Hyperparameter tuning is performed to optimize classification performance.

For model evaluation, standard performance metrics such as accuracy, precision, recall, and F1 score are used to assess the classifier's effectiveness in distinguishing ADHD from TD participants. This pipeline provides an objective and data-driven approach for ADHD detection, using eye-tracking data as a non-invasive biomarker. By applying machine learning techniques to gaze behavior, this methodology contributes to more reliable and automated diagnostic support for ADHD classification.

### **3.3.1 Feature Extraction**

Feature extraction is a critical step in the analysis of eye-tracking data, particularly for distinguishing between individuals with ADHD and TD. The goal of feature extraction is to transform raw eye-tracking data into meaningful quantitative metrics that can effectively characterize oculomotor behavior. These features serve as the foundation for downstream machine learning models used for the classification of ADHD. Eye-tracking data consists of various measurements, including gaze coordinates, pupil size, and fixation/saccade/blink classification. However, raw data are often noisy and contain irrelevant information, necessitating a robust feature extraction process. The extracted features can be broadly categorized into three main domains: velocity-based features, angle-based features, and pupilometry.

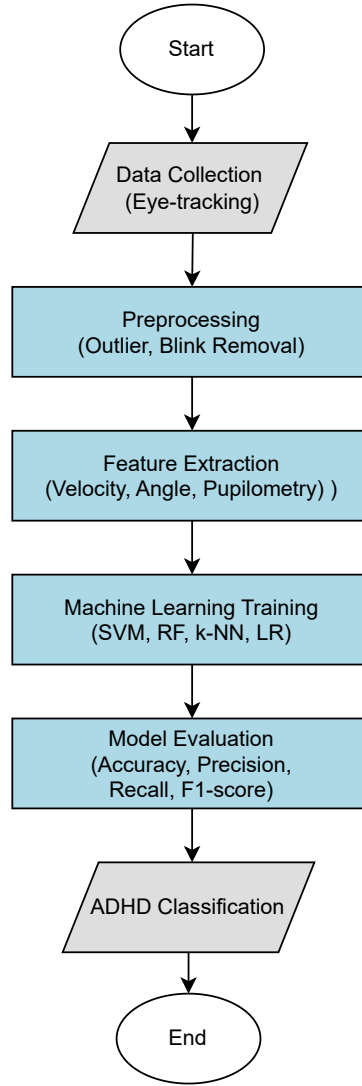


Figure 3.3. Flowchart of the ADHD detection methodology.

Angle-based features focus on capturing directional changes in gaze movement, providing insights into visual scanning behavior. The angle between consecutive gaze points can be computed using the arctangent function, which calculates the angular deviation based on the change in horizontal and vertical gaze coordinates. Figure 3.4 shows the angle between different points where  $\alpha$  and  $\beta$  are the angles between the gaze points (P1, P2) and (P2, P3), respectively.

Given two consecutive gaze points,  $(P1_x, P1_y)$  and  $(P2_x, P2_y)$ , the movement between these points can be described using basic trigonometry. The angle  $\theta$  between the gaze direction and the horizontal axis is calculated as:

$$\theta = \tan^{-1} \left( \frac{P2_y - P1_y}{P2_x - P1_x} \right) \quad (3.1)$$

This method enables the construction of an angle histogram that reflects the frequency and

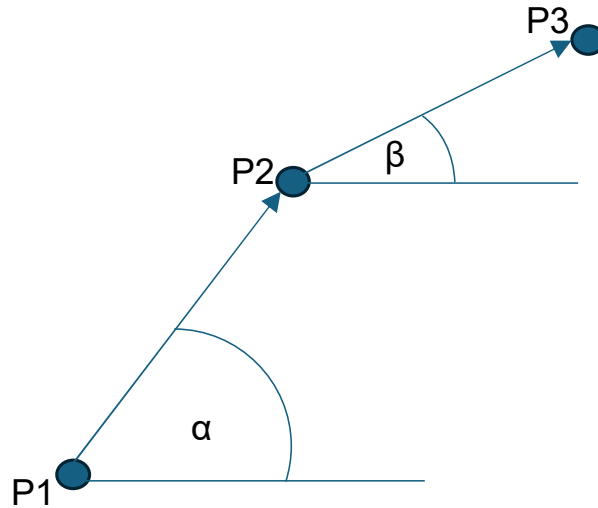


Figure 3.4. Angle between gaze points.

distribution of directional shifts throughout the task. Such histograms help identify patterns in eye movements, distinguishing between systematic exploration and erratic scanning behavior. In the context of ADHD detection, participants with ADHD often exhibit a broader distribution in their angle histograms, characterized by frequent and abrupt changes in gaze direction. This indicates less controlled and more impulsive visual exploration compared to Typically Developing (TD) participants, who generally display smoother and more consistent gaze trajectories. The angle-based features extracted include metrics such as the mean angle and the angular variance, which can be useful to describe the participant's visual scanning strategy. Figure 3.5 shows the angle histograms, which reveal variations in directional changes.

The velocity of consecutive gaze points is a critical feature in analyzing the dynamics of eye movements. It is computed by measuring the Euclidean distance between consecutive gaze points and dividing it by the time interval between samples, providing a continuous measure of gaze dynamics that captures both subtle shifts and rapid movements. The velocity between the two points is given by:

$$d = \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2} \quad (3.2)$$

To better visualize the wide range of velocities and reduce the influence of extreme values, a logarithmic transformation is applied to the velocity data before generating the histogram. This logarithmic transformation velocity histogram highlights distribution patterns more clearly, making it easier to compare gaze behaviors between participants.

In participants with ADHD, the log-transformed velocity histogram often exhibits a broader distribution, with frequent instances of higher velocities, indicating impulsive gaze patterns and

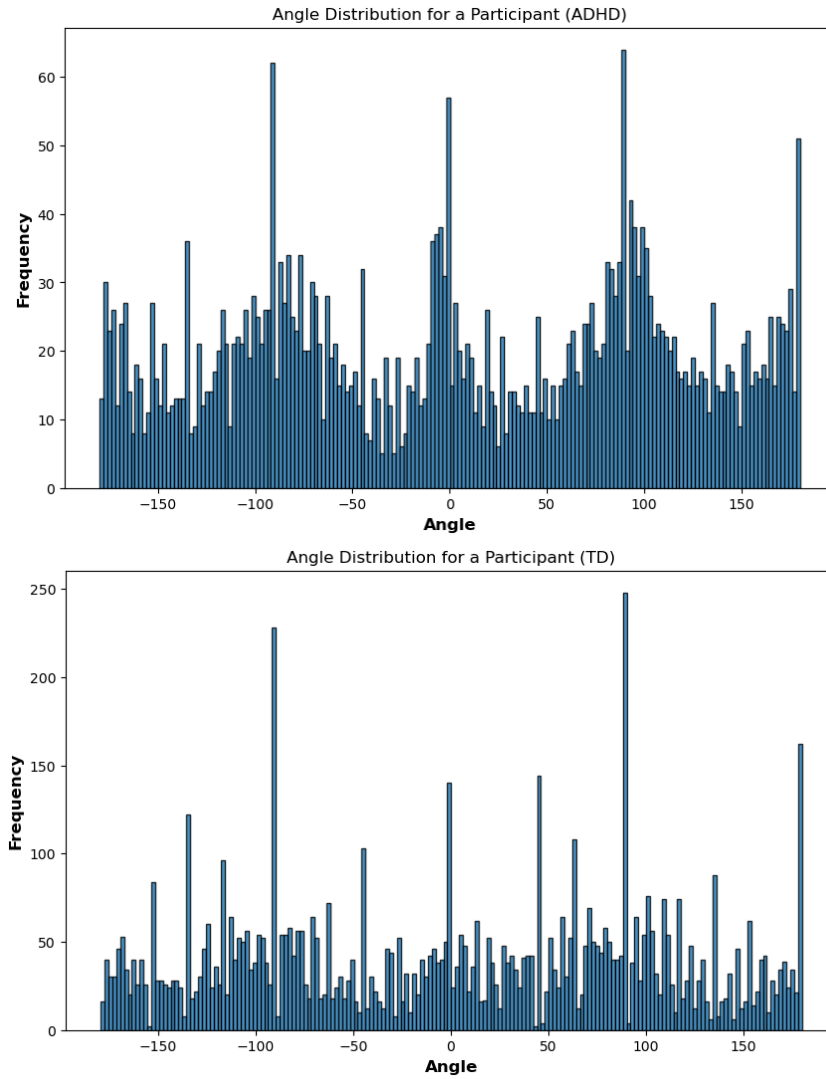


Figure 3.5. Angle histogram of a sample ADHD (top) and TD (bottom) participant.

reduced attentional control. In contrast, typically developing (TD) participants tend to show a more concentrated distribution, reflecting smoother and more controlled eye movements. The use of logarithmic scaling improves the visibility of these differences, especially in the lower and middle velocity ranges, where subtle variations can be significant. Key statistical features extracted from velocity data include mean, standard deviation, and variance. These metrics provide a comprehensive understanding of gaze dynamics and contribute to the differentiation between ADHD and TD participants.

Figure 3.6 presents the log-transformed velocity histograms for an ADHD participant and a TD participant, clearly illustrating the differences in gaze movement patterns between the two groups.

Pupil size serves as a valuable physiological marker of cognitive load, emotional arousal,

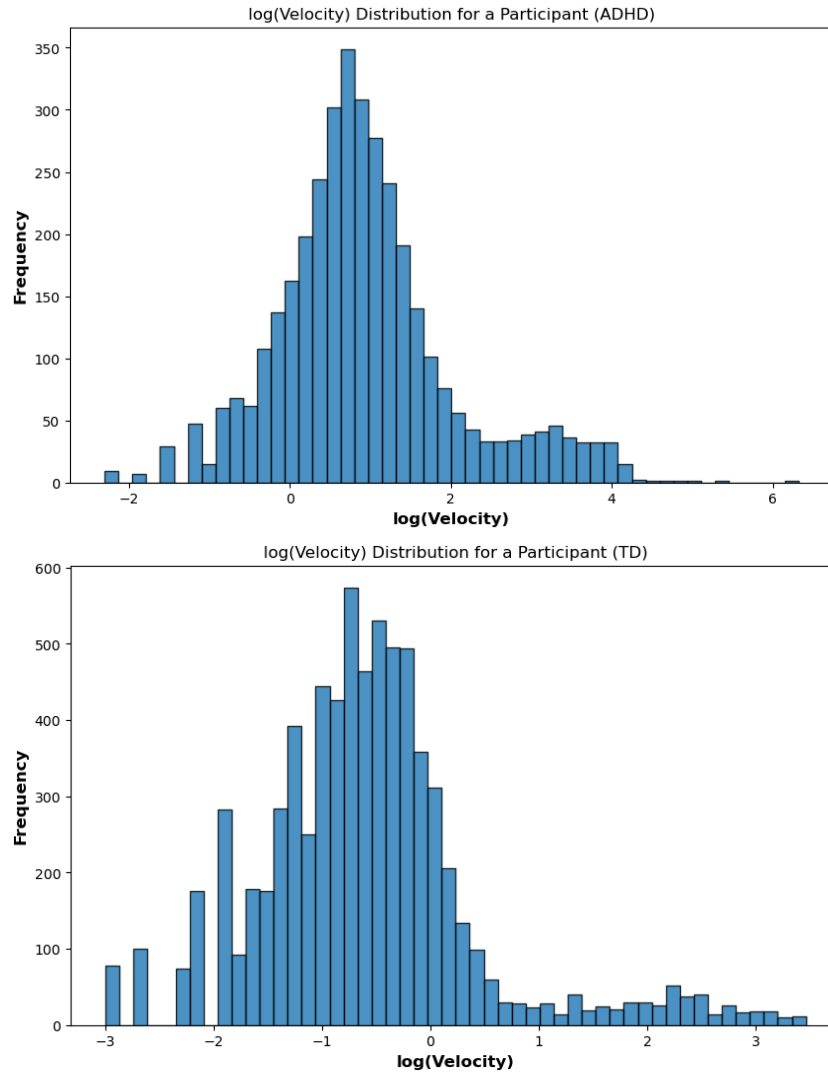


Figure 3.6. Log velocity histogram of a sample ADHD (top) and TD (bottom) participant.

and attentional engagement. Variability in pupil size during a task can reveal differences in cognitive processing and focus levels. Features such as mean pupil size, standard deviation, and variance can be used as features. Pupil size histograms are created to visualize the distribution of pupil diameters over time. ADHD participants often exhibit greater fluctuations in pupil size, suggesting inconsistent attentional engagement, while TD participants tend to show more stable patterns. These differences in pupil dynamics provide additional cues for distinguishing between the two groups. Figure 3.7 depicts the pupil size histograms, showing fluctuations in pupil diameter. Histograms offer a comprehensive view of eye movement characteristics, helping to classify participants with ADHD and TD. They can also be used as features.

The cross-correlation between log-transformed eye movement velocity and pupil size helps us understand differences in attention and cognitive control in ADHD and TD (Typically

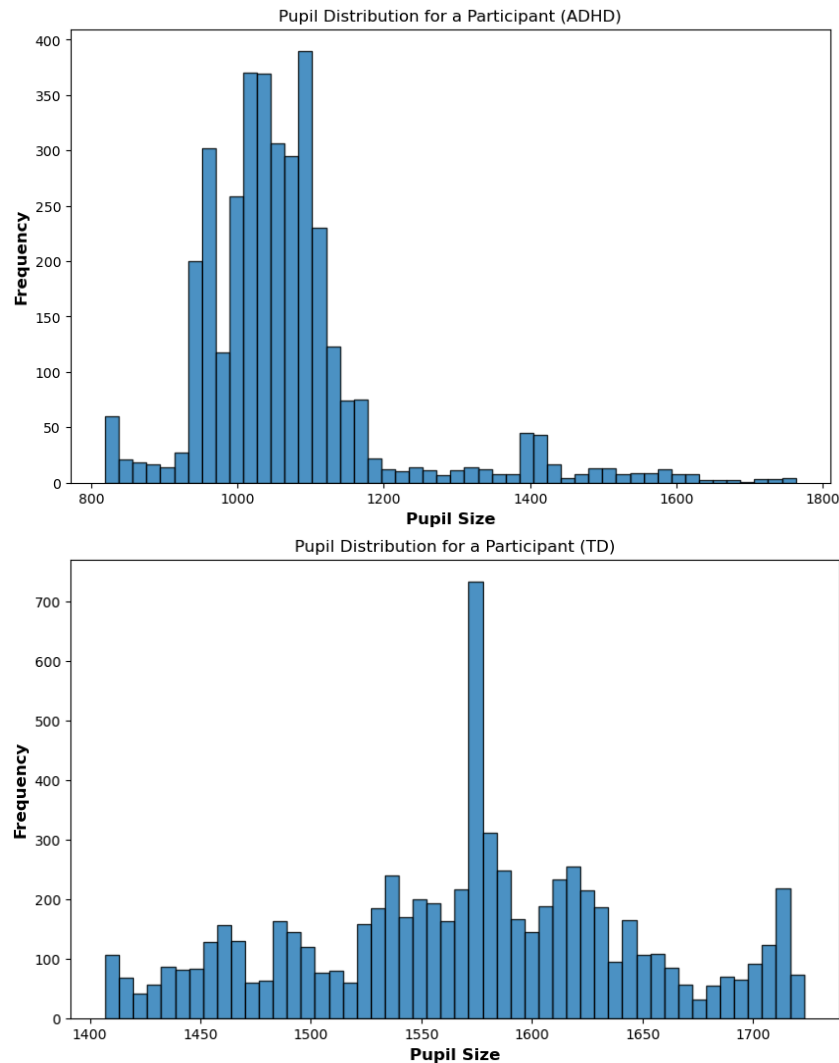


Figure 3.7. Pupil histogram of a sample ADHD (top) and TD (bottom) participant.

Developing) participants. As shown in Figure 3.8, the top plot represents an ADHD participant, while the bottom plot represents a TD participant.

In the top plot (ADHD participant), the cross-correlation coefficient is 0.0827, which means that the relationship between eye movement velocity and pupil size is weak. The peak appears to be near zero lag, which shows that changes in eye velocity and pupil size sometimes occur together. However, the plot is irregular and fluctuates a lot, suggesting inconsistent attention and cognitive control. This fits what we know about ADHD: People with ADHD often shift between focus and distraction, which may explain why the connection between velocity and pupil size is unstable. The extra peaks and dips in the plot suggest that the pupil's response to velocity changes is not always consistent. This could be due to impulsive attention shifts, processing delays, or difficulty maintaining focus. The lower correlation suggests that eye movement and

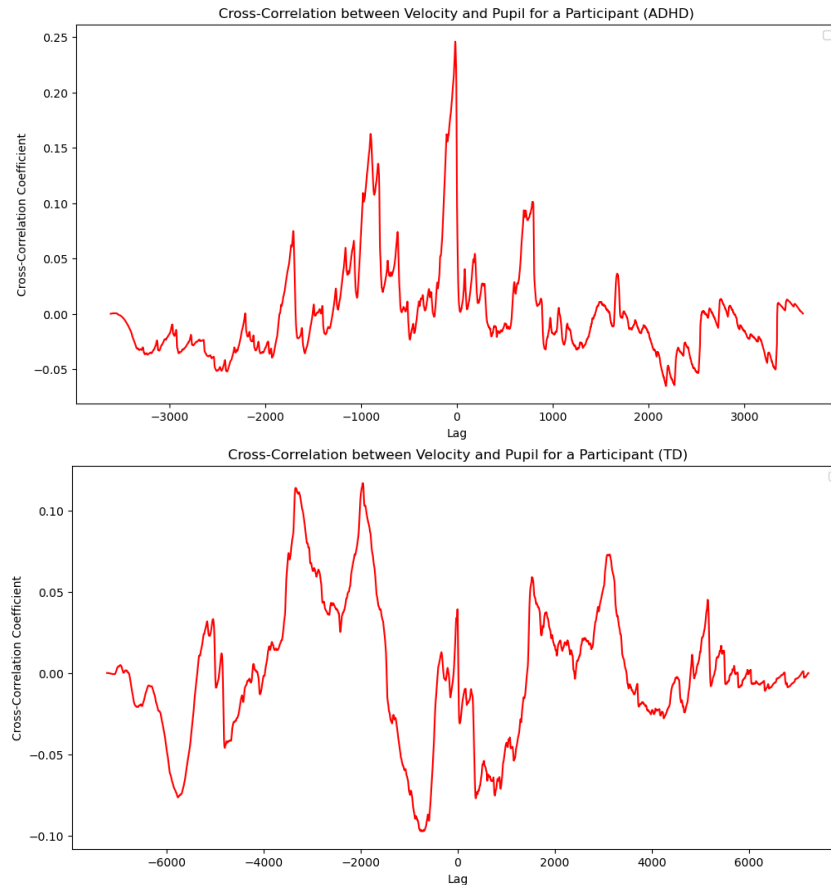


Figure 3.8. Cross-correlation between log velocity and pupil size of a sample ADHD (top) and TD (bottom) participant.

pupil dilation are not strongly linked in ADHD, probably due to attention lapses and varying levels of focus.

In the bottom plot (TD participant), the cross-correlation coefficient is 0.1009, which is stronger than in the ADHD case. The peak is also centered near zero lag, meaning that pupil size and eye velocity change together more reliably. Unlike the ADHD plot, this one is smoother and more stable, suggesting that TD participants have more controlled attention and better cognitive processing. The absence of strong fluctuations means that TD individuals have more consistent coordination between eye movement and cognitive effort. The higher correlation value further confirms that pupil size and eye movement velocity are better linked in participants with TD. This suggests that their attention and cognitive responses are steady and well-regulated.

Comparing the two plots shows clear differences between participants with ADHD and TD. Both groups show a positive relationship between eye movement velocity and pupil size, but the ADHD participant has a weaker and more unstable connection. This likely reflects inconsistent focus and attention shifts. However, the TD participant has a stronger and more



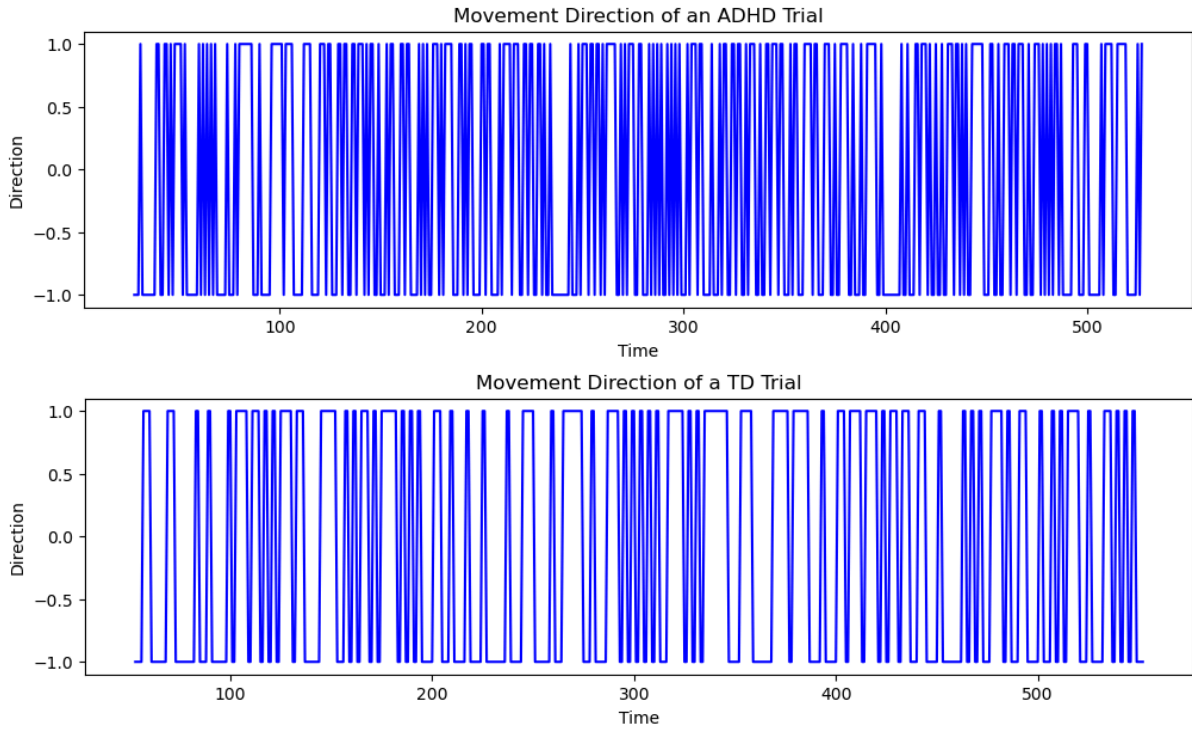


Figure 3.9. Horizontal or vertical movement dominance using  $\cos(2\theta)$ .

stable relationship, suggesting better control of attention and cognitive processes. These results match existing ADHD research, which shows that people with ADHD often have less stable attention regulation. Using cross-correlation in eye-tracking studies is a useful way to measure differences in cognitive control, which could help develop better ADHD assessment tools and targeted interventions.

The next feature explored here is horizontal movement. In eye-tracking analysis, it is often necessary to determine the horizontal movement of the gaze points. One approach is to use the squared cosine function,  $\cos(2\theta)$ , to isolate the horizontal component of movement between consecutive gaze points. This method ensures that only the horizontal displacement is considered while minimizing vertical influence.

The horizontal or vertical dominance can be computed by:

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta \quad (3.3)$$

where  $\cos(2\theta) > 0$  indicates horizontal dominance and  $\cos(2\theta) < 0$  indicates vertical dominance. In Figure 3.9, the top plot represents the data of an ADHD participant with dominance of directional movement, while the bottom plot represents the same for a participant with TD. It can be noticed in the figure that the selected TD participant has fewer fluctuations than the selected ADHD one.

### 3.3.2 Variance Inflation Factor (VIF)

In feature analysis to be applied for machine learning models for classifications, multicollinearity arises when independent variables exhibit strong correlations, making it difficult to determine their individual effects on the dependent variable. The Variance Inflation Factor (VIF) serves as a diagnostic tool to assess the severity of multicollinearity by measuring the extent to which the variance of an estimated regression coefficient increases due to correlation among explanatory variables.

Mathematically, the VIF for a given predictor  $X_i$  is expressed as:

$$VIF(X_i) = \frac{1}{1 - R_i^2} \quad (3.4)$$

where  $R_i^2$  represents the coefficient of determination obtained when  $X_i$  is regressed on all the other independent variables in the model. A high VIF indicates that the predictor shares significant information with other variables, leading to instability in the estimated coefficients. A commonly used rule of thumb suggests that VIF values exceeding 10 signal severe multicollinearity, although some analyses consider a threshold of 5 to be more conservative. To determine the VIF values, each predictor in the dataset, denoted as  $X_1, X_2, \dots, X_n$ , is treated as a dependent variable in a regression model where the remaining predictors serve as independent variables. The resulting  $R_i^2$  value for each model is then substituted into the VIF equation. This process is repeated for all predictors.

# Chapter 4

## Results

This chapter presents an overview of the input features used to differentiate individuals with ADHD from those who are typically developing, based on eye-tracking data. The features encompass a variety of behavioral indicators related to gaze patterns, pupil responses, and visual attention dynamics, as well as psychological measures derived from standardized assessments. Visualizations such as scatter plots and box plots are used to explore how these features vary across the two groups, highlighting which attributes may hold discriminative power.

### 4.1 Visualization of features

Initially, a total of 66 features were used for classifying individuals as ADHD or TD. Among them, the first 60 are extracted features derived from eye-tracking data. These features quantify various aspects of gaze behavior, pupil dynamics, and visual attention. For instance, variability in gaze and pupil signals is captured by features like `Pupil_Std` (standard deviation of pupil size), `Angle_Std` (standard deviation of gaze angle), and `Y_Std` (standard deviation of vertical gaze). Measures related to spatial and directional properties include `Cos2Theta_Min` (minimum cosine of angular direction), `Fixation_Length` (duration of visual fixations), and `Diff_Y_Kurtosis` (kurtosis of vertical gaze differences), among others. In contrast, the last six features are considered metafeatures because they are obtained from standardized psychological assessments rather than extracted from the eye-tracking signal itself. These metafeatures include `PartA_Score` (raw ADHD symptom score from ASRS-v1.1), `z.ASRS_score` (standardized ASRS score), `SymSpan.PartialScore` (raw working memory score from the Symmetry Span task), `z.sspan_score` (scaled working memory score), `ravens_score` (raw fluid intelligence score from Raven's matrices), and

|                |                       |                       |
|----------------|-----------------------|-----------------------|
| X_Mean         | Diff_X_Median         | Angle_Std             |
| X_Median       | Diff_X_Std            | Angle_Min             |
| X_Std          | Diff_X_Min            | Angle_Max             |
| X_Min          | Diff_X_Max            | Angle_Skewness        |
| X_Max          | Diff_X_Skewness       | Angle_Kurtosis        |
| X_Skewness     | Diff_X_Kurtosis       | Cos2Theta_Mean        |
| X_Kurtosis     | Diff_Y_Mean           | Cos2Theta_Median      |
| Y_Mean         | Diff_Y_Median         | Cos2Theta_Std         |
| Y_Median       | Diff_Y_Std            | Cos2Theta_Min         |
| Y_Std          | Diff_Y_Min            | Cos2Theta_Max         |
| Y_Min          | Diff_Y_Max            | Cos2Theta_Skewness    |
| Y_Max          | Diff_Y_Skewness       | Cos2Theta_Kurtosis    |
| Y_Skewness     | Diff_Y_Kurtosis       | Max_Cross_Correlation |
| Y_Kurtosis     | Log_Velocity_Mean     | Min_Cross_Correlation |
| Pupil_Mean     | Log_Velocity_Median   | Fixation_Length       |
| Pupil_Median   | Log_Velocity_Std      | Succade_Length        |
| Pupil_Std      | Log_Velocity_Min      | PartA_score           |
| Pupil_Min      | Log_Velocity_Max      | z.ASRS_score          |
| Pupil_Max      | Log_Velocity_Skewness | SymSpan.PartialScore  |
| Pupil_Skewness | Log_Velocity_Kurtosis | z.sspan_score         |
| Pupil_Kurtosis | Angle_Mean            | ravens_score          |
| Diff_X_Mean    | Angle_Median          | z.ravens_score        |

Figure 4.1. The feature set for ADHD and TD participants.

`z.ravens_score` (scaled Raven's score).

These high-level features offer additional behavioral and cognitive context, enriching the gaze data and improving classification performance. Figure 4.1 shows the initial set of 66 features to be extracted from the data of the participants in ADHD and TD. Figure A.1 (see the Appendix) shows the description of all 66 features. Figures 4.2 to 4.4 represents some feature graphs (that is, scatter and box plots) for participants in ADHD and TD. Plots such as Figure 4.3 shows good clustering of samples from ADHD and TD, whereas other plots are not that well clustered. The rest of the features are shown in the Appendix from Figure A.2 to Figure A.61.

## 4.2 VIF on Features

By analyzing VIF values, we can identify problematic variables and refine the model to improve interpretability and reliability. Addressing multicollinearity ensures that coefficient estimates remain stable, enabling more accurate predictions and meaningful statistical inference.

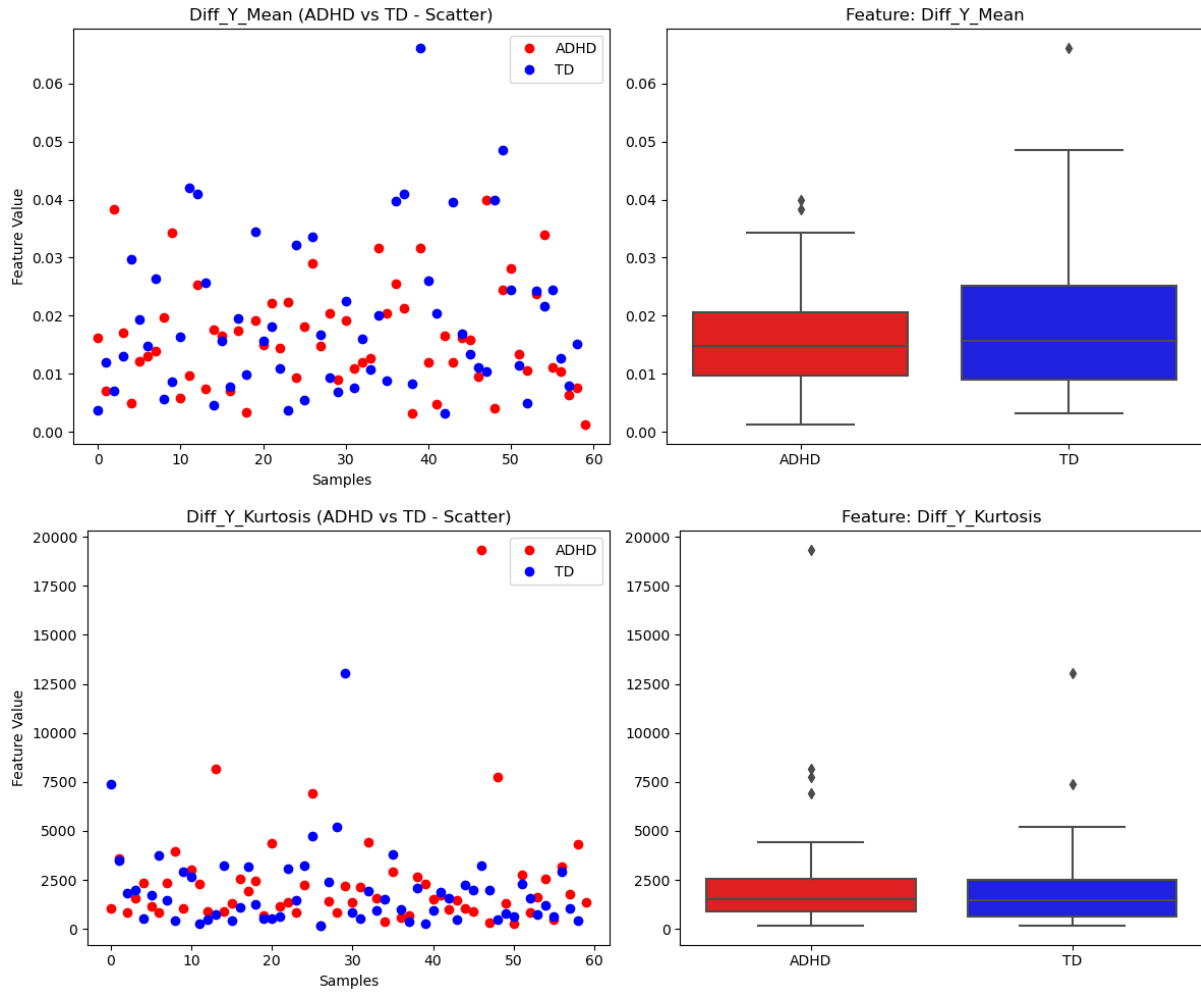


Figure 4.2. Scatter and box plots of Diff\_Y\_Mean and Diff\_Y\_Kurtosis.

Although VIF thresholds of 5 or 10 are mostly used, the optimal threshold is context-dependent. In this study, a threshold of 20 was used based on empirical performance, as it retained a feature set yielding higher classification accuracy. Figure 4.5 shows different numbers of features for different VIF thresholds. For example, after applying VIF in Figure 4.1 and considering the threshold value of 20, 49 features were obtained as shown in Figure 4.6.

## 4.3 Logistic Regression

A logistic regression model was used as the initial classifier for the dataset considered. The primary objective of this experiment was to evaluate the performance of logistic regression in a binary classification setting. Prior to model training, the dataset was divided into training and testing subsets using a random 70-30 split. The input features were provided in matrix form as  $X$ , and the binary class labels as  $Y$ . The training subset was used to fit the logistic regression

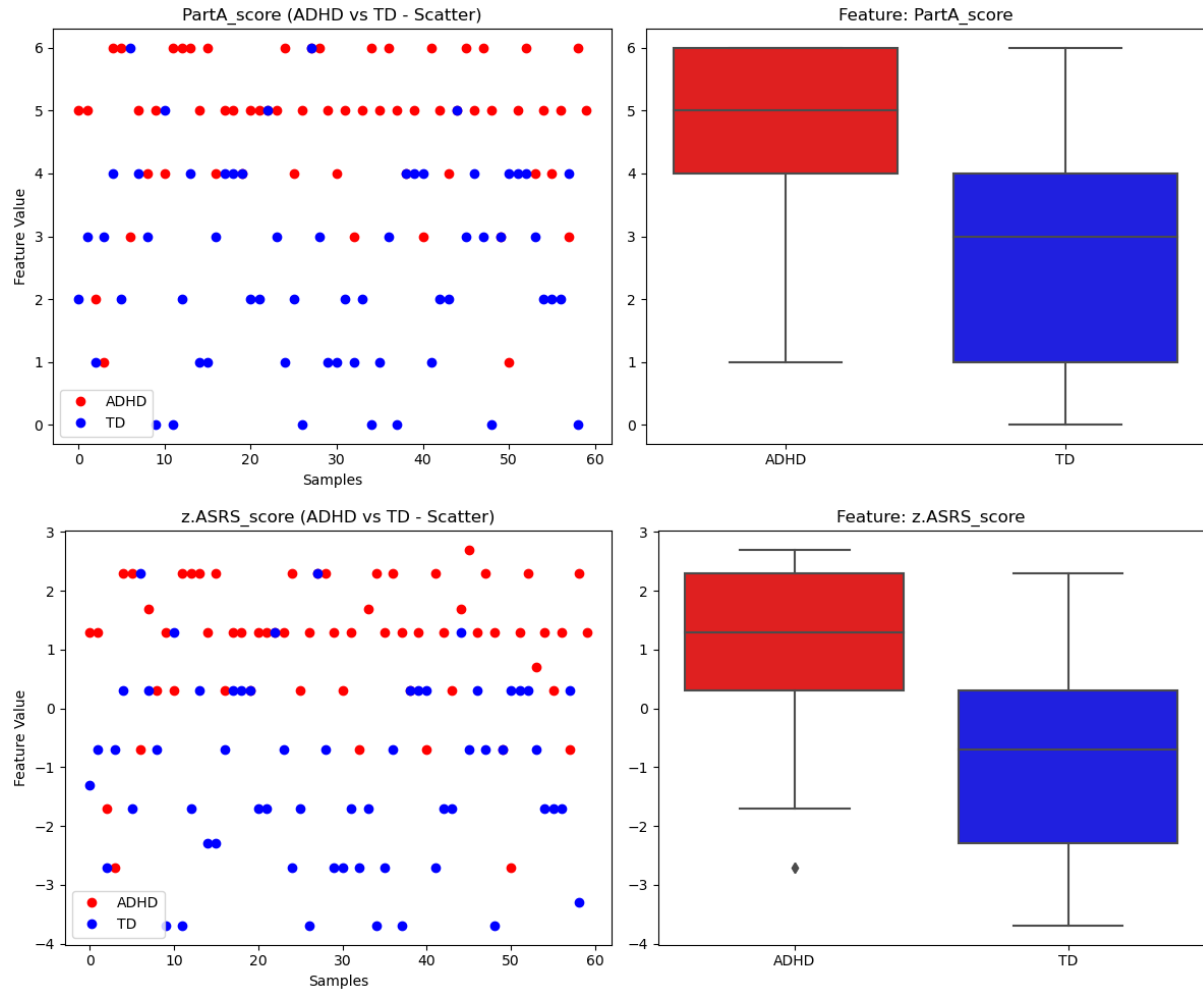


Figure 4.3. Scatter and box plots of PartA\_score and z.ASRS\_score.

model, while the testing subset was reserved for evaluating the model's generalization ability. After training, predictions were generated on the testing set, and the performance of the model was assessed using multiple evaluation metrics. The accuracy score, defined as the ratio of correct predictions to the total number of predictions, was calculated as 72%. Additionally, the confusion matrix was computed, presenting the true positives, false positives, true negatives, and false negatives. This matrix provided a more comprehensive understanding of the model's classification tendencies.

To further interpret the confusion matrix, a generalized confusion matrix was plotted as well as shown in Figure 4.7. This visual representation helped to easily identify the proportion of correctly and incorrectly classified instances in relation to each class. The confusion matrix offered valuable insights into the types of errors the model made and was particularly useful in identifying any bias toward a particular class

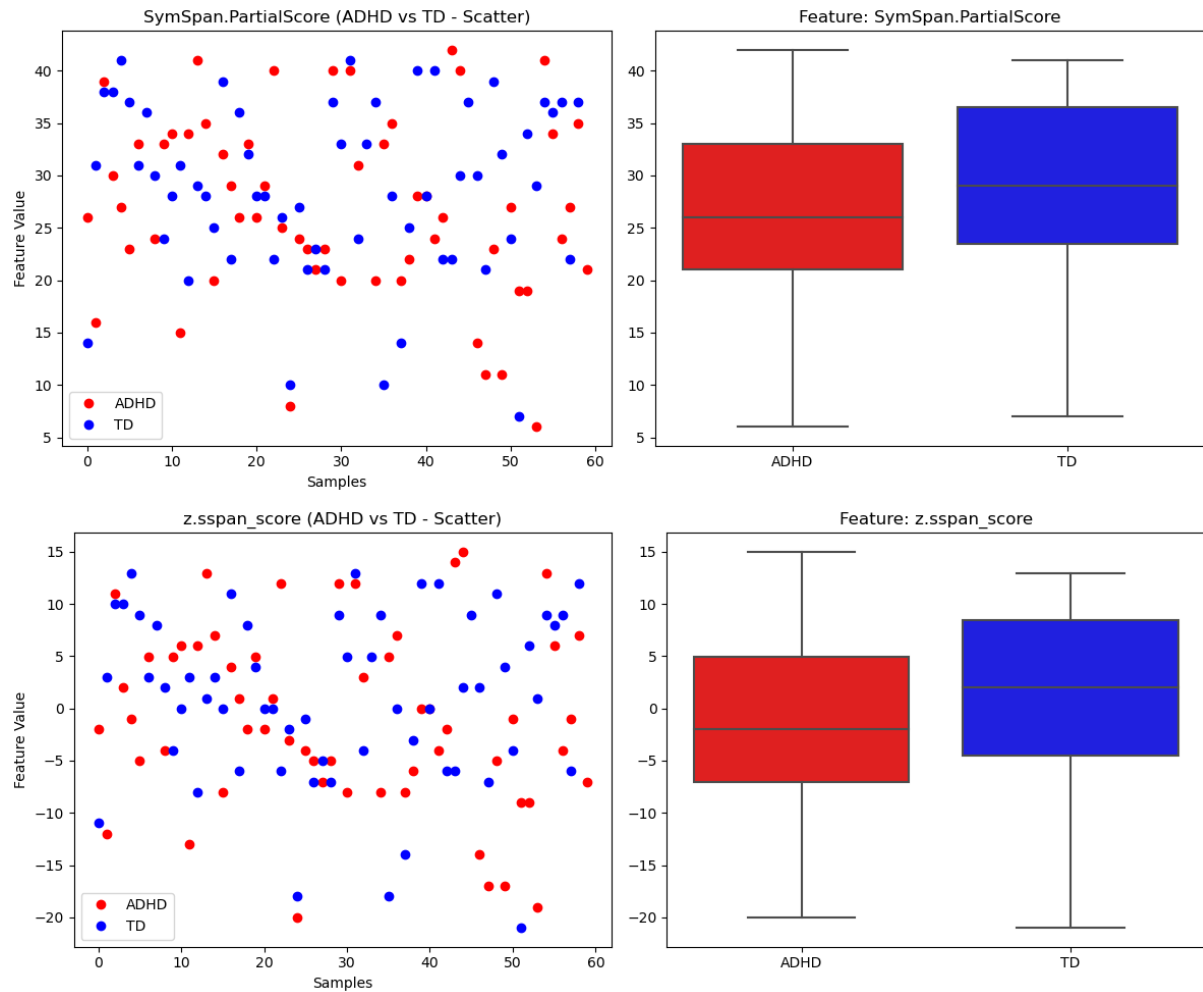


Figure 4.4. Scatter and box plots of SymSpan.PartialScore and z.sspan\_score.

Figure 4.8 shows the feature coefficients after using logistic regression. The feature importance from the logistic regression model shows that `z.ASRS_score` is by far the strongest predictor in the classification, with an importance more than 75% higher than the next most important feature. This indicates that self-reported ADHD symptoms have a dominant influence on the model's decisions. The second tier of features includes `X_Min`, `Pupil_Std`, and `Angle_Std`, all with relatively high importance values. These features suggest that the minimum horizontal gaze position and variability in pupil size and gaze angle are strong behavioral indicators. Other features such as `Diff_X_Skewness`, `Pupil_Skewness`, and `Y_Std` follow closely, pointing to the relevance of asymmetry and variability in both horizontal and vertical gaze patterns. Features like `Log_Velocity_Max`, `X_Std`, and `Diff_Y_Kurtosis` continue to contribute moderately, with a gradual tapering seen across the middle of the list. Cognitive scores such as `z.ravens_score` and `z.sspan_score` provide some predictive value but are notably less influential than gaze

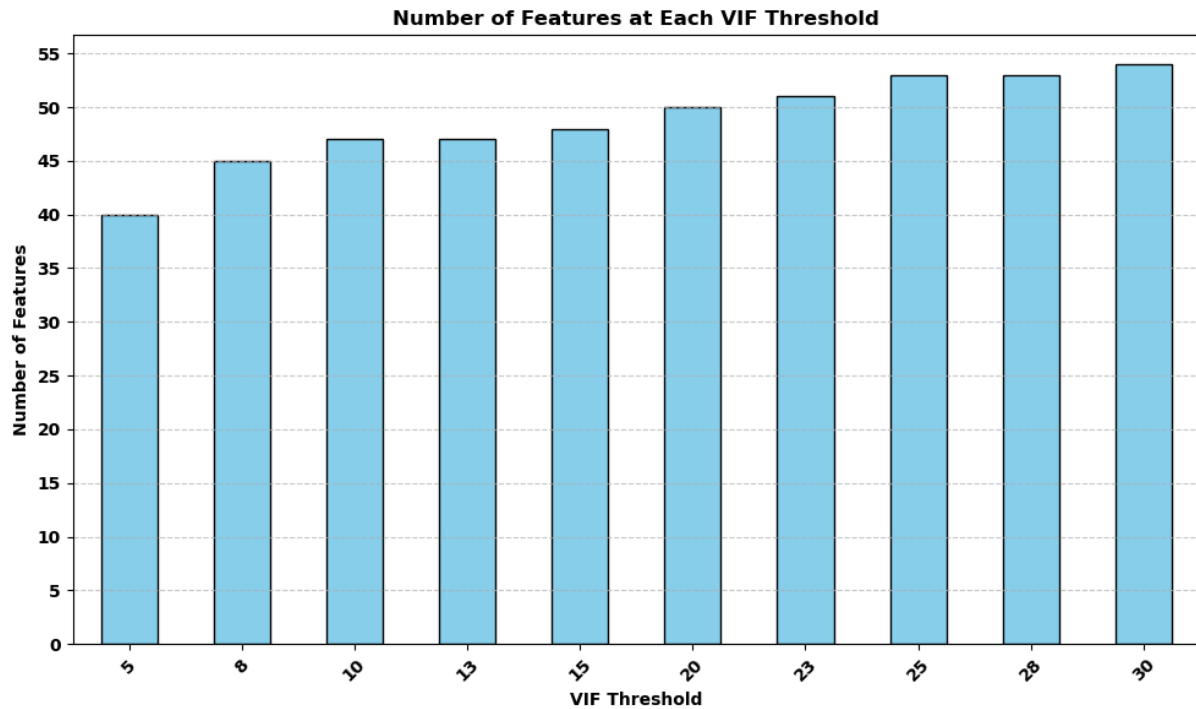


Figure 4.5. Features vs different VIF thresholds.

and pupil-based features. Toward the bottom, features like `Cos2Theta_Min`, `Angle_Mean`, `Fixation_Length`, and `Diff_Y_Skewness` have limited importance. The least important are `Cos2Theta_Max` and `Angle_Max`, which do not contribute at all. Thus, the model heavily prioritizes ADHD symptom scores and gaze-based dynamics over cognitive or less variable features. In addition to the initial train-test evaluation, cross-validation was applied to ensure a more robust and reliable assessment of model performance. Specifically, 5-fold, 10-fold, and 15-fold cross-validation schemes were implemented. In each approach, the dataset was divided into equal-sized subsets or “folds.” During each round of cross-validation, one fold was held out as the validation set while the remaining folds were used to train the model. This process was repeated such that every fold served once as the validation set.

For the 5-fold cross-validation, the dataset was divided into five parts. The model was trained and validated five times, each time with a different partition held out for testing. The classification metrics, including precision, recall, F1-score, and accuracy, were computed for each iteration and then averaged to obtain an overall performance estimate. This approach helped in reducing the potential variability introduced by random train-test splits.

The 10-fold cross-validation followed a similar procedure, with the dataset partitioned into ten subsets. The increase in the number of folds resulted in a larger proportion of data being used for training in each iteration, which typically leads to a more accurate and less biased estimate of the



|                |                       |                       |
|----------------|-----------------------|-----------------------|
| X_Mean         | Pupil_Kurtosis        | Angle_Std             |
| X_Std          | Diff_X_Mean           | Angle_Min             |
| X_Min          | Diff_X_Median         | Angle_Max             |
| X_Max          | Diff_X_Std            | Angle_Skewness        |
| X_Skewness     | Diff_X_Skewness       | Angle_Kurtosis        |
| X_Kurtosis     | Diff_X_Kurtosis       | Cos2Theta_Mean        |
| Y_Mean         | Diff_Y_Mean           | Cos2Theta_Min         |
| Y_Std          | Diff_Y_Median         | Cos2Theta_Max         |
| Y_Min          | Diff_Y_Max            | Cos2Theta_Kurtosis    |
| Y_Max          | Diff_Y_Skewness       | Max_Cross_Correlation |
| Y_Skewness     | Diff_Y_Kurtosis       | Min_Cross_Correlation |
| Y_Kurtosis     | Log_Velocity_Std      | Fixation_Length       |
| Pupil_Median   | Log_Velocity_Min      | Succade_Length        |
| Pupil_Std      | Log_Velocity_Max      | z.ASRS_score          |
| Pupil_Min      | Log_Velocity_Kurtosis | z.sspan_score         |
| Pupil_Skewness | Angle_Mean            | z.ravens_score        |
|                | Angle_Median          |                       |

Figure 4.6. The feature set of 49 features after applying the VIF threshold of 20.

model's performance. The average classification report was derived by computing mean values across the ten validation results, and accuracy was reported as the mean of the ten accuracy scores.

Likewise, a 15-fold cross-validation was conducted. This higher number of folds further reduced bias in estimating the generalization error. However, it also increased the computational load, as the model had to be trained and evaluated fifteen times. Despite the added cost, this method provided an even more granular and stable estimate of the logistic regression model's effectiveness. Once again, the classification reports and accuracy were averaged to summarize the results.

Logistic Regression achieved an accuracy of  $70.54\% \pm 4.08$  in 5-fold cross-validation, indicating a stable and fairly accurate performance. When evaluated with 10-fold cross-validation, its accuracy dropped slightly to  $64.77\% \pm 13.18$ , with a noticeable increase in variability. In the 15-fold setting, the model reached  $65.60\% \pm 17.85$ , showing some consistency

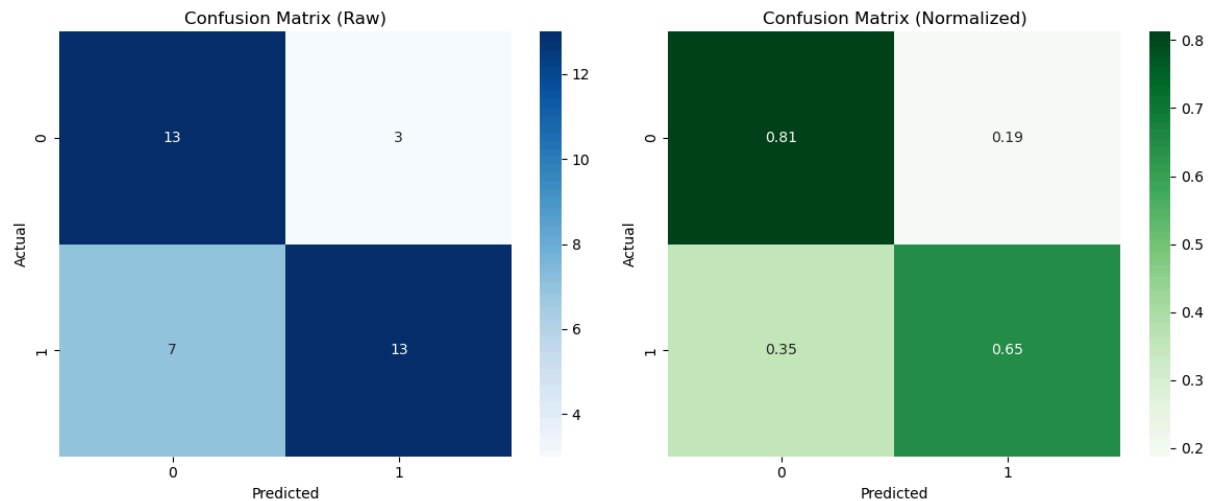


Figure 4.7. Confusion matrix applying logistic regression.

in average performance but higher variation across folds. These results suggest that Logistic Regression performs steadily with fewer folds but becomes more variable with increased partitioning.

Overall, this experiment confirmed that logistic regression can serve as a baseline model for binary classification. Its interpretability, efficiency, and simplicity make it a suitable choice, especially in the early stages of model evaluation. The use of cross-validation provided a comprehensive understanding of the model's stability and helped to mitigate the risk of overfitting by ensuring that the performance metrics were not overly dependent on a specific data split. The findings from this evaluation will serve as a reference for comparing more advanced models in subsequent experiments.

## 4.4 Decision Trees

The next experiment on the dataset was based on Decision trees. Decision trees are widely used in machine learning for tasks involving classification and regression due to their straightforward structure and interpretability. These models work by dividing the dataset into smaller, more homogenous groups based on input feature values, eventually leading to a specific outcome. The process resembles a flowchart, where each decision point or node splits the data based on the most informative feature at that stage. Initially, the complete dataset is placed at the top node, referred to as the root. The algorithm then selects a feature that effectively separates the data according to a predefined criterion, such as Gini impurity or information gain. This feature is used to divide the dataset, forming branches that extend to new nodes. This splitting continues

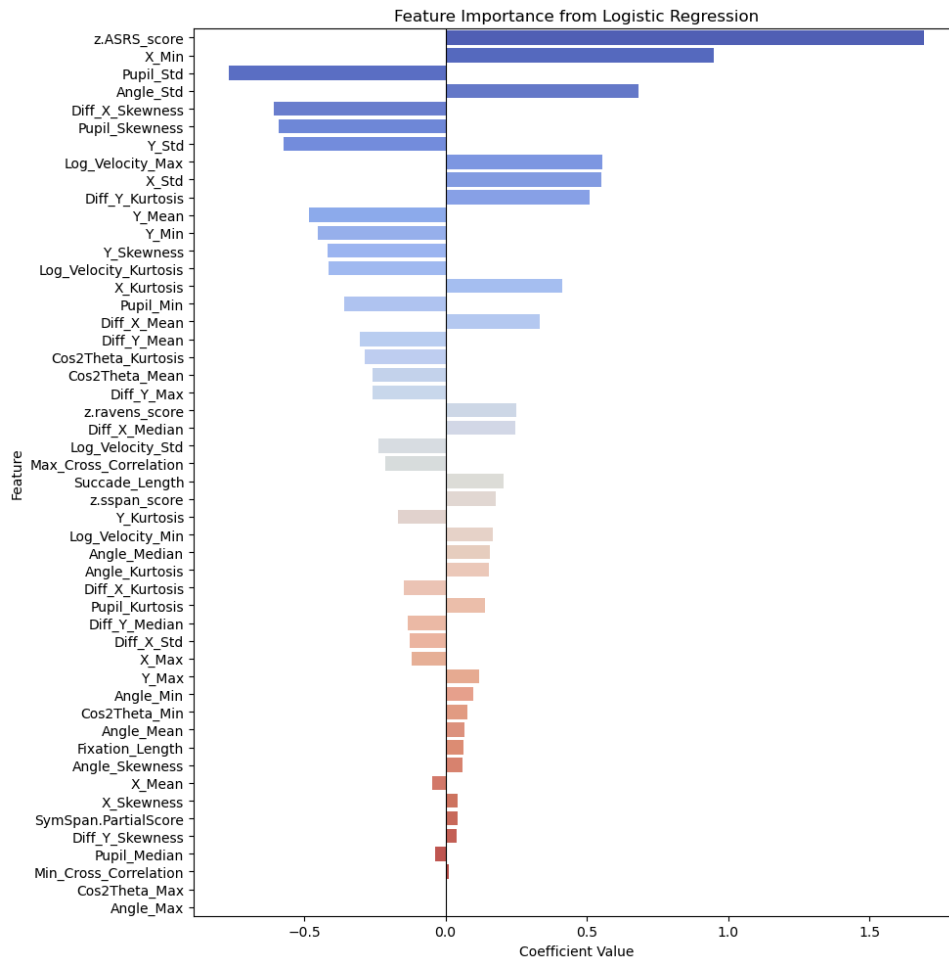


Figure 4.8. Feature coefficients after applying logistic regression.

recursively, creating a tree structure. The recursion halts when a node reaches a stopping condition—for instance, when all samples in a node belong to the same class, a maximum depth is reached, or further splits do not significantly improve the model.

Each internal node in the tree corresponds to a decision rule based on a particular feature and value, whereas each leaf node provides the final output or prediction. In classification tasks, for example, a leaf node might represent a class label. The model navigates from the root to a leaf based on the values of input features, producing the prediction. One of the key strengths of decision trees is their ease of interpretation. They allow users to trace the decision-making path and understand which features influence the outcome most. However, they can be sensitive to fluctuations in the data and may overfit if not properly controlled. To address this, constraints like limiting the depth or applying pruning techniques are commonly used. Moreover, decision trees serve as foundational elements in advanced ensemble models like random forests and gradient boosting systems. The split of 70% training and 30% testing data to be applied DT gives 75% accuracy. Figure 4.9 shows the confusion matrix applying DT.

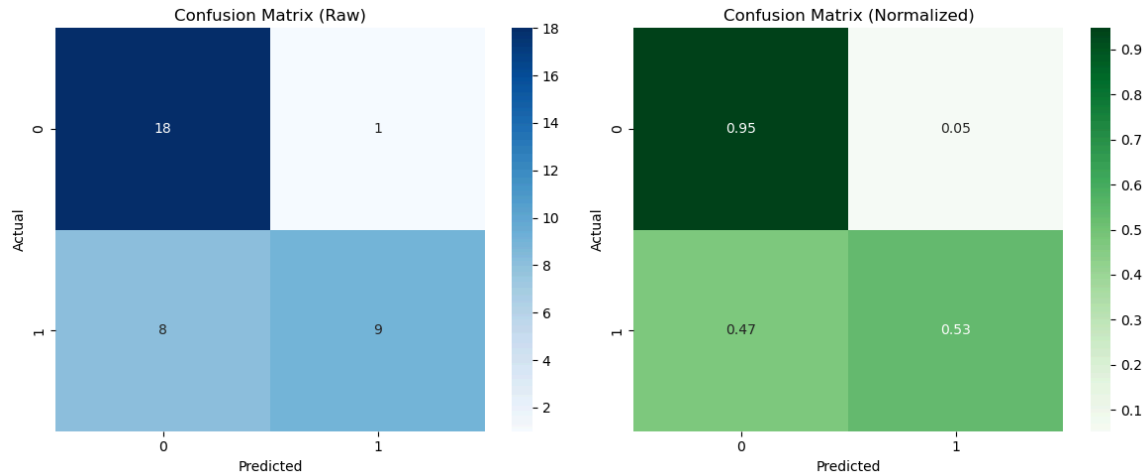


Figure 4.9. Confusion matrix after applying decision trees.

Figure 4.10 shows the important features using DT. The feature importance results from the decision tree model show that the most influential predictor is `z.ASRS_score`, indicating that self-reported ADHD symptoms play the most significant role in classification. This feature alone contributes over one-third of the total importance, far more than any other variable. The next most important feature is `Diff_Y_Max`, reflecting the maximum change in vertical gaze position. This suggests that variability in vertical eye movement is a strong behavioral indicator. Other gaze-related features such as `Cos2Theta_Kurtosis` and `Angle_Min` also hold considerable importance, indicating that aspects of gaze angle and its statistical distribution are relevant to the model. Mid-level contributors include `Pupil_Kurtosis` and `Fixation_Length`, pointing to the role of pupil dynamics and fixation behavior. `X_Mean` and `Cos2Theta_Mean` add further insight into average gaze position and angular behavior. `Diff_Y_Skewness` also contributes modestly, reflecting asymmetry in vertical gaze changes. `z.ravens_score`, a cognitive measure, shows the least importance among the selected features. Overall, the model relies most heavily on symptom severity and eye movement behavior, with cognitive metrics offering minimal additional predictive value. The decision tree structure clearly prioritizes behavioral patterns captured through eye tracking in making its classifications. The Decision Tree classifier showed moderate performance with  $67.17\% \pm 5.22$  accuracy in the 5-fold evaluation. Its 10-fold results were nearly the same at  $67.27\% \pm 11.92$ , showing similar average accuracy but greater variability. With 15-fold cross-validation, the performance slightly improved to  $68.10\% \pm 18.68$ , though variability increased further. Overall, Decision Trees maintained consistent mean accuracy, but their stability reduced as the count of folds increased.

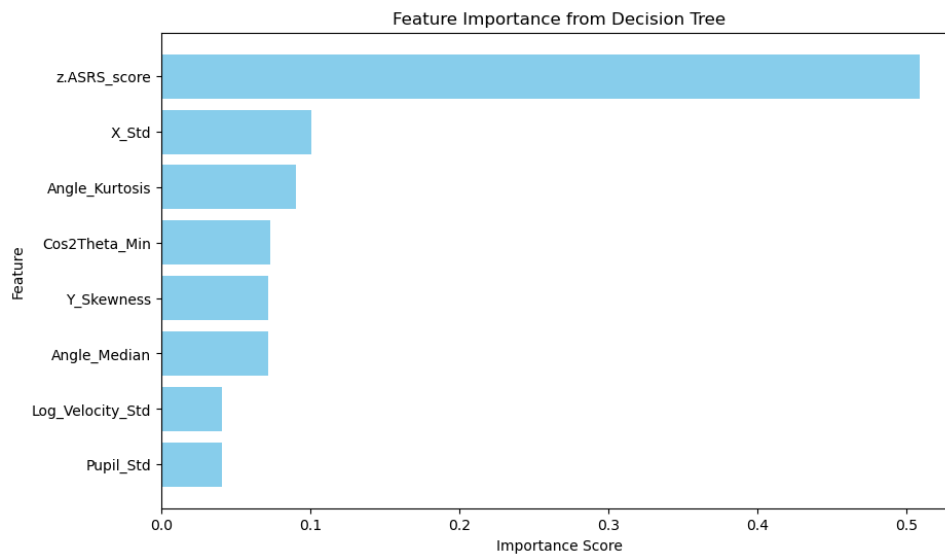


Figure 4.10. Feature importance after applying decision trees.

## 4.5 Random Forests

Random Forest is a widely used ensemble learning method known for its robustness and high predictive accuracy. It constructs a collection of decision trees during the training phase, each trained on a randomly selected subset of the training data and a randomly chosen subset of features. This randomness contributes to the diversity among the trees, which, in turn, enhances the overall generalization ability of the model.

In classification tasks, each decision tree casts a vote for a predicted class, and the class with the majority vote becomes the final prediction. This ensemble strategy helps reduce the risk of overfitting that is often associated with individual decision trees. As a result, Random Forest provides more stable and reliable performance, particularly on complex datasets with intricate feature interactions.

In the conducted experiment, the Random Forest classifier was applied to a dataset using a 70%-30% train-test split. The model achieved an accuracy of 86% on the test data, indicating strong performance on unseen data. Figure 4.11 shows the confusion matrix applying RF.

Figure 4.12 shows the importance of features using RF. The RF model identifies `z.ASRS_score` as the most important feature, indicating that self-reported ADHD symptoms play the leading role in classification. While its influence is strongest, the model also places moderate importance on features like `Pupil_Std`, `Angle_Std`, and `X_Min`, highlighting the relevance of variability in pupil size, gaze angles, and horizontal eye positions. Additional gaze-related metrics such as `Angle_Kurtosis`, `Pupil_Kurtosis`,

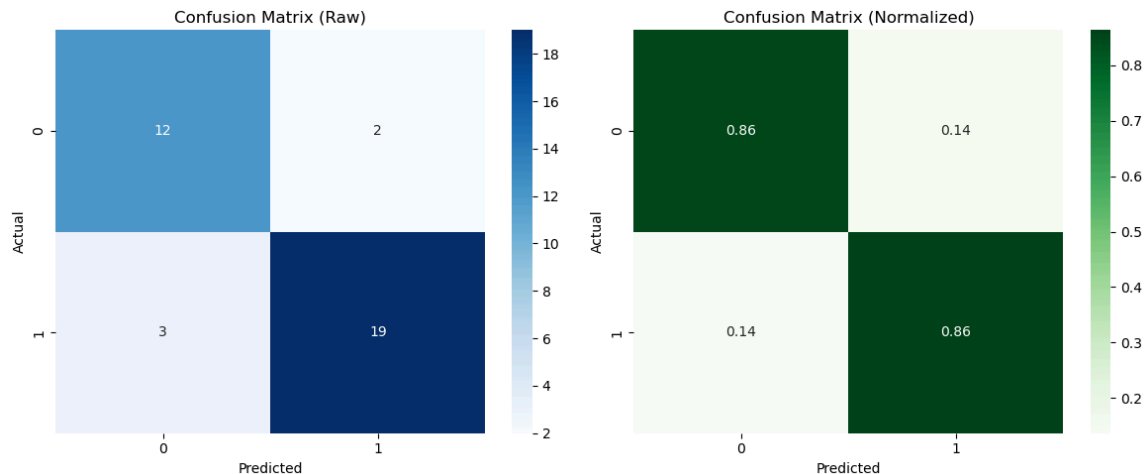


Figure 4.11. Confusion matrix using random forest.

and Angle\_Skewness further contribute to the model's decisions. Mid-ranked features include X\_Kurtosis, Log\_Velocity\_Kurtosis, Y\_Std, and angle statistics like Angle\_Mean and Angle\_Median, showing that the model captures multiple aspects of eye movement dynamics. Cognitive measures like SymSpan.PartialScore and z.ravens\_score are included but play a smaller role, suggesting that while working memory and fluid intelligence provide some predictive value, they are secondary to oculomotor and symptom-based features. Features like Diff\_Y\_Skewness, X\_Std, and Diff\_Y\_Median have little effect on the model, and Log\_Velocity\_Min contributes the least. Overall, the model relies most heavily on ADHD symptom severity and gaze-related behavioral patterns, with minimal dependence on lower-ranked or less variable features.

Random Forest consistently performed the best among all classifiers. It achieved  $76.45\% \pm 6.84$  accuracy in 5-fold,  $76.52\% \pm 12.17$  in 10-fold, and  $78.21\% \pm 13.21$  in 15-fold validation. These results show that the model not only has the highest average accuracy but also maintains relatively low standard deviation, especially in the 5-fold case. Its reliability and robustness across all folds make it a top-performing choice.

These results confirm that Random Forest is capable of delivering high classification performance and remains dependable even when subjected to more rigorous evaluation methods like cross-validation. Its ensemble nature allows it to generalize well, making it a suitable choice for various classification tasks where model reliability is a critical concern.

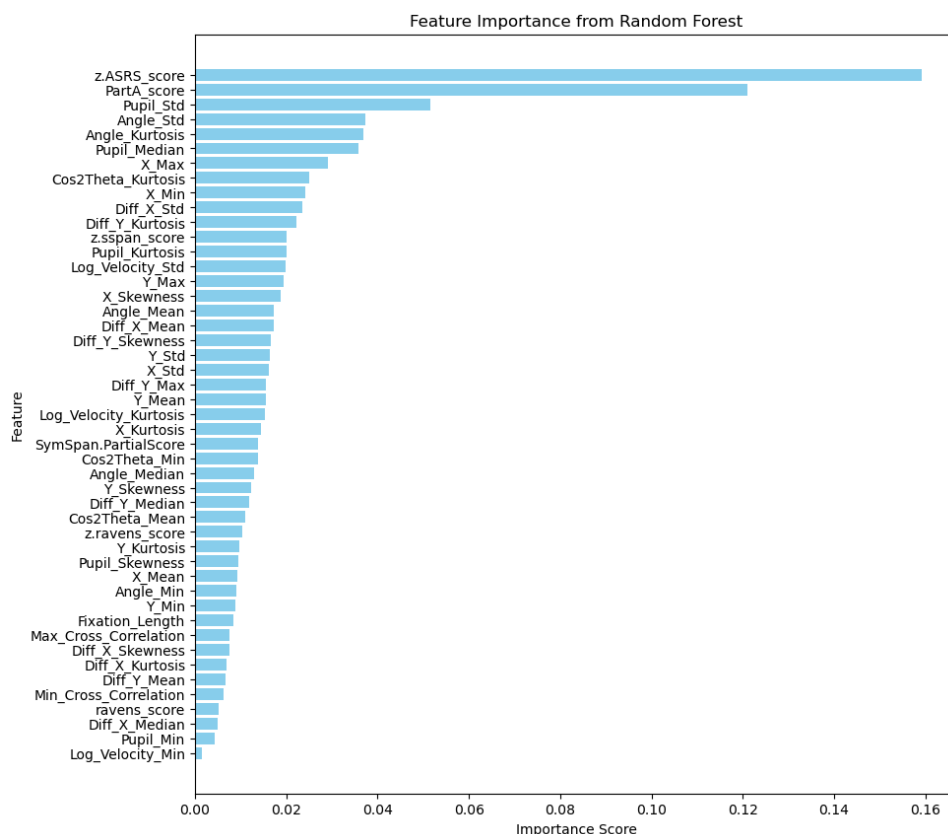


Figure 4.12. Feature importance after applying random forests.

## 4.6 Gradient Boosting

In this experiment, a Gradient Boosting (GB) classification model was applied to a dataset containing eye-tracking data. The Gradient Boosting model was trained on 70% of the data, and testing was carried out on the remaining 30%. From this setup, an accuracy of 77% was obtained. This initial result indicated that the model was capable of learning meaningful patterns from the training data and generalizing them to unseen data with a reasonable level of accuracy. Figure 4.13 shows the confusion matrix applying GB.

Figure 4.14 shows the importance of features using GB. The GB model shows a clear dominance of `z.ASRS_score`, which has the highest importance by a wide margin, indicating that self-reported ADHD symptoms are the most critical feature in prediction. Following this, `Cos2Theta_Kurtosis`, `Cos2Theta_Min`, and `Angle_Std` rank notably high, suggesting that angular gaze dynamics and variability are also important behavioral indicators. Features such as `Diff_Y_Median`, `Angle_Kurtosis`, `X_Std`, and `Pupil_Std` contribute moderately, pointing to the role of both eye movement characteristics and pupil responses. Mid-level contributors include `Diff_Y_Max`, `Y_Mean`, `Pupil_Median`, and

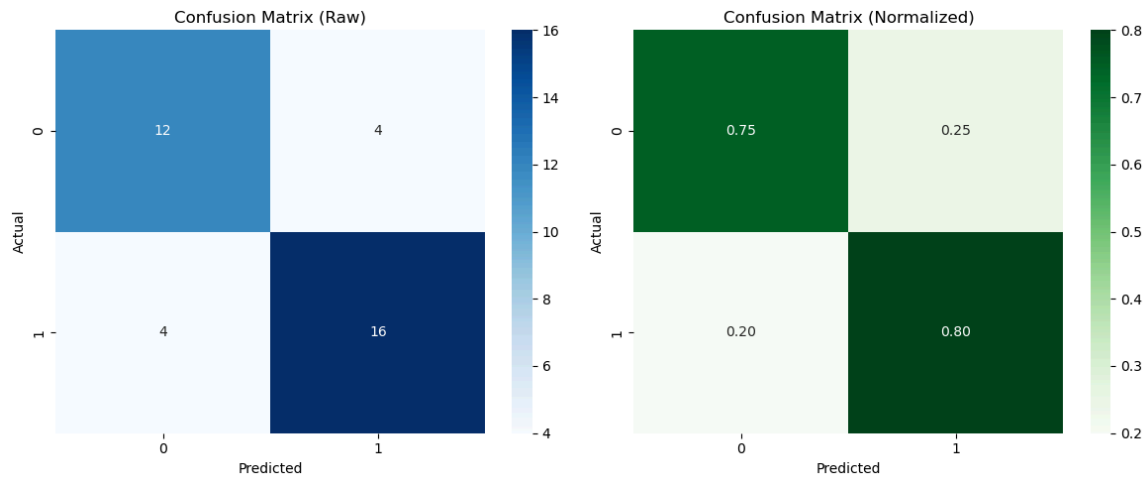


Figure 4.13. Feature importance after applying gradient boosting.

Angle\_Mean, while scaled working memory and cognitive scores like `z.sspan_score` and `SymSpan.PartialScore` also provide some influence. A large number of features—including those related to velocity, skewness, and kurtosis of gaze and pupil signals—have smaller yet consistent contributions, reflecting the model’s ability to integrate subtle variations. However, a long tail of very low-importance features such as `X_Mean`, `Pupil_Min`, and `Diff_X_Skewness` indicates that not all behavioral metrics carry meaningful predictive power. Overall, the model strongly emphasizes ADHD symptom scores while also leveraging a diverse set of eye-tracking features to support its predictions.

To obtain a more reliable estimate of model performance and reduce potential biases introduced by a single split, k-fold cross-validation was also applied. Gradient Boost delivered solid results with  $72.36\% \pm 11.83$  in 5-fold cross-validation. Its performance remained stable in 10-fold ( $71.59\% \pm 12.35$ ) and slightly increased in 15-fold ( $72.38\% \pm 15.16$ ). Although the standard deviation was higher compared to Random Forest, the mean accuracy stayed consistent. This suggests that Gradient Boost is a dependable classifier, though slightly more sensitive to changes in data partitioning.

## 4.7 Naive Bayes

A Naive Bayes classifier was also applied to the dataset as one of the baseline models for binary classification. The primary aim of this experiment was to assess the suitability of Naive Bayes for the given task by evaluating its predictive performance on unseen data. The dataset was split into training and testing subsets using a random 70-30 division as training and testing parts. Once trained, the model was used to generate predictions on the test subset. The accuracy



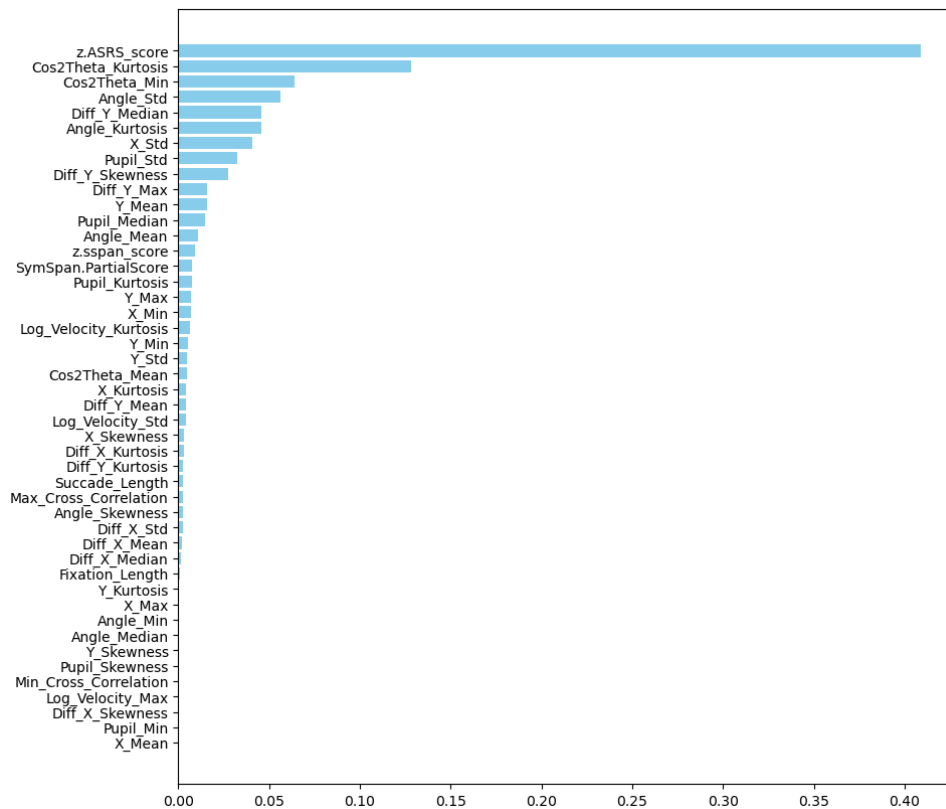


Figure 4.14. Feature importance after applying gradient boosting.

of the model, calculated as the proportion of correctly predicted labels to the total number of predictions, was found to be 55%.

To better interpret the performance, a visual representation of the confusion matrix was generated, as shown in Figure 4.15. This heatmap allowed for a more intuitive understanding of where the model succeeded and where it tended to misclassify. It was especially useful in identifying any potential skew or class imbalance in the model's predictions.

To further evaluate the performance of the Naive Bayes classifier, k-fold cross-validation was applied with varying fold sizes. Naive Bayes had the lowest 5-fold performance at  $62.32\% \pm 9.56$ , but improved significantly in the 10-fold and 15-fold validations, reaching  $70.76\% \pm 11.20$  and  $71.43\% \pm 13.25$ , respectively. This trend indicates that Naive Bayes benefits from more training data in each fold, becoming more accurate and effective as the number of folds increases.

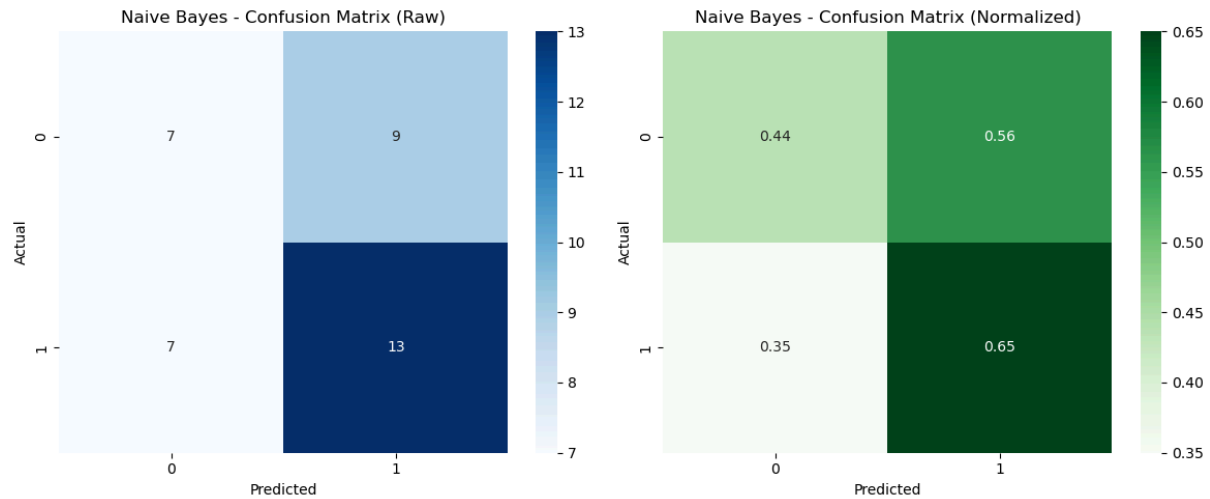


Figure 4.15. Confusion matrix applying Naive Bayes.

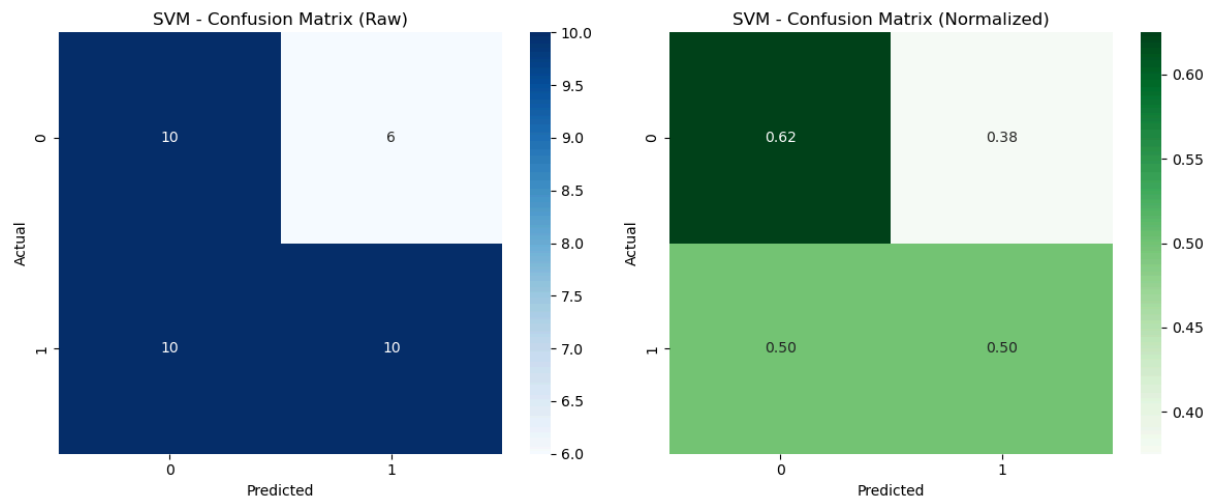


Figure 4.16. Confusion matrix applying Support Vector Machine.

## 4.8 Support Vector Machine

A Support Vector Machine (SVM) classifier was applied to the dataset. For a 70-30 split, the accuracy of the model, calculated as the proportion of correctly predicted labels to the total number of predictions, was found to be 58%. A visual representation of the confusion matrix was generated, as shown in Figure 4.16.

To further evaluate the performance of the SVM classifier, k-fold cross-validation was applied with varying fold sizes. It performed well across all settings, starting with  $69.75\% \pm 8.49$  in 5-fold cross-validation. Its 10-fold accuracy rose slightly to  $70.61\% \pm 11.27$ , while 15-fold results remained consistent at  $70.60\% \pm 18.60$ . Although the average performance was strong,

Table 4.1. Model accuracy comparison using 70-30 train-test split

| Model               | Accuracy (%) |
|---------------------|--------------|
| Naive Bayes         | 55           |
| SVM                 | 55           |
| Logistic Regression | 72           |
| Decision Tree       | 75           |
| Gradient Boosting   | 77           |
| Random Forest       | 86           |

Table 4.2. Mean accuracy and standard deviation of classifiers with different cross-validation folds

| Classifier          | 5-Fold Accuracy | 10-Fold Accuracy | 15-Fold Accuracy |
|---------------------|-----------------|------------------|------------------|
| Naive Bayes         | $0.62 \pm 0.10$ | $0.71 \pm 0.11$  | $0.71 \pm 0.13$  |
| Decision Tree       | $0.67 \pm 0.05$ | $0.67 \pm 0.12$  | $0.68 \pm 0.19$  |
| SVM                 | $0.70 \pm 0.08$ | $0.71 \pm 0.11$  | $0.71 \pm 0.19$  |
| Logistic Regression | $0.71 \pm 0.04$ | $0.65 \pm 0.13$  | $0.66 \pm 0.18$  |
| Gradient Boost      | $0.72 \pm 0.12$ | $0.72 \pm 0.12$  | $0.72 \pm 0.15$  |
| Random Forest       | $0.76 \pm 0.07$ | $0.77 \pm 0.12$  | $0.78 \pm 0.13$  |

the high variation in 15-fold suggests sensitivity to data splits. Overall, SVM is a reliable model with generally stable performance.

## 4.9 Comparison of Model Performance

To show a comparison of model performance, Tables 4.1 and 4.2 summarize the classification accuracies obtained through two evaluation strategies. Table 4.1 presents the accuracy results from a 70-30% train-test split, offering a snapshot of how well each model performs on unseen data in a single run, where GB shows the highest accuracy of 86%.

Table 4.2 shows the results from k-fold cross-validation (with  $k = 5, 10, 15$ ), which provides a more robust and stable estimate of model generalization by averaging performance across multiple data partitions. Figure 4.17 shows the plot of different accuracies obtained for different Cross-validation performances of different classifiers, including standard deviation.

Furthermore, different VIF thresholds were checked for random forest since it showed the highest performance in both 70-30% and different fold-based experiments. Figure 4.18 shows

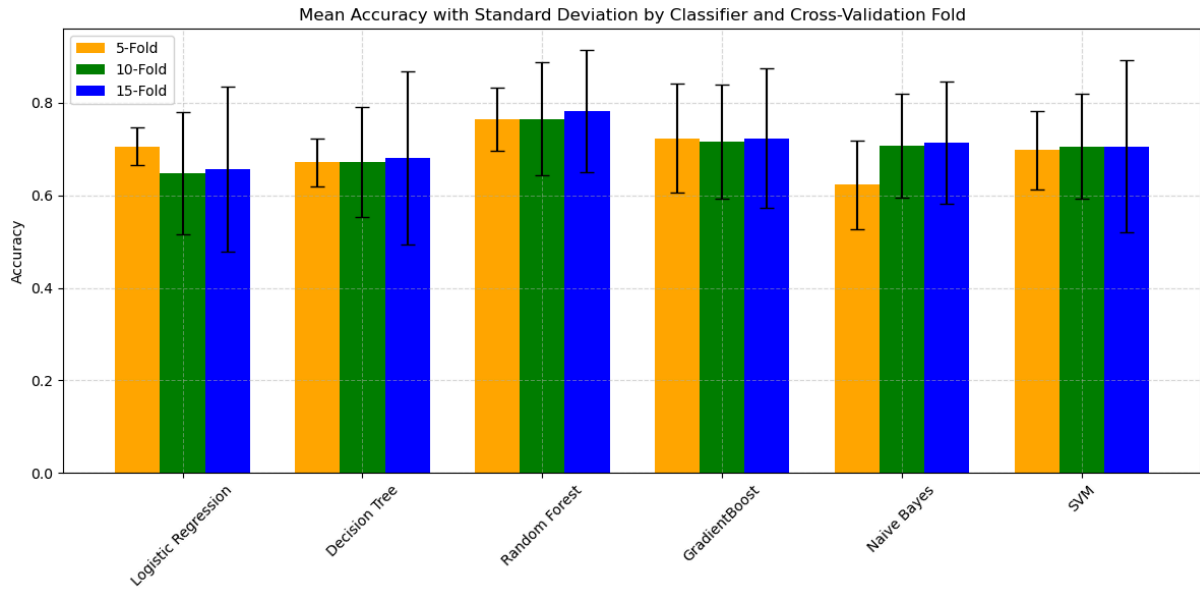


Figure 4.17. Cross-validation performances of different classifiers, including standard deviation.

different accuracies obtained for different thresholds for 5-fold cross-validation of random forests experiments.

## 4.10 AIC and BIC in Model Selection

The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are commonly used for model selection by balancing goodness-of-fit with model complexity, helping prevent overfitting. AIC is defined as  $AIC = 2k - 2\ln(L)$ , where  $k$  is the number of parameters and  $L$  is the likelihood of the model, with lower AIC values indicating better models. BIC, defined as  $BIC = \ln(n)k - 2\ln(L)$ , where  $n$  is the sample size, also penalizes model complexity, but more strongly than AIC due to the  $\ln(n)$  term, making BIC particularly useful for larger datasets. Both AIC and BIC are widely used in model comparison to identify the model that best balances fit and complexity.

The AIC and BIC were applied to Naive Bayes and SVM models, where for Logistic Regression, the AIC is 208.32 and the BIC is 289.08, suggesting a relatively better fit with fewer penalties for model complexity. In contrast, Naive Bayes has a much higher AIC of 534.68 and BIC of 693.03, which indicates that it might be a less optimal model in comparison, likely due to overfitting or unnecessary complexity. In general, lower AIC and BIC values suggest a better model fit, so Logistic Regression seems to be the preferred model in this case.

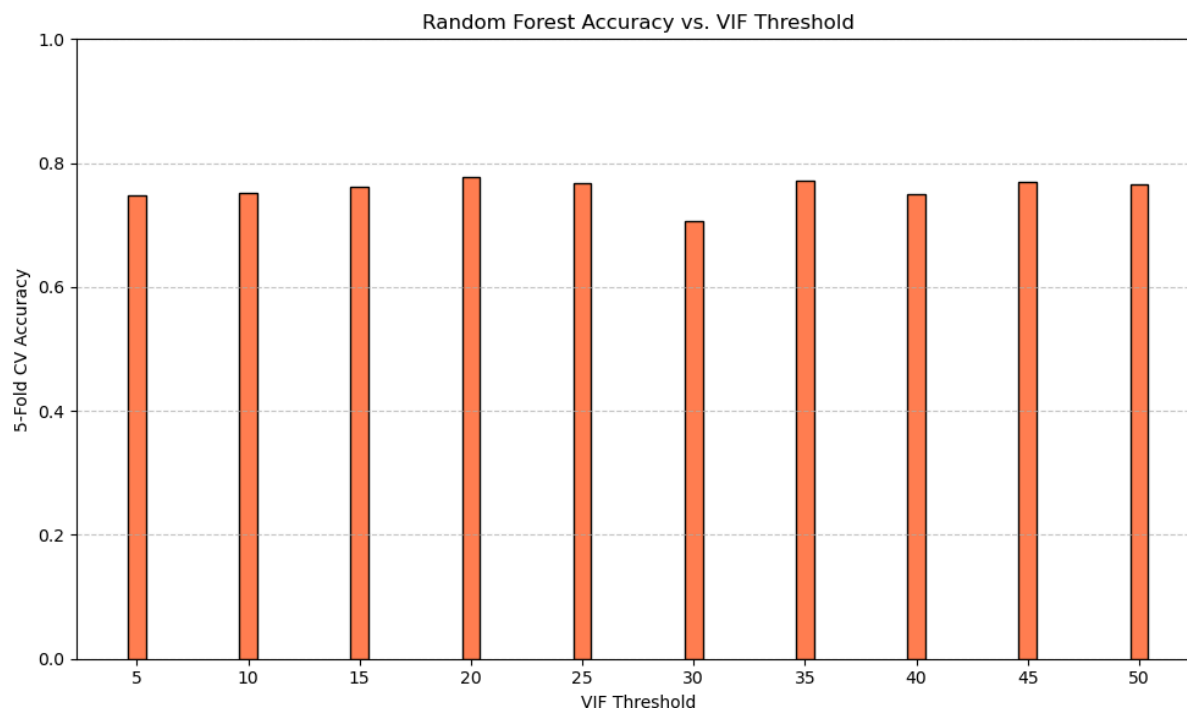


Figure 4.18. VIF versus accuracies for 5-fold cross-validation of RF experiments.

## 4.11 Model Parameters

In this study, the hyperparameters of each machine learning classifier were retrieved using the `get_params()` method provided by the scikit-learn Python library for classical machine learning. This method returns a dictionary containing the configuration settings of a model instance, either default values or those that were manually specified. Understanding these parameters is important for interpreting how each model was set up and operated during training and evaluation. For example, the logistic regression classifier was configured with L2 regularization (`penalty='l2'`) and a regularization strength `C=1.0`, using the `'lbfgs'` optimization solver.

The Naive Bayes model (GaussianNB) used default settings with a small smoothing parameter (`var_smoothing=1e-09`) to handle numerical stability. The decision tree model was based on the Gini impurity criterion and had no maximum depth limit, allowing it to grow until all leaves were pure or other stopping conditions were met. The random forest classifier included 100 trees (`n_estimators=100`), used feature sampling (`max_features='sqrt'`), and was built using bootstrap aggregation (`bootstrap=True`).

Finally, the gradient boosting classifier also used 100 estimators and a learning rate of 0.1, optimizing performance with the `'log_loss'` loss function and `'friedman_mse'`

Table 4.3. Model accuracy comparison using 70-30 train-test split without considering metafeatures

| Model               | Accuracy (%) |
|---------------------|--------------|
| Gradient Boosting   | 47           |
| Naive Bayes         | 55           |
| SVM                 | 56           |
| Logistic Regression | 64           |
| Decision Tree       | 63           |
| Random Forest       | 78           |

splitting criterion. These parameter settings influenced how each algorithm learned from the data, and retrieving them helps document and justify the modeling choices made throughout the study.

## 4.12 Results Excluding Metafeatures

The experimental results without incorporating metafeatures provide valuable insights into the standalone predictive performance of different classifiers in distinguishing between ADHD and typically developed (TD) participants. Using a 70-30 train-test split, the Random Forest model demonstrated the highest accuracy of 78%, significantly outperforming other classifiers. Logistic Regression and Decision Tree models showed moderate accuracy at 64% and 63%, respectively, while Support Vector Machine (SVM) and Naive Bayes achieved 56% and 55%. Gradient Boosting, surprisingly, yielded the lowest performance with an accuracy of only 47%, suggesting its potential inefficiency without the context of metafeatures in this dataset.

Figure 4.19 presents the feature importance values derived from logistic regression. Since logistic regression is a linear model, importance is based on the coefficients of the input features. In the experiment without metafeatures, the logistic regression model depended entirely on eye-tracking features. The most important were `Y_Std` and `Cos2Theta_Min`, indicating that vertical gaze variability and angular gaze metrics played key roles. `Angle_Mean`, `Pupil_Std`, and `Angle_Std` also ranked highly, showing the relevance of gaze dynamics and pupil size changes. Features like `Fixation_Length`, `Angle_Min`, and `Pupil_Skewness` contributed moderately, along with measures of gaze asymmetry and variability. Lower-ranked features, including velocity and correlation metrics, provided some influence, while a few, such as `Cos2Theta_Max` and `Angle_Max`, had no impact. Overall, the model effectively used gaze and pupil behavior to classify data without relying on cognitive or symptom-based scores.

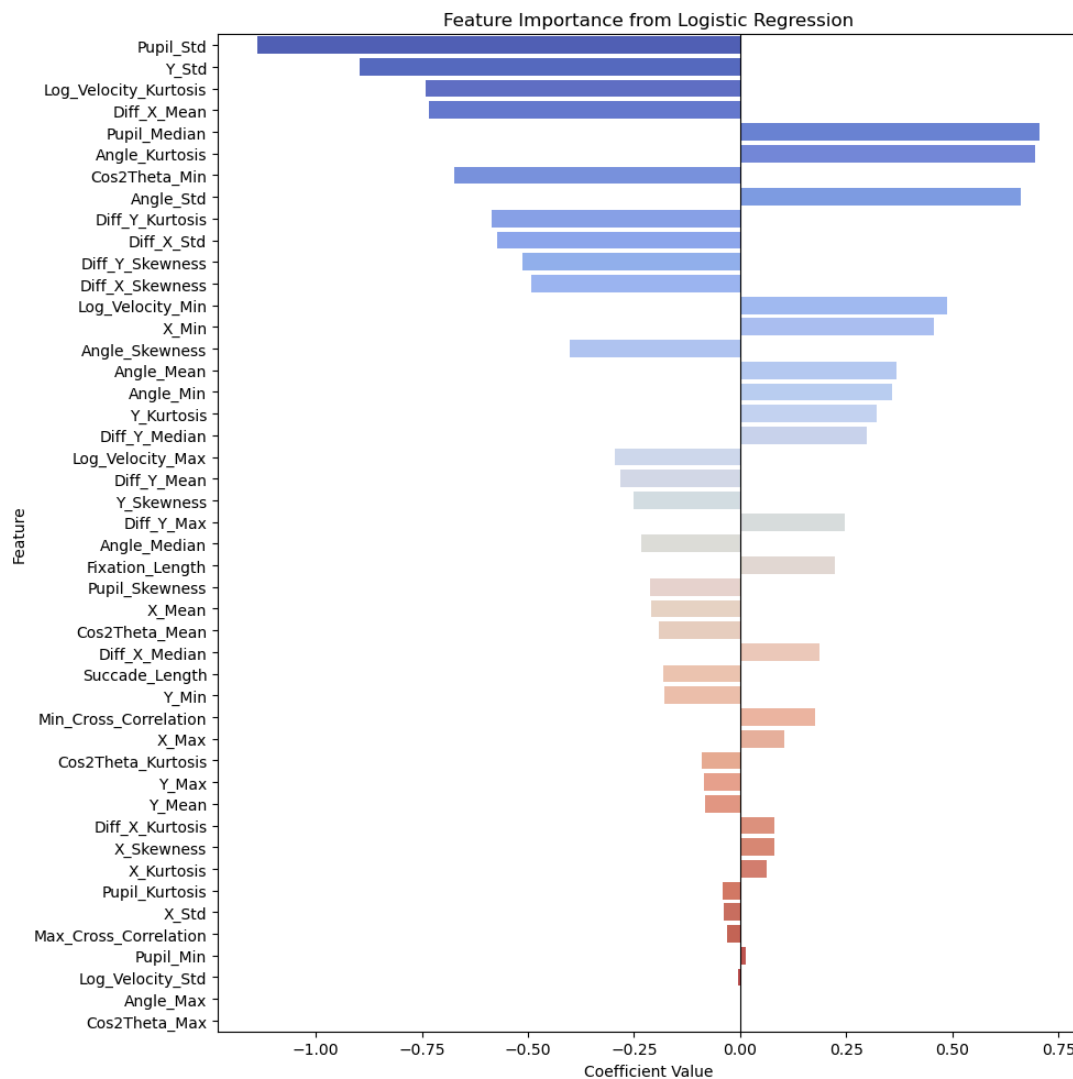


Figure 4.19. Feature importance after applying logistic regression without metafeatures.

Figure 4.20 shows feature importance from decision trees, where importance is determined by the reduction in impurity achieved by each feature across all splits. The most important feature was *Cos2Theta\_Kurtosis*, indicating that the distribution of angular gaze shifts played a key role in classification. This was followed closely by *Pupil\_Std* and *Angle\_Std*, highlighting the importance of pupil size variability and gaze angle variation. *Cos2Theta\_Min* and *Log\_Velocity\_Kurtosis* also ranked high, suggesting that specific gaze angle values and velocity patterns contribute significantly. *X\_Min* and *Angle\_Kurtosis* showed moderate importance, reflecting the role of eye position and gaze shape. Lower-ranked features like *Pupil\_Min*, *Saccade\_Length*, and *Y\_Skewness* still added some value, reinforcing the model's reliance on oculomotor behavior.

Figure 4.21 displays the feature importance scores computed by the random forest classifier.

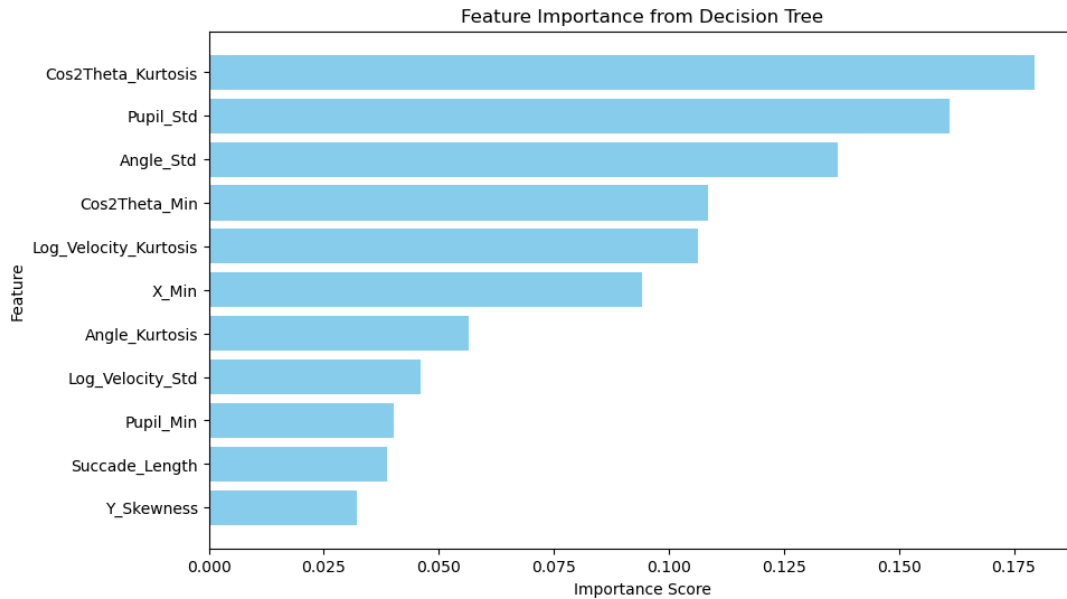


Figure 4.20. Feature importance after applying decision trees without metafeatures.

This ensemble model aggregates importance over many decision trees, offering a more robust estimation of feature relevance. In the random forest model without metafeatures, the most important features were `Angle_Std` and `Y_Std`, indicating that variability in gaze angles and vertical eye movement played a key role in classification. `Pupil_Std` and `Pupil_Median` followed, highlighting the relevance of pupil size variation. Other angle-related metrics like `Angle_Kurtosis` and `Angle_Skewness`, along with `Log_Velocity_Std` and `Cos2Theta_Kurtosis`, contributed moderately, showing the model's reliance on gaze dynamics. Features such as `X_Min`, `X_Max`, and various statistical measures related to gaze and pupil signals added additional value. Lower-ranked features like `Diff_Y_Skewness`, `Diff_X_Mean`, and `Log_Velocity_Min` had minimal impact. Overall, the model captured a broad range of eye movement characteristics to support prediction.

Figure 4.22 illustrates the importance values from gradient boosting, which evaluates features based on their contribution to improving the model's predictions through successive iterations. Together, these visualizations help identify which eye-tracking features are most predictive in distinguishing between ADHD and TD classes when no meta-features are used. In the gradient boosting model without metafeatures, the most important features were `Pupil_Std` and `Angle_Std`, indicating that variability in pupil size and gaze angles had the strongest influence on classification. `Succade_Length` and `Cos2Theta_Min` followed, suggesting that eye movement stability and minimum angular values were also key. Other significant contributors included `Diff_X_Std`, `Angle_Kurtosis`, and `Log_Velocity_Kurtosis`, showing the model's sensitivity to statistical patterns in eye movement and velocity. Mid-ranked



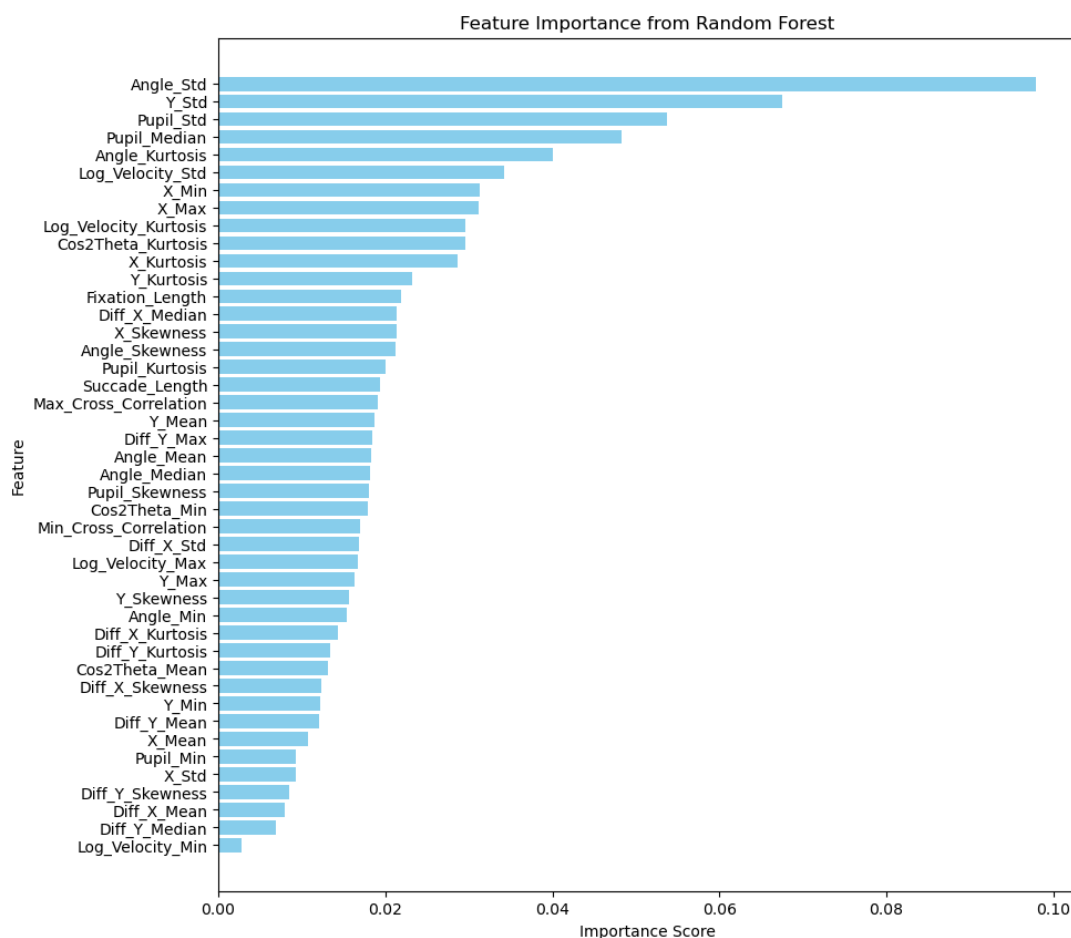


Figure 4.21. Feature importance after applying random forests without metafeatures.

features like `Y_Min`, `Pupil_Skewness`, `Diff_Y_Kurtosis`, and `Diff_Y_Max` added further behavioral insight. Lower-ranked features such as `Angle_Median`, `X_Mean`, and `Pupil_Median` had minimal impact. Overall, the model relied heavily on fine-grained eye-tracking metrics, especially those that capture variability and asymmetry.

Figure 4.23 (left) shows the relationship between the number of features retained and their corresponding VIF (Variance Inflation Factor) scores. Figure 4.23 (right) illustrates how the classification accuracy of the random forest model changes as features are progressively removed based on VIF. This helps assess whether reducing multicollinearity can improve or maintain classification performance without meta-features. It can be noticed that after the threshold of 20, there was no improvement in accuracy. Hence, the VIF threshold was chosen as 20.

Table 4.4 presents the performance of six different classifiers evaluated using 5-fold, 10-fold, and 15-fold cross-validation, without the inclusion of any metafeatures. Each value represents the mean accuracy across folds, accompanied by its standard deviation (shown as the accuracy

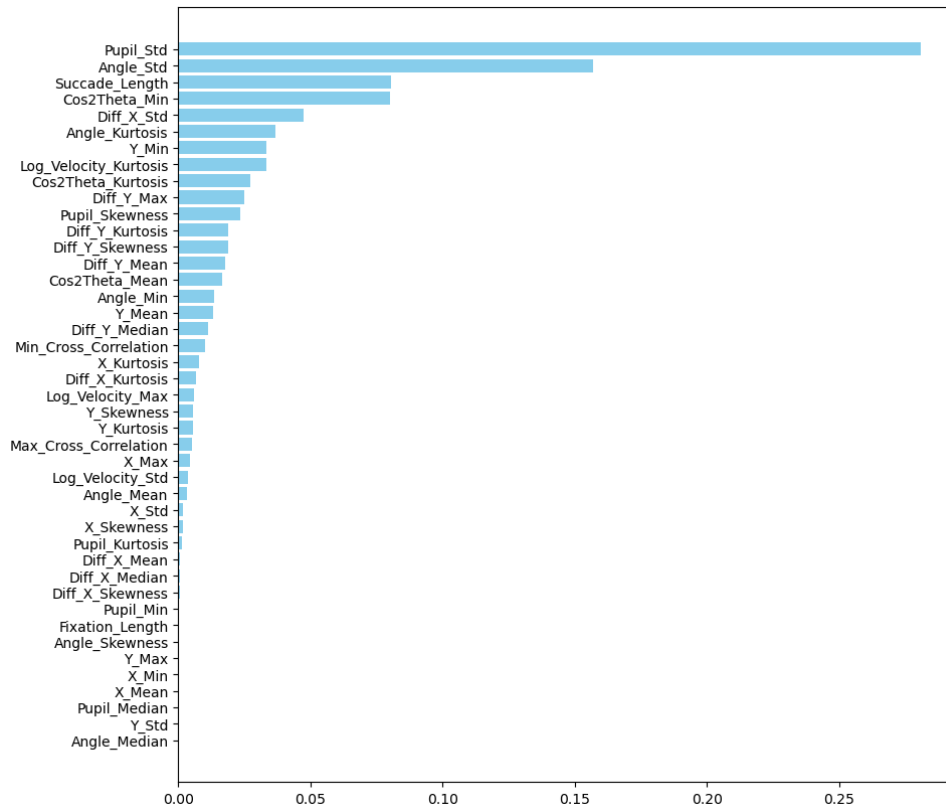


Figure 4.22. Feature importance after applying gradient boosting without metafeatures.

$\pm$  SD), both rounded to two decimal places.

From the results, it is evident that ensemble models such as Random Forest and Gradient Boost outperform the other classifiers in terms of average accuracy across all fold settings. Random Forest achieved the highest 5-fold accuracy at 0.66 and maintained strong performance in both 10-fold (0.66) and 15-fold (0.65) evaluations, despite moderate variability (standard deviations around 0.17–0.18). Gradient Boost closely followed with consistent improvement across folds, reaching 0.67 in 15-fold validation. In contrast, traditional models like Logistic Regression and Decision Trees showed lower performance, both averaging around 0.56 in 5-fold. While Decision Tree showed a notable increase in 10-fold accuracy (0.66), its performance dropped again in 15-fold, suggesting inconsistency. Logistic Regression remained relatively flat and stable, but did not improve with more folds. SVM demonstrated improvement with larger fold sizes, going from 0.56 in 5-fold to 0.66 in 15-fold, showing that it benefits from more data per training fold. Naive Bayes performed moderately, gradually increasing from 0.57 to 0.62 across folds with low variability. Figure 4.24 shows the plot of different accuracies obtained for different Cross-validation performances of different classifiers, including standard deviation, without metafeatures.

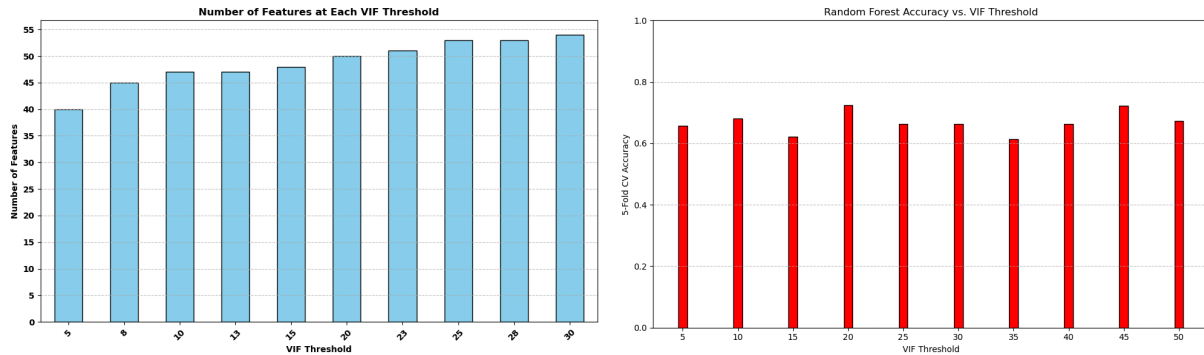


Figure 4.23. VIF versus number of features (left) and VIF versus accuracies using random forests (right), both without metafeatures.

Table 4.4. Mean accuracy and standard deviation of classifiers using cross-validation without metaFeatures

| Classifier          | 5-Fold Accuracy | 10-Fold Accuracy | 15-Fold Accuracy |
|---------------------|-----------------|------------------|------------------|
| Decision Tree       | $0.56 \pm 0.09$ | $0.66 \pm 0.17$  | $0.58 \pm 0.12$  |
| Logistic Regression | $0.56 \pm 0.04$ | $0.59 \pm 0.17$  | $0.56 \pm 0.14$  |
| SVM                 | $0.56 \pm 0.13$ | $0.64 \pm 0.17$  | $0.66 \pm 0.15$  |
| Naive Bayes         | $0.57 \pm 0.07$ | $0.61 \pm 0.09$  | $0.62 \pm 0.11$  |
| Gradient Boost      | $0.64 \pm 0.10$ | $0.63 \pm 0.14$  | $0.67 \pm 0.18$  |
| Random Forest       | $0.66 \pm 0.10$ | $0.66 \pm 0.17$  | $0.65 \pm 0.18$  |

The results indicate that ensemble models are more robust and generalize better across validation strategies, while simpler models show higher variance and lower average performance when no meta-features are included. The results also highlight the Random Forest model as the most robust and generalizable classifier in the absence of metafeatures. While other models show varying degrees of performance, the integration of metafeatures is likely to further differentiate their effectiveness. These baseline evaluations are essential for understanding how models behave with raw eye-tracking data alone and serve as a comparison point for subsequent experiments involving additional feature enhancements.

The results without metafeatures indicate that including metafeatures is better due to their contribution to generating better performance in terms of accuracy.

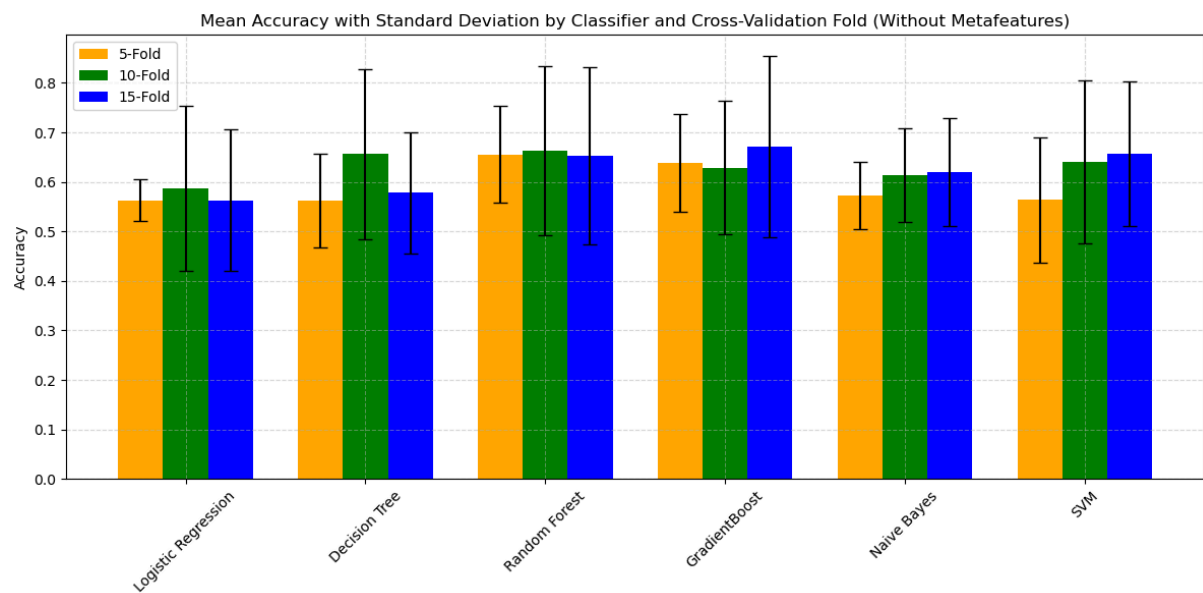


Figure 4.24. Cross-validation performances of different classifiers, including standard deviation without metafeatures.

# Chapter 5

## Discussion and Conclusions

The primary goal of this study was to investigate the applicability of classical machine learning techniques to distinguish between individuals with Attention Deficit/Hyperactivity Disorder (ADHD) and those with typical development (TD), using gaze-tracking eye data. A structured methodology was followed, beginning with feature extraction, visualization and multicollinearity reduction, followed by the application and evaluation of several traditional classification models.

### 5.1 Summary of Key Findings

The initial inspection of the dataset provided a preliminary understanding of data distribution and class separability. Certain features, such as *PartA\_score* (raw ADHD symptom score from ASRS) and *z.ASRS\_score* (standardized ADHD symptom score), showed strong discrimination between the ADHD and TD groups. However, other features exhibited considerable overlap, suggesting that not all predictors carry the same discriminative power. These insights emphasized the importance of feature selection in improving model performance.

To address multicollinearity among features, a Variance Inflation Factor (VIF) analysis was conducted. Features with VIF values exceeding the threshold of 10 were excluded, resulting in a refined set of 49 predictors from an initial 66. This step was crucial in ensuring model stability—especially for algorithms that rely on linear assumptions—by reducing redundancy and improving coefficient precision.

Among the models evaluated, Random Forest demonstrated the best performance, achieving the highest classification accuracy of 86% using a 70–30 train-test split. The ensemble nature of this algorithm, which aggregates predictions from multiple decision trees trained on random

subsets of features and data, enabled robust performance across different data splits and cross-validation settings. This consistency highlights Random Forest as a strong candidate for ADHD classification using gaze data.

Gradient Boosting also performed well, reaching 77% accuracy on the test set. Although slightly lower than Random Forest, its performance remained stable, particularly under 10-fold cross-validation, suggesting effective modeling of complex feature interactions.

Decision Trees offered intuitive and interpretable model structures but were more sensitive to data partitioning. Although a maximum accuracy of 75% was observed during the 10-fold cross-validation, the performance fluctuated between folds, indicating a higher variance and susceptibility to overfitting without regularization.

Logistic Regression, which served as the baseline model, achieved 72% accuracy. Despite being a simple linear method, it performed reasonably well and yielded consistent results across folds. Its strengths lie in its ease of interpretation and computational efficiency, making it valuable for preliminary analyses and benchmarking.

With 5-fold cross-validation, the Naive Bayes model reached 63.03% accuracy. Increasing the number of folds to 10 improved the accuracy to 72.27%, while the 15-fold validation slightly reduced it to 70.59%. These results suggest that the model benefits from larger training splits, although performance stabilizes beyond 10 folds.

The SVM model achieved 67.23% accuracy with a 5-fold cross-validation, increasing to 73.11% with 10 folds, and then slightly decreasing to 69.75% with 15 folds. Thus, SVM demonstrated its strongest generalization performance under 10-fold cross-validation, indicating this as the most balanced setting for the dataset.

Overall, cross-validation with 5, 10, and 15 folds was instrumental in evaluating the robustness of the model. Ten-fold cross-validation generally offered the best trade-off between training data size and evaluation stability.

Models were also evaluated without the inclusion of meta-features to assess their added value. In the 70–30 train-test split, Random Forest remained the top performer with 78% accuracy, followed by Logistic Regression and Decision Tree with 64% and 63% respectively. SVM and Naive Bayes delivered moderate performance (56% and 55%), while Gradient Boosting performed the worst at 47%. In cross-validation, Random Forest maintained its lead with up to 68.15% accuracy, closely followed by Gradient Boosting and SVM. Logistic Regression showed the lowest performance, highlighting the importance of meta-features in enhancing linear model effectiveness. These results confirm that even without engineered features,

ensemble models are capable of capturing complex data patterns and serve as strong baselines for the classification of ADHD.

Random Forest consistently delivered the strongest performance across both evaluation setups. Using a 70–30 train-test split, it achieved 86% accuracy with meta-features, compared to 78% without. Similarly, under k-fold cross-validation, it reached up to 78% accuracy with meta-features and 67% without. These findings suggest that meta-features significantly enhance model performance, especially for ensemble methods, by providing additional high-level information that improves classification between ADHD and TD groups.

Including meta-features in the classification pipeline also contributed to more consistent results. For instance, Random Forest exhibited standard deviations around 0.12 to 0.13 across 5-, 10-, and 15-fold validations, indicating reliable generalization. Gradient Boosting showed slightly more variation but remained stable overall. The relatively low variability suggests that meta-features reduce model sensitivity to data partitioning—crucial in clinical contexts where consistency across data splits is essential.

In contrast, excluding meta-features increased standard deviations across most models. For example, both Gradient Boosting and Random Forest showed standard deviations exceeding 0.17 in several folds, indicating greater sensitivity to training/test set composition. Simpler models such as Logistic Regression and SVM also exhibited higher variance without meta-features, especially in 10- and 15-fold validations. These outcomes indicate that the absence of contextual meta-features makes classifiers more susceptible to data imbalance or noise, resulting in less stable predictions.

## 5.2 Interpretation and Implications

As reported in the *Background and State of the Art* section of this thesis, various prior studies have employed structured gaze-based tasks to classify ADHD. These include reading tasks, memory recall, smooth pursuit, visual search, and immersive environments such as virtual reality. By leveraging machine learning techniques such as Support Vector Machines (SVM), decision trees, logistic regression, and random forests, these works achieved classification accuracies ranging between 75% and 86%. The features used often stem from task-specific conditions, including fixation duration, saccade length, and visual response timing, collected under controlled experimental setups. Furthermore, these studies frequently relied on contextual behavioral tasks—such as attentional blink or audiovisual integration—to enhance the predictive power of their models.

In contrast, this thesis applied classical machine learning models to statistical features derived directly from gaze reading data, without the use of specialized experimental tasks. Using a dataset enriched with both behavioral features and meta-features (such as symptom scores and cognitive assessments), Random Forest achieved a classification accuracy of 86% when meta-features were included, and 78% when excluded. Other models like logistic regression and gradient boosting also performed strongly, especially when augmented with meta-features. These results demonstrate that even without complex task designs, general gaze data, when paired with classical classifiers, can yield competitive or superior performance. This approach emphasizes practical applicability and scalability, offering an efficient and reliable framework for distinguishing ADHD from typically developing individuals.

Across all experiments, with and without meta-features, eye-tracking features played a key role in classification. Features such as `Pupil_Std` (pupil size variability), `Angle_Std` (gaze angle variability), `Y_Std` (vertical gaze variability), and `Cos2Theta_Min` (minimum cosine-squared of angle) were consistently among the most important. When meta-features such as `z.ASRS_score` (standardized ADHD symptom score), `z.ravens_score` (standardized fluid intelligence), `z.sspan_score` (standardized working memory score), and `SymSpan.PartialScore` (raw working memory score) were included, classification performance improved significantly. These high-level features provided contextual information that the raw gaze data could not fully capture. Their integration contributed to more accurate and robust classification results.

### 5.3 Limitations and Future Directions

Despite promising results, this study has several limitations. The dataset size was modest, which could limit the generalizability of the findings. While cross-validation helped mitigate this issue, future research should involve larger and more diverse samples. Furthermore, all models were based on classical machine learning techniques. Although these methods offer transparency and ease of interpretation, they may not effectively capture complex non-linear patterns in the data.

Future work should explore deep learning approaches, such as convolutional neural networks or transformers, to better model gaze dynamics. Incorporating temporal sequences of gaze data may also enhance classification by capturing time-based patterns. Furthermore, extending the feature set to include task-specific eye-tracking metrics or multimodal inputs (e.g., EEG or behavioral data) could improve diagnostic accuracy. These avenues offer promising directions for the development of more advanced and clinically relevant ADHD detection models [76].



# **Appendix A**

## **Feature Lists and Plots**

Figure A.1 shows the description of all the 66 features extracted to apply classical machine learning approaches to classify the ADHD and TD categories. Figure A.2 to Figure A.61 visualize the rest of the features that were not displayed in visualization of features section of Chapter 4.

| Feature Name    | Description                                             | Feature Name          | Description                                     |
|-----------------|---------------------------------------------------------|-----------------------|-------------------------------------------------|
| X_Mean          | Mean of horizontal coordinates                          | Diff_Y_Skewness       | Skewness of differences between vertical points |
| X_Median        | Median of horizontal coordinates                        | Diff_Y_Kurtosis       | Kurtosis of differences between vertical points |
| X_Std           | Std. deviation of horizontal coordinates                | Log_Velocity_Mean     | Mean of log-transformed velocity                |
| X_Min           | Minimum horizontal value                                | Log_Velocity_Median   | Median of log-transformed velocity              |
| X_Max           | Maximum horizontal value                                | Log_Velocity_Std      | Std. deviation of log-transformed velocity      |
| X_Skewness      | Skewness of horizontal distribution                     | Log_Velocity_Min      | Minimum of log-transformed velocity             |
| X_Kurtosis      | Kurtosis of horizontal distribution                     | Log_Velocity_Max      | Maximum of log-transformed velocity             |
| Y_Mean          | Mean of vertical coordinates                            | Log_Velocity_Skewness | Skewness of log-transformed velocity            |
| Y_Median        | Median of vertical coordinates                          | Log_Velocity_Kurtosis | Kurtosis of log-transformed velocity            |
| Y_Std           | Std. deviation of vertical coordinates                  | Angle_Mean            | Mean of direction angle                         |
| Y_Min           | Minimum vertical value                                  | Angle_Median          | Median of direction angle                       |
| Y_Max           | Maximum vertical value                                  | Angle_Std             | Std. deviation of direction angle               |
| Y_Skewness      | Skewness of vertical distribution                       | Angle_Min             | Minimum direction angle                         |
| Y_Kurtosis      | Kurtosis of vertical distribution                       | Angle_Max             | Maximum direction angle                         |
| Pupil_Mean      | Mean pupil diameter                                     | Angle_Skewness        | Skewness of direction angle                     |
| Pupil_Median    | Median pupil diameter                                   | Angle_Kurtosis        | Kurtosis of direction angle                     |
| Pupil_Std       | Std. deviation of pupil diameter                        | Cos2Theta_Mean        | Mean of squared cosine of angle                 |
| Pupil_Min       | Minimum pupil diameter                                  | Cos2Theta_Median      | Median of squared cosine of angle               |
| Pupil_Max       | Maximum pupil diameter                                  | Cos2Theta_Std         | Std. deviation of squared cosine of angle       |
| Pupil_Skewness  | Skewness of pupil diameter distribution                 | Cos2Theta_Min         | Minimum squared cosine of angle                 |
| Pupil_Kurtosis  | Kurtosis of pupil diameter distribution                 | Cos2Theta_Max         | Maximum squared cosine of angle                 |
| Diff_X_Mean     | Mean of differences between horizontal points           | Cos2Theta_Skewness    | Skewness of squared cosine of angle             |
| Diff_X_Median   | Median of differences between horizontal points         | Cos2Theta_Kurtosis    | Kurtosis of squared cosine of angle             |
| Diff_X_Std      | Std. deviation of differences between horizontal points | Max_Cross_Correlation | Maximum cross-correlation between X and Y       |
| Diff_X_Min      | Minimum difference between horizontal points            | Min_Cross_Correlation | Minimum cross-correlation between X and Y       |
| Diff_X_Max      | Maximum difference between horizontal points            | Fixation_Length       | Average duration of fixation periods            |
| Diff_X_Skewness | Skewness of differences between horizontal points       | Succade_Length        | Average duration of saccadic movements          |
| Diff_X_Kurtosis | Kurtosis of differences between horizontal points       | PartA_score           | Raw ADHD symptom score from ASRS Part A         |
| Diff_Y_Mean     | Mean of differences between vertical points             | z.ASRS_score          | Std.ized ADHD symptom score                     |
| Diff_Y_Median   | Median of differences between vertical points           | SymSpan.PartialScore  | Raw memory score from Symmetry Span task        |
| Diff_Y_Std      | Std. deviation of differences between vertical points   | z.sspan_score         | Std.ized working memory score                   |
| Diff_Y_Min      | Minimum difference between vertical points              | ravens_score          | Raw score from Raven's Progressive Matrices     |
| Diff_Y_Max      | Maximum difference between vertical points              | z.ravens_score        | Std.ized fluid intelligence score               |

Figure A.1. The set of all 66 features to classify ADHD and TD participants

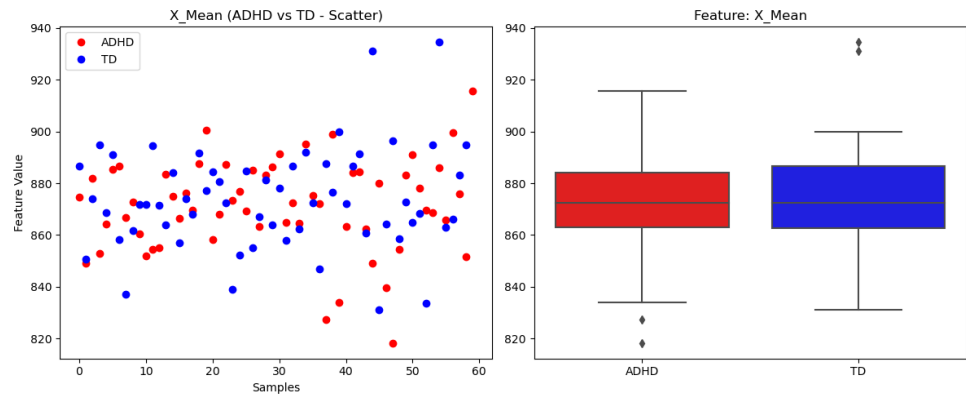
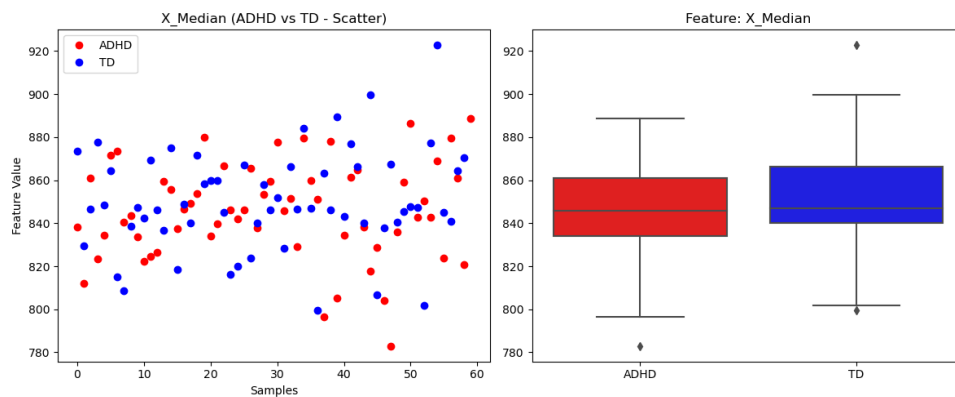
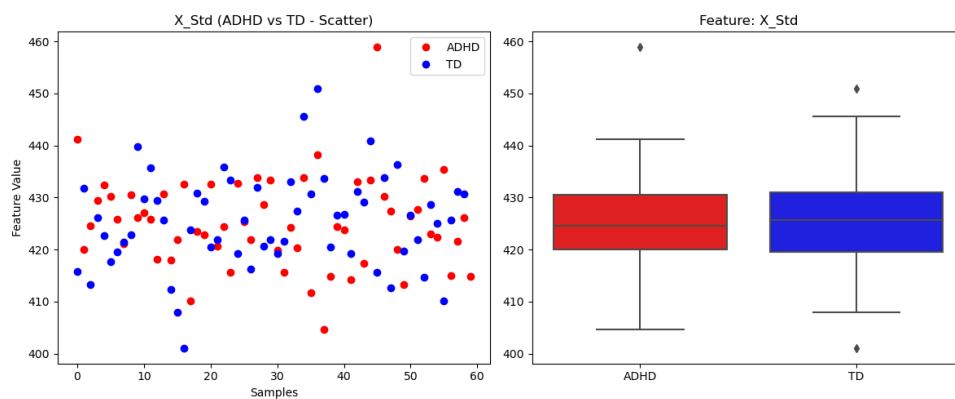
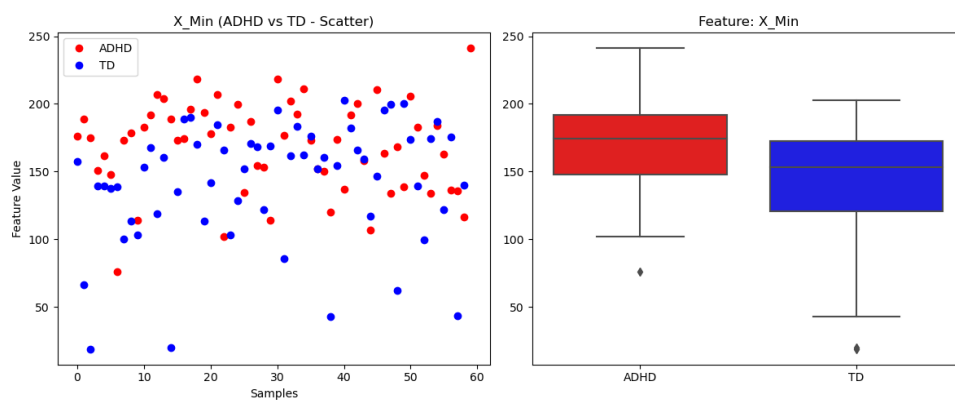


Figure A.2. Scatter and box plots of X\_Mean

Figure A.3. Scatter and box plots of  $X\_Median$ Figure A.4. Scatter and box plots of  $X\_Std$ Figure A.5. Scatter and box plots of  $X\_Min$

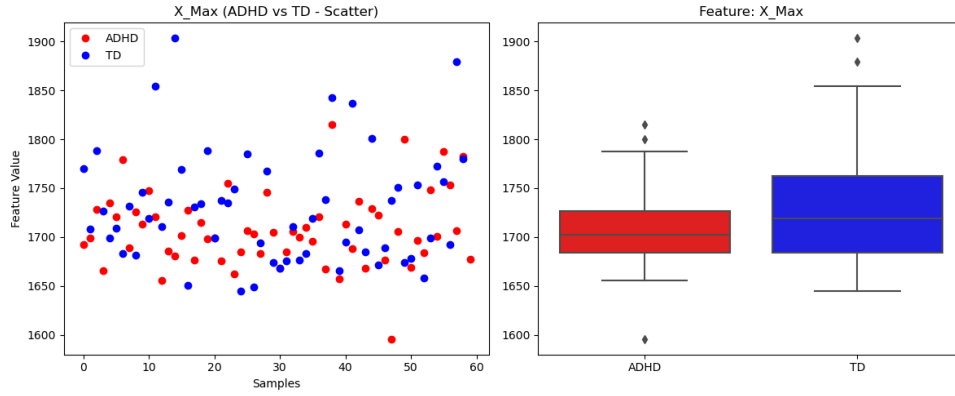


Figure A.6. Scatter and box plots of X\_Max

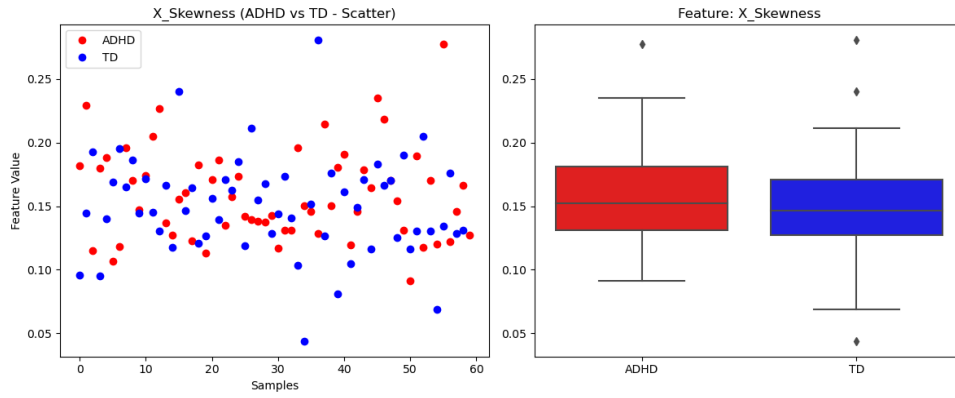


Figure A.7. Scatter and box plots of X\_Skewness

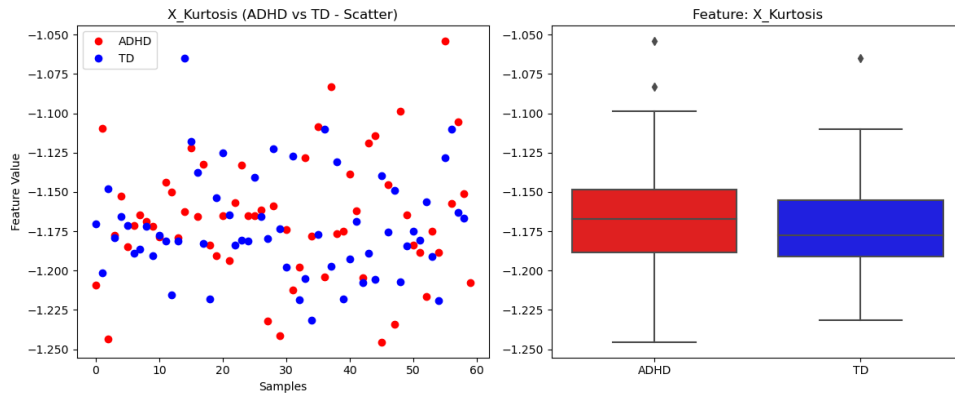


Figure A.8. Scatter and box plots of X\_Kurtosis

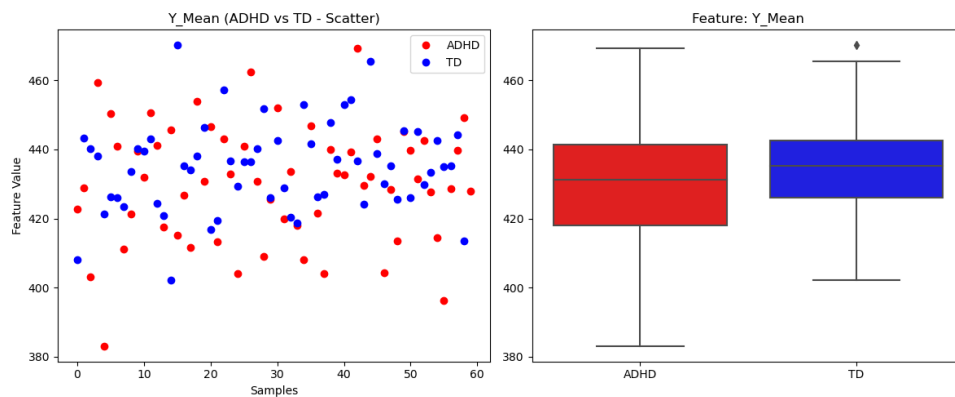


Figure A.9. Scatter and box plots of Y\_Mean

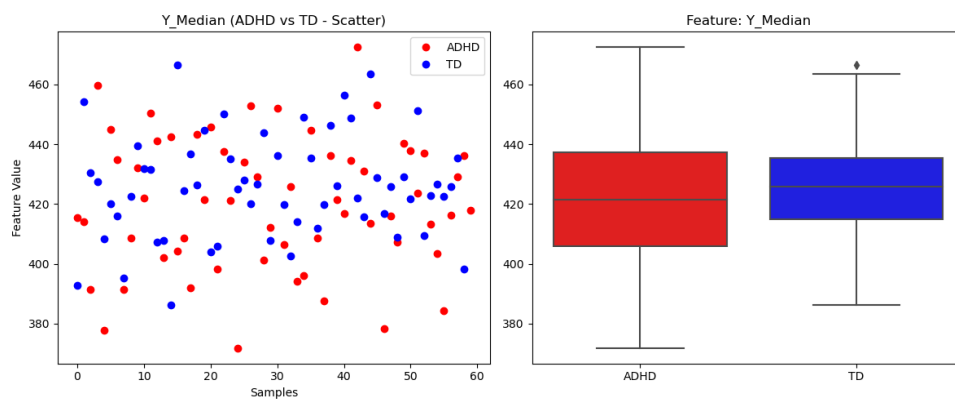


Figure A.10. Scatter and box plots of Y\_Median

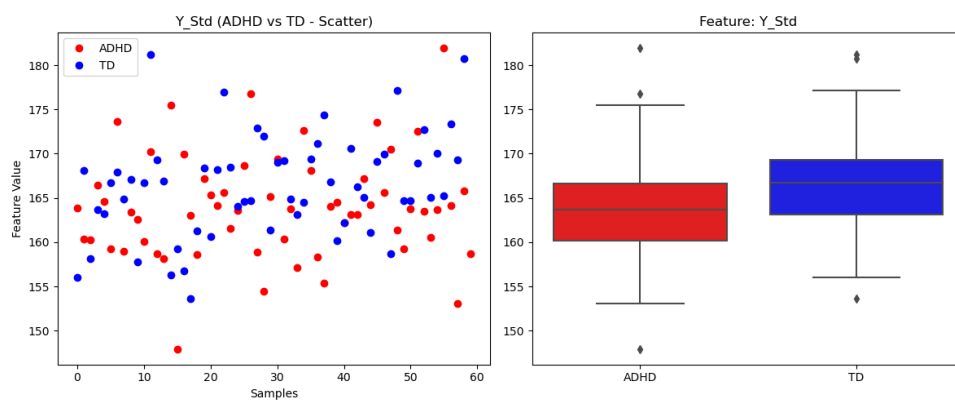


Figure A.11. Scatter and box plots of Y\_Std

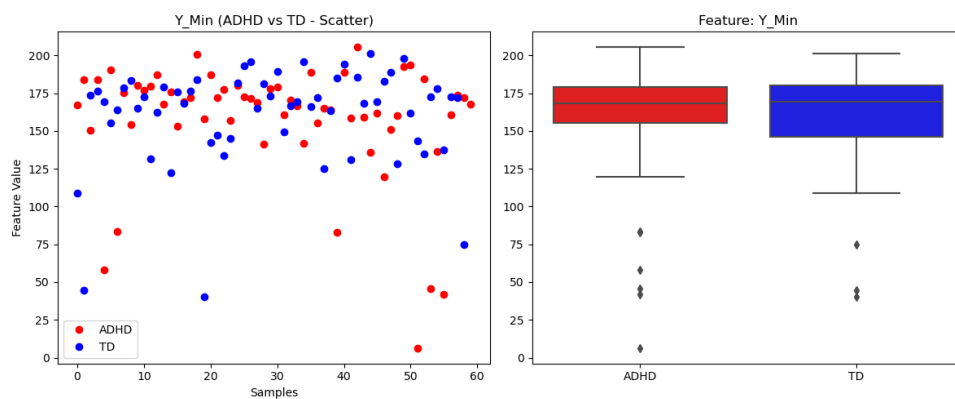


Figure A.12. Scatter and box plots of Y\_Min

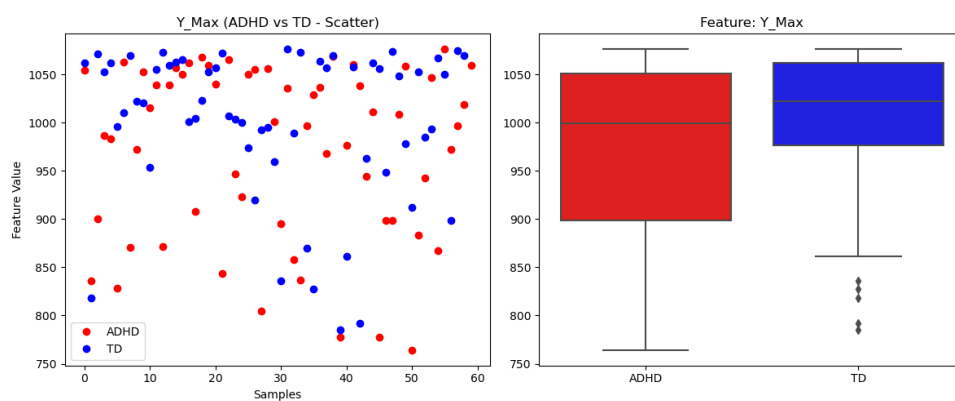


Figure A.13. Scatter and box plots of Y\_Max

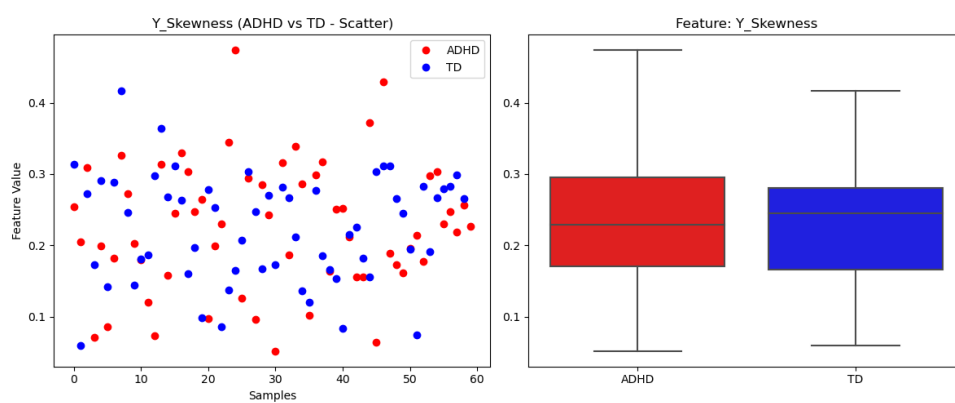


Figure A.14. Scatter and box plots of Y\_Skewness

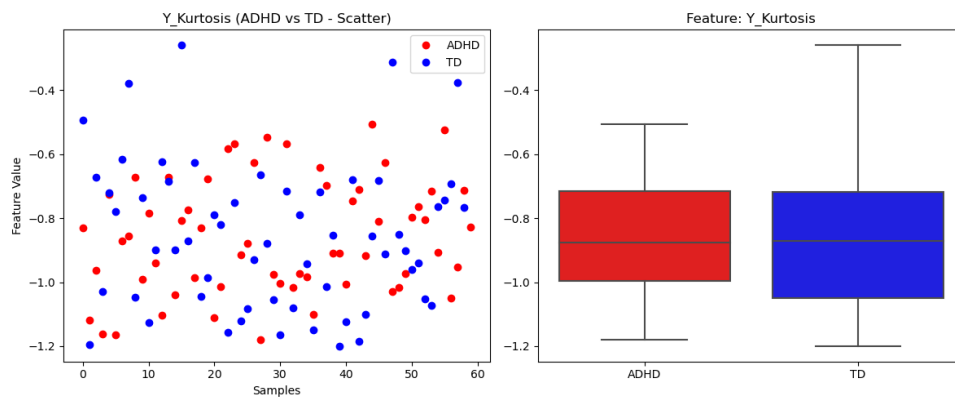


Figure A.15. Scatter and box plots of Y\_Kurtosis

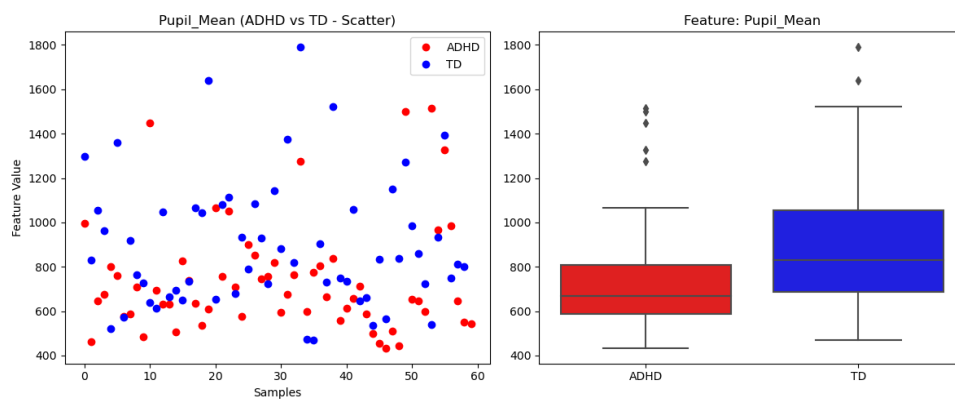


Figure A.16. Scatter and box plots of Pupil\_Mean

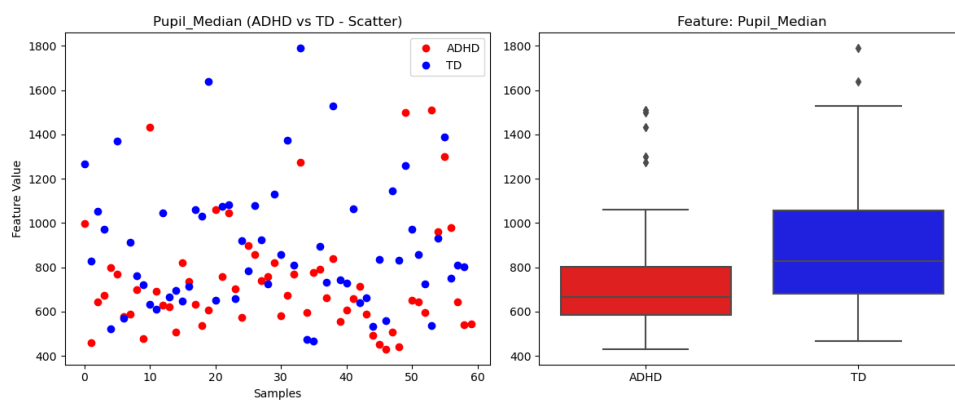


Figure A.17. Scatter and box plots of Pupil\_Median

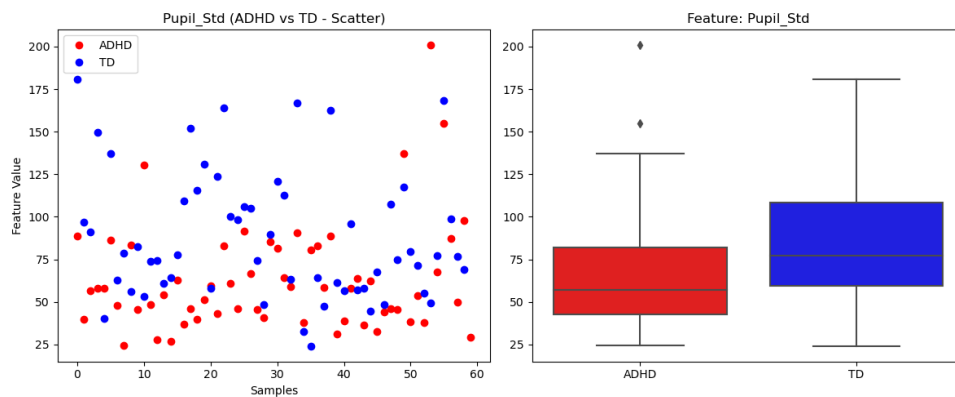


Figure A.18. Scatter and box plots of Pupil\_Std

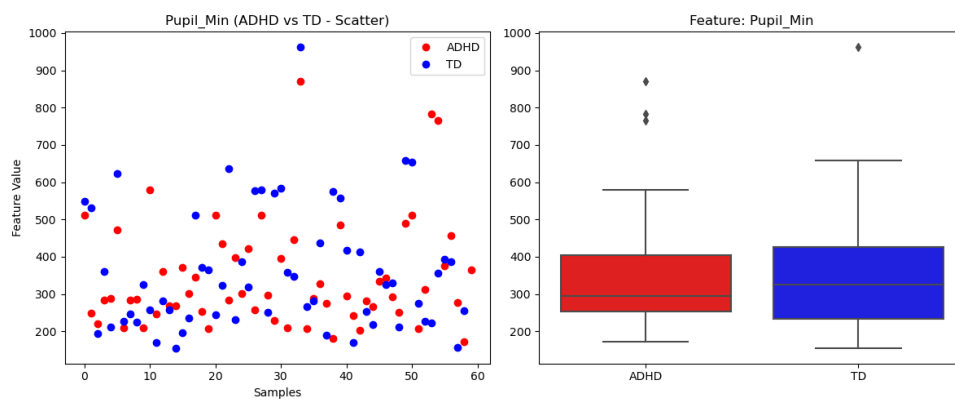


Figure A.19. Scatter and box plots of Pupil\_Min

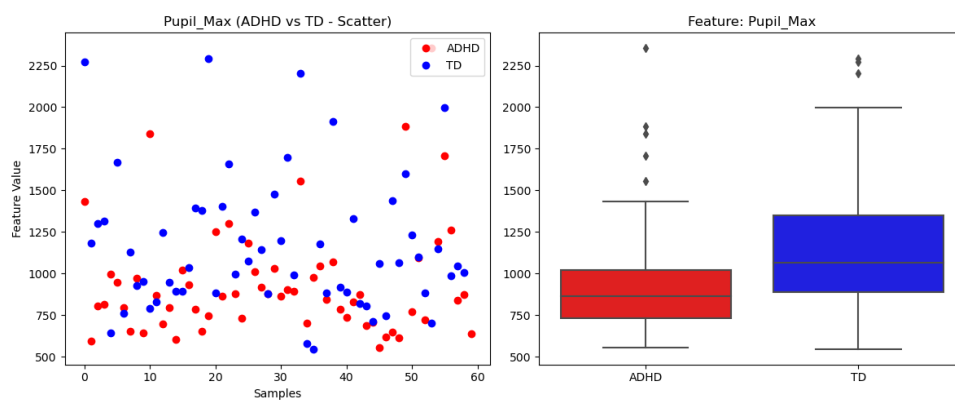


Figure A.20. Scatter and box plots of Pupil\_Max



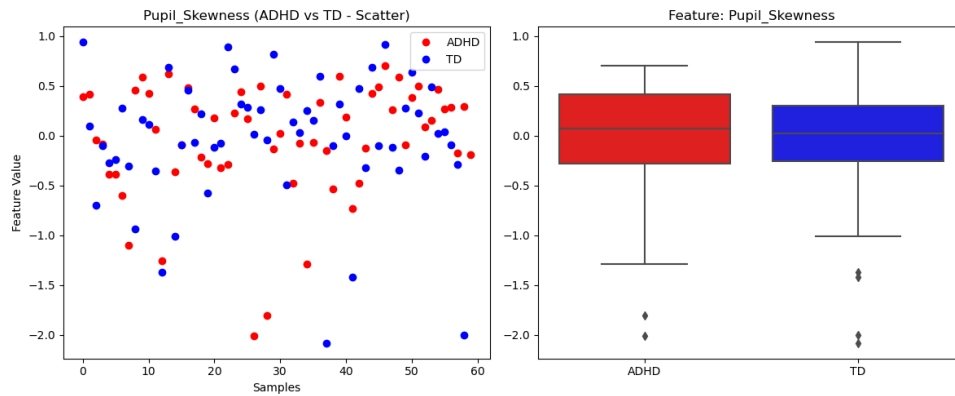


Figure A.21. Scatter and box plots of Pupil\_Skewness

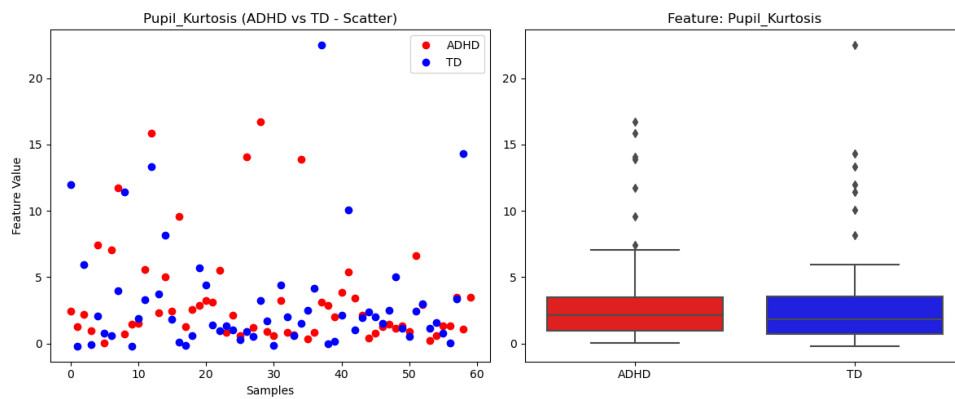


Figure A.22. Scatter and box plots of Pupil\_Kurtosis

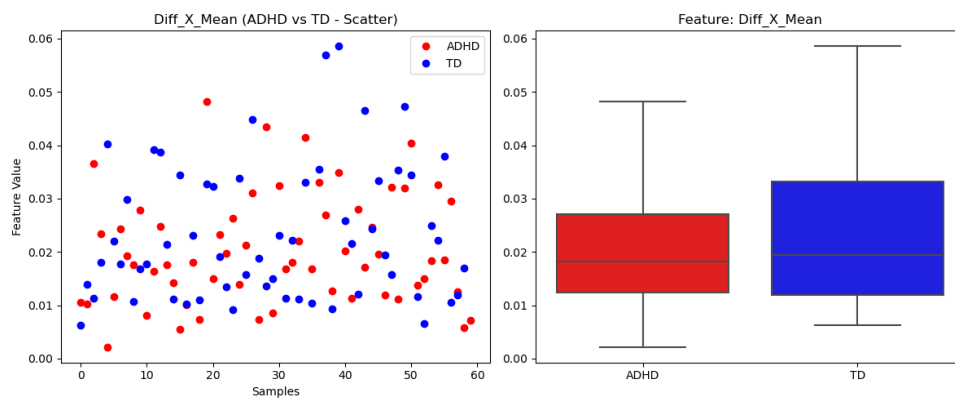


Figure A.23. Scatter and box plots of Diff\_X\_Mean

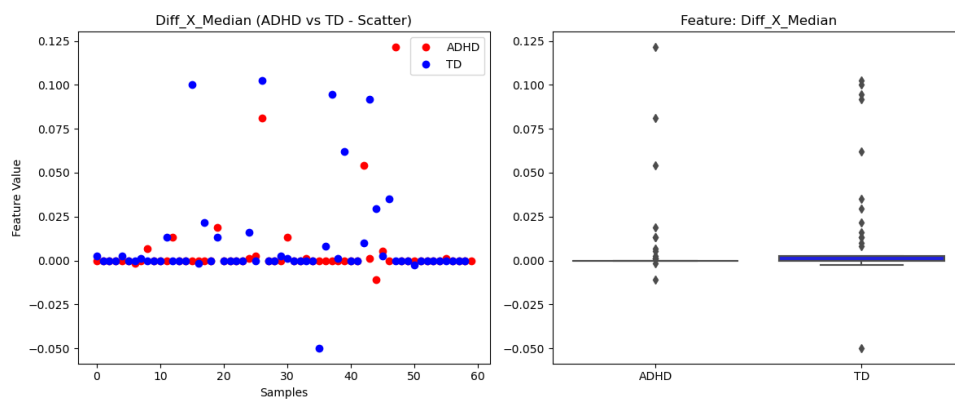


Figure A.24. Scatter and box plots of Diff\_X\_Median

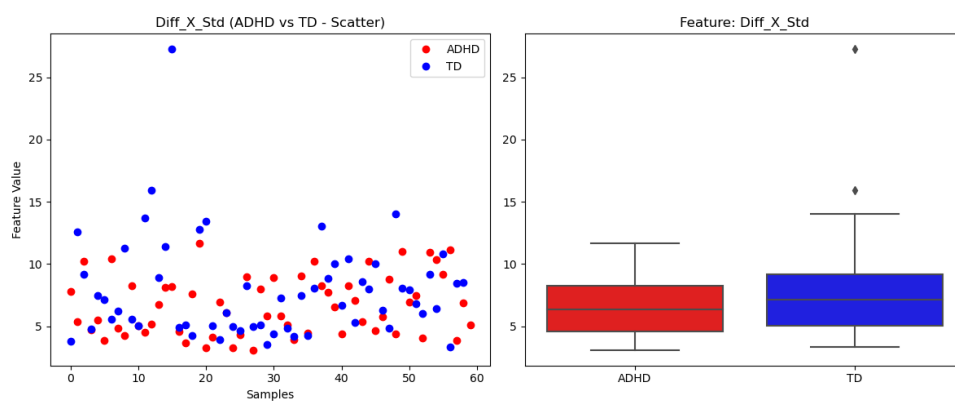


Figure A.25. Scatter and box plots of Diff\_X\_Std

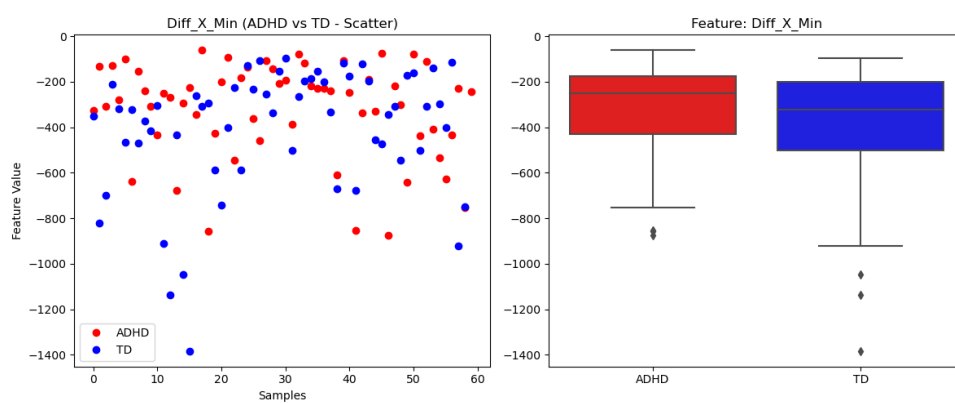


Figure A.26. Scatter and box plots of Diff\_X\_Min

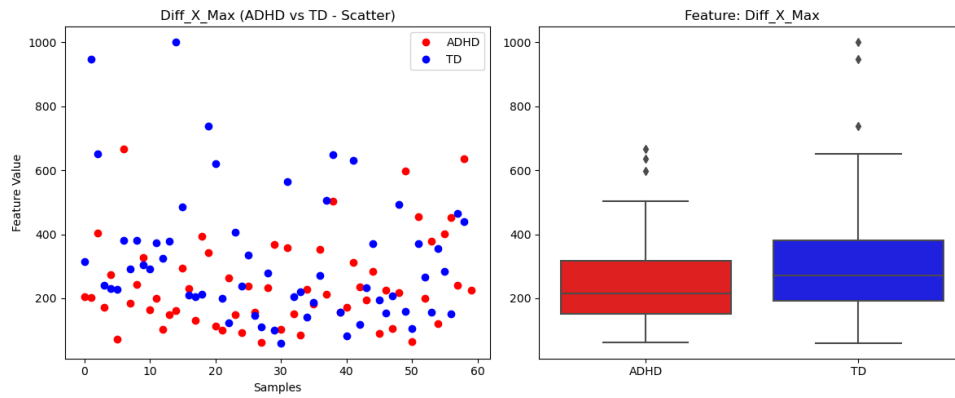


Figure A.27. Scatter and box plots of Diff\_X\_Max

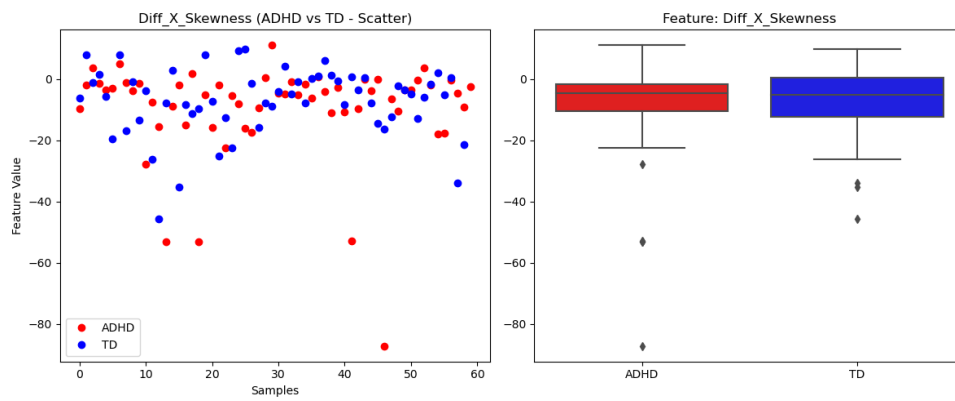


Figure A.28. Scatter and box plots of Diff\_X\_Skewness

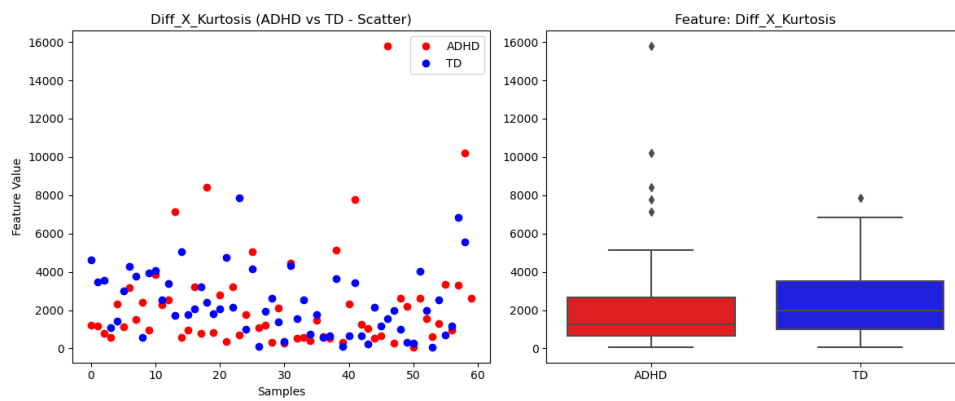


Figure A.29. Scatter and box plots of Diff\_X\_Kurtosis

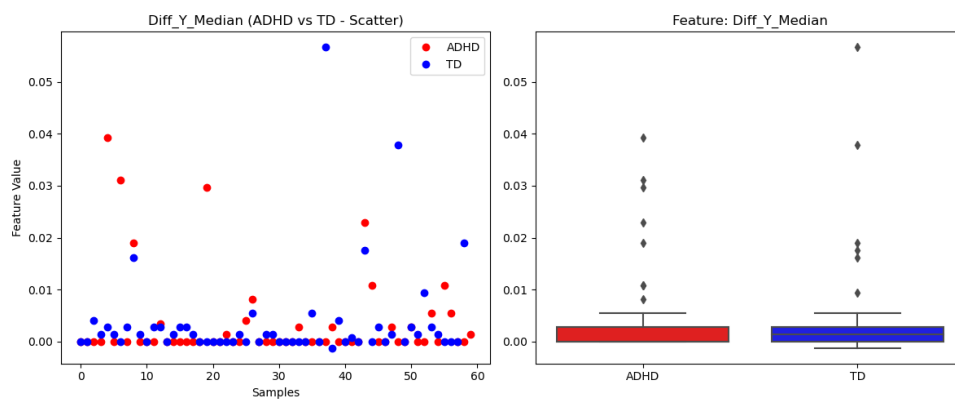


Figure A.30. Scatter and box plots of Diff\_Y\_Median

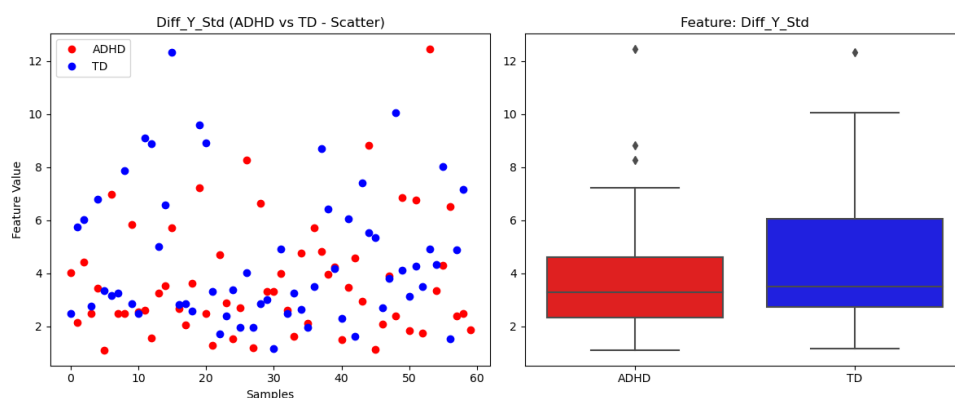


Figure A.31. Scatter and box plots of Diff\_Y\_Std

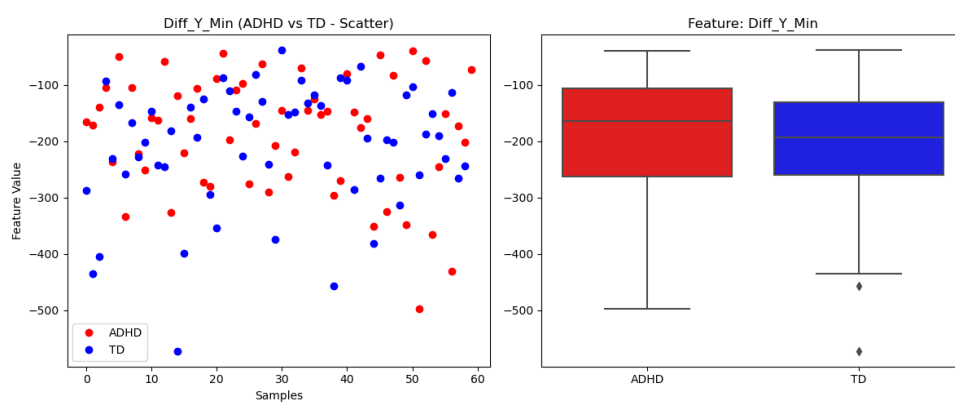


Figure A.32. Scatter and box plots of Diff\_Y\_Min

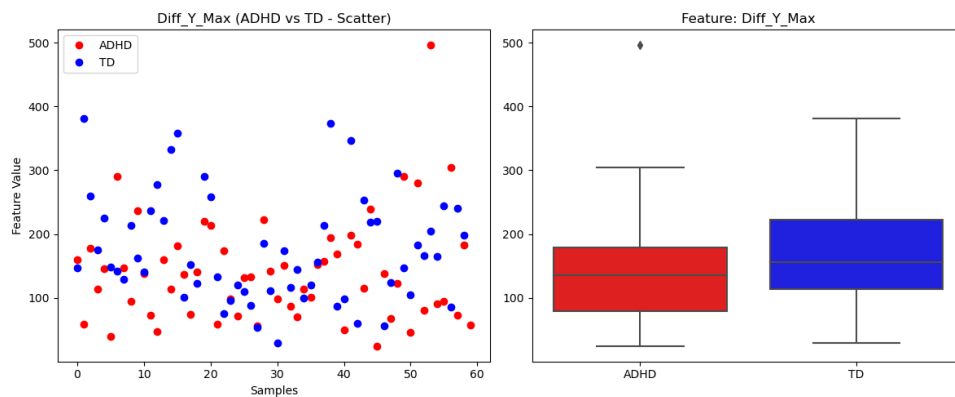


Figure A.33. Scatter and box plots of Diff\_Y\_Max

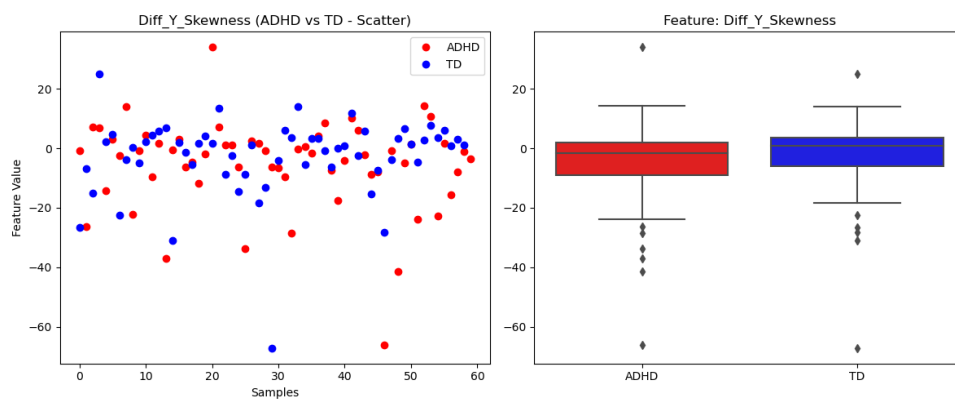


Figure A.34. Scatter and box plots of Diff\_Y\_Skewness

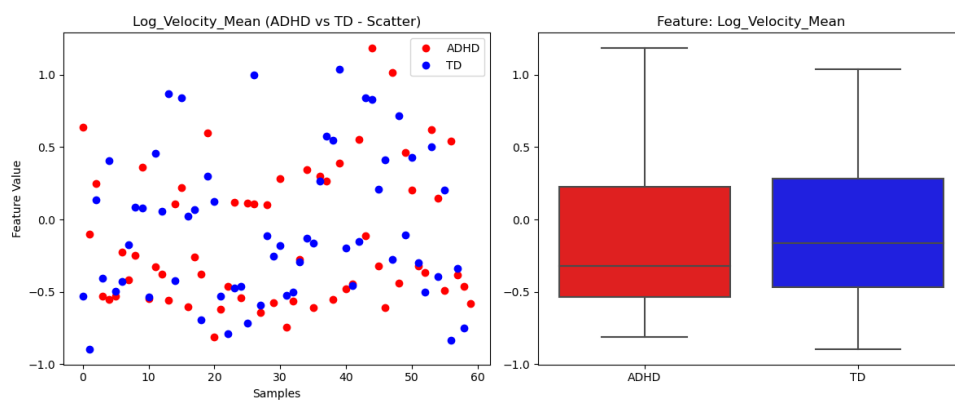


Figure A.35. Scatter and box plots of Log\_Velocity\_Mean

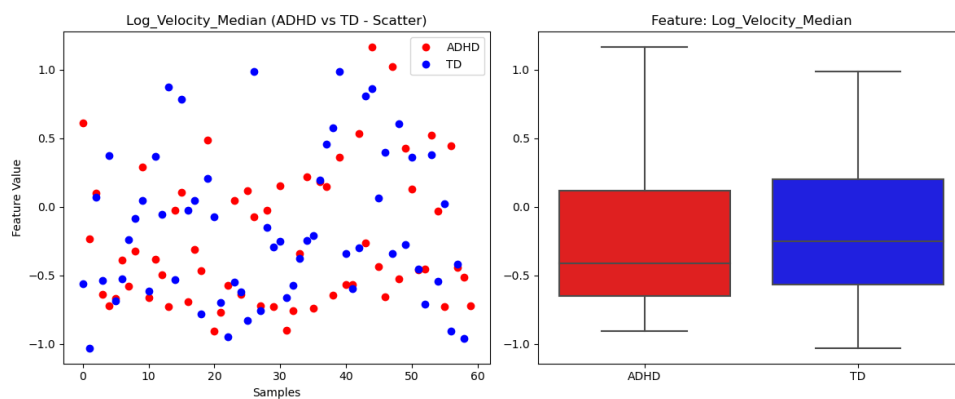


Figure A.36. Scatter and box plots of Log\_Velocity\_Median

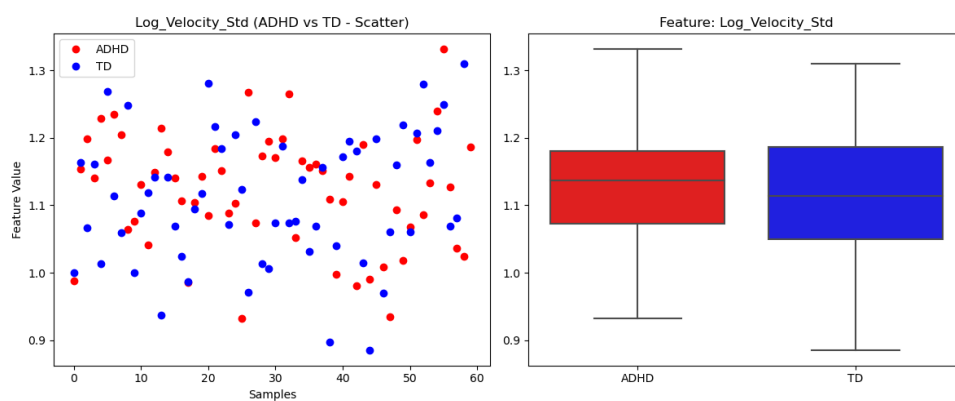


Figure A.37. Scatter and box plots of Log\_Velocity\_Std

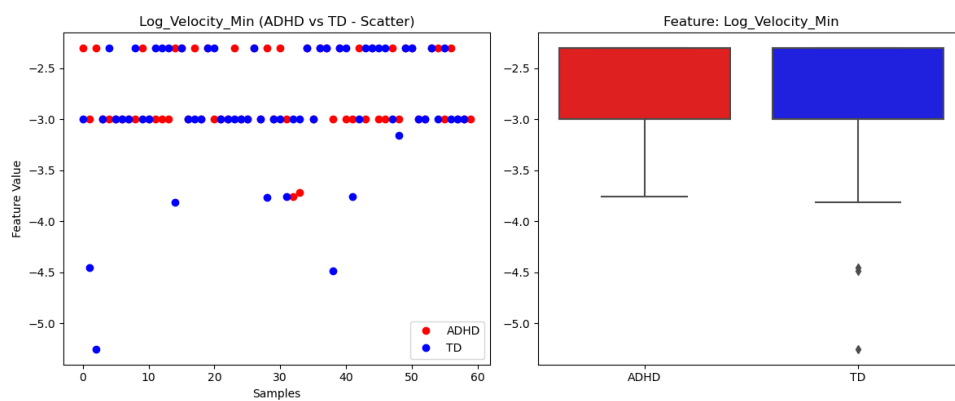


Figure A.38. Scatter and box plots of Log\_Velocity\_Min

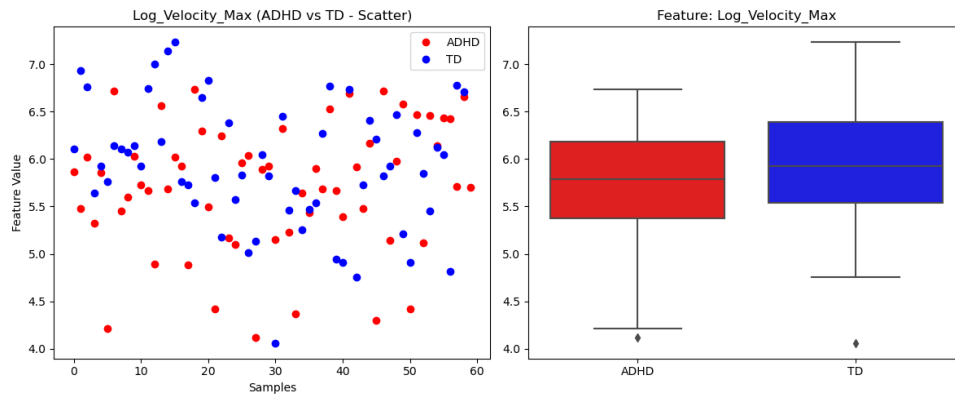


Figure A.39. Scatter and box plots of Log\_Velocity\_Max

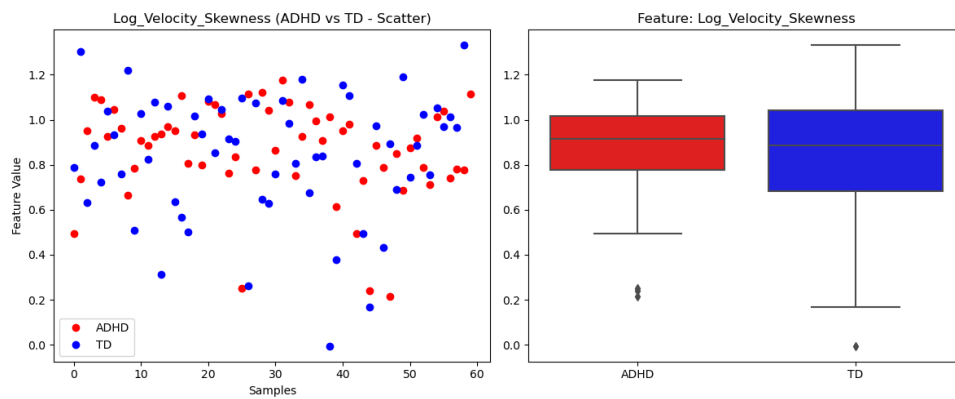


Figure A.40. Scatter and box plots of Log\_Velocity\_Skewness

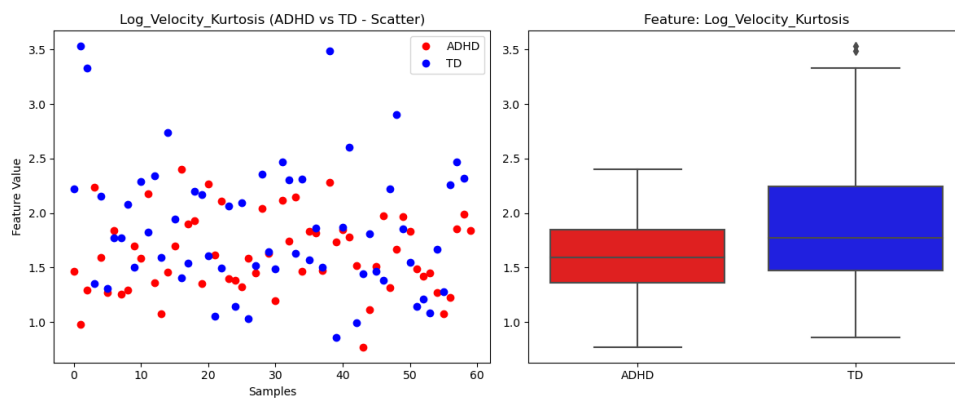


Figure A.41. Scatter and box plots of Log\_Velocity\_Kurtosis

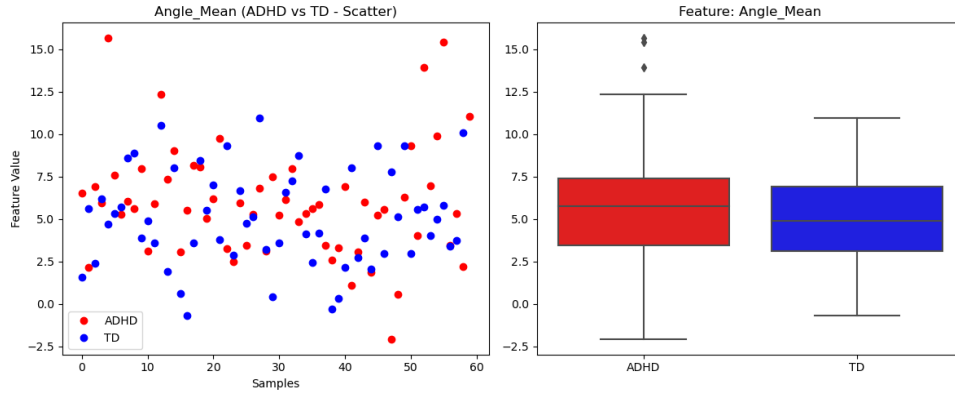


Figure A.42. Scatter and box plots of Angle\_Mean

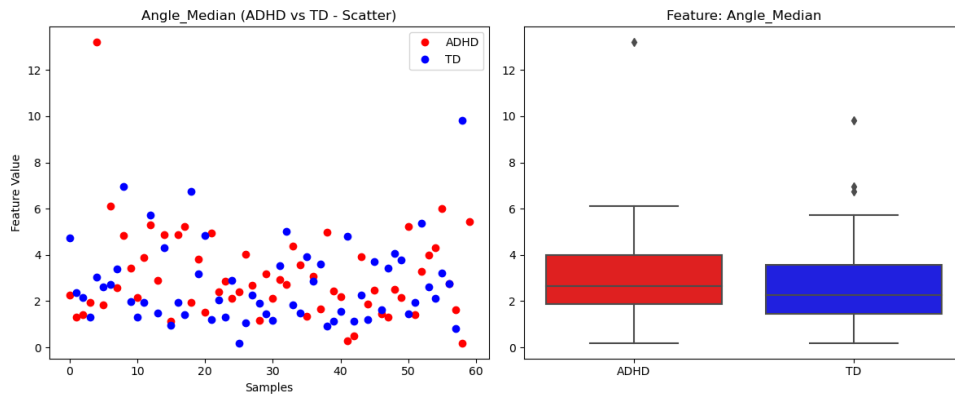


Figure A.43. Scatter and box plots of Angle\_Median

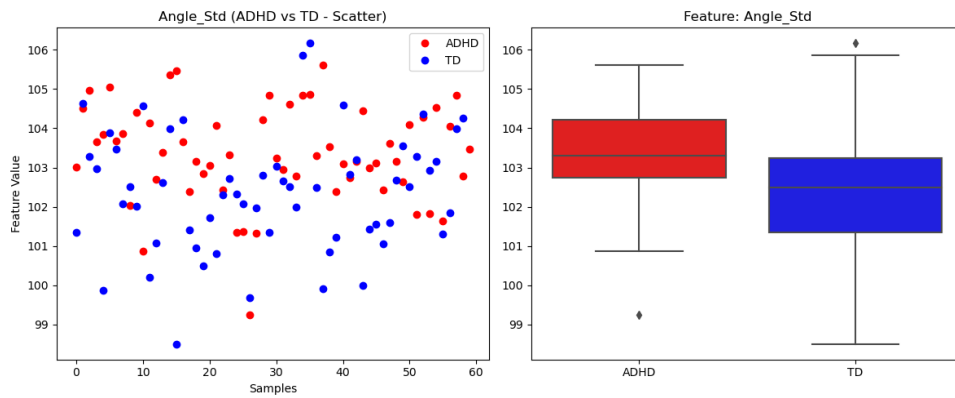


Figure A.44. Scatter and box plots of Angle\_Std



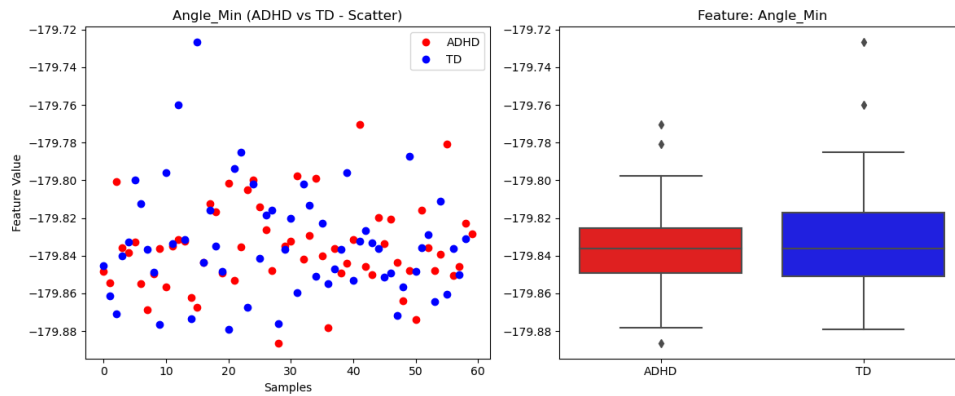


Figure A.45. Scatter and box plots of Angle\_Min

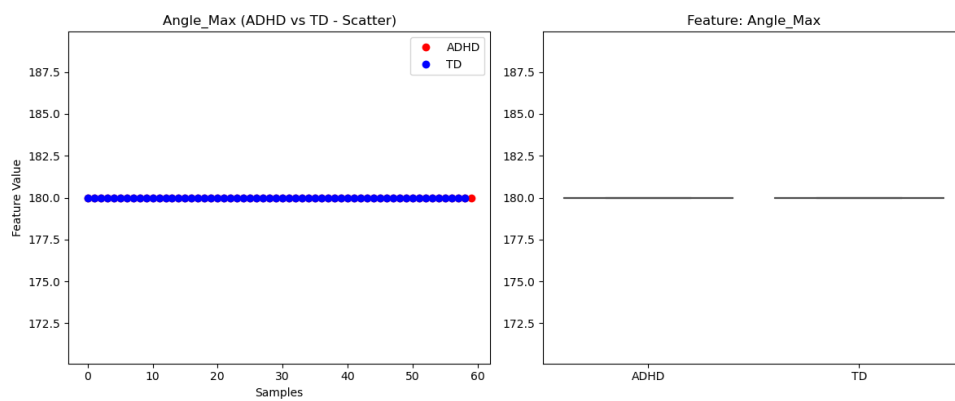


Figure A.46. Scatter and box plots of Angle\_Max

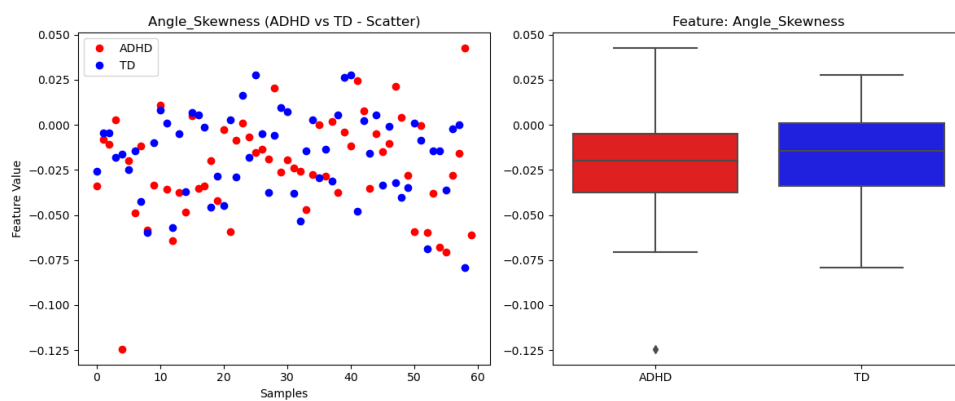


Figure A.47. Scatter and box plots of Angle\_Skewness

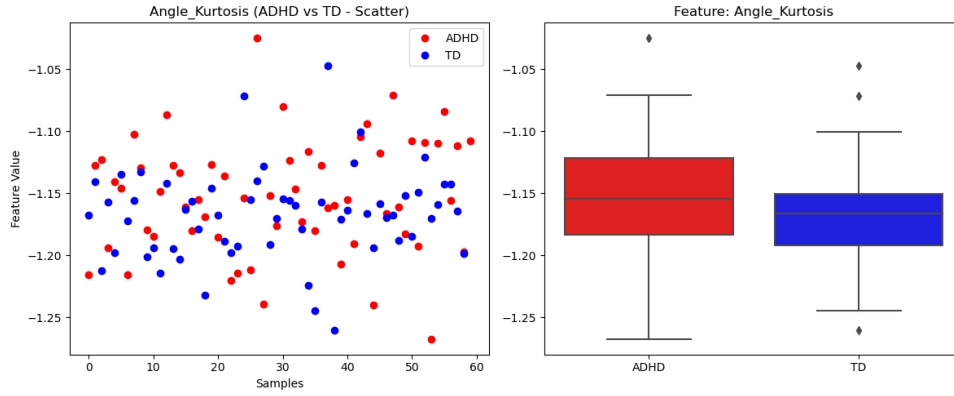
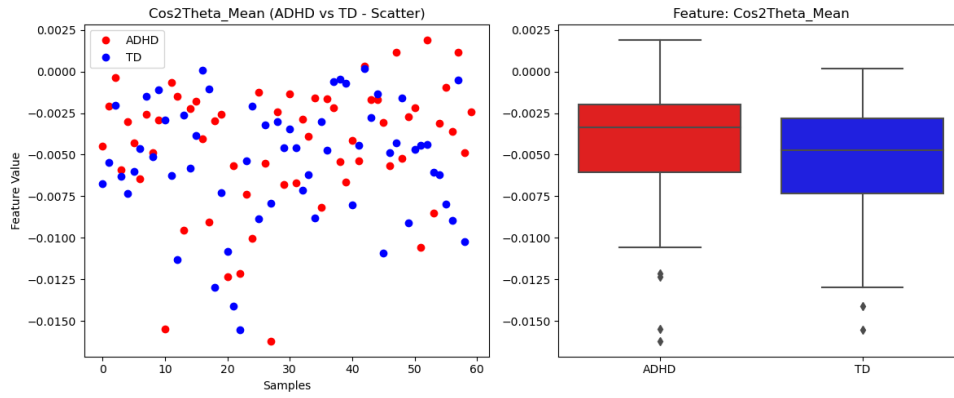
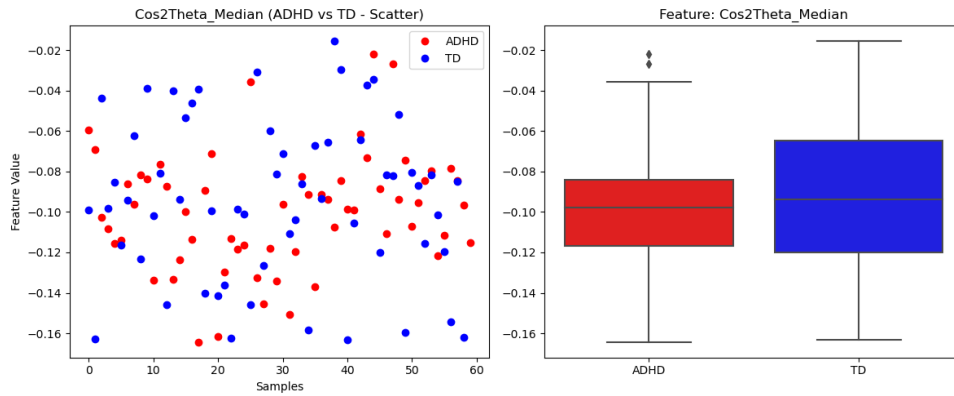
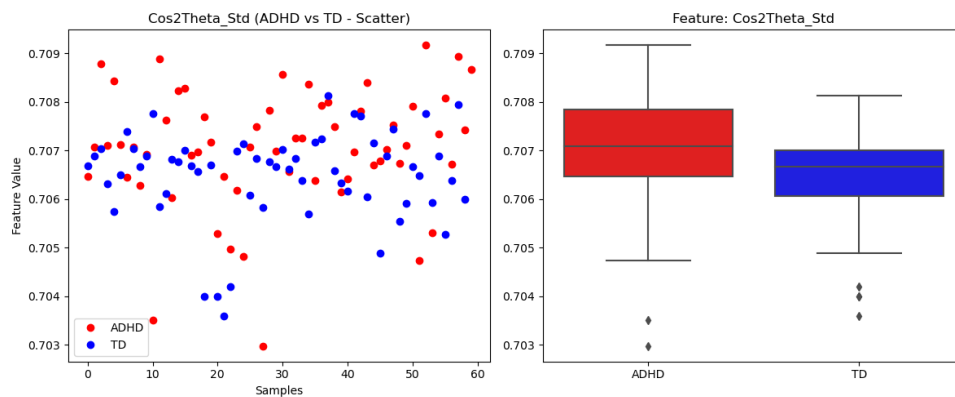
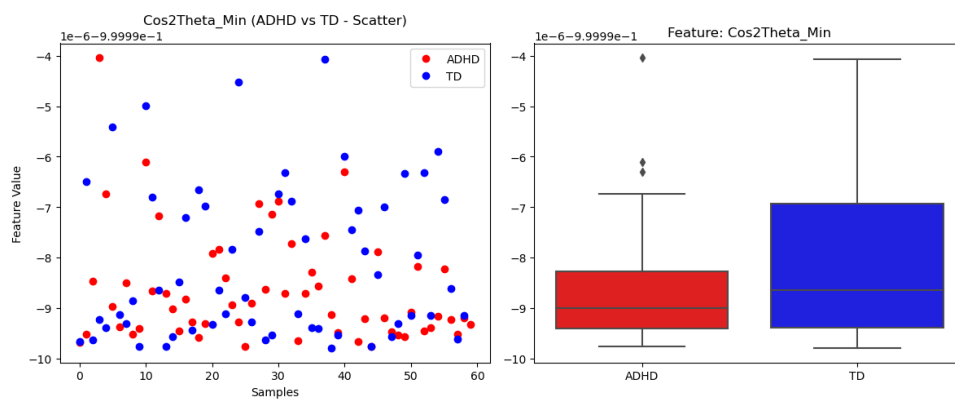
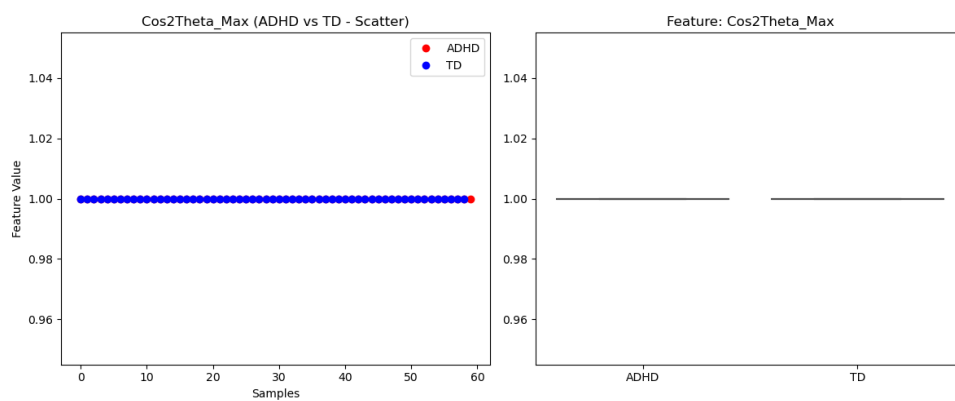


Figure A.48. Scatter and box plots of Angle\_Kurtosis

Figure A.49. Scatter and box plots of  $\cos(2\theta)$ \_MeanFigure A.50. Scatter and box plots of  $\cos(2\theta)$ \_Median

Figure A.51. Scatter and box plots of  $\cos(2\theta)_{Std}$ Figure A.52. Scatter and box plots of  $\cos(2\theta)_{Min}$ Figure A.53. Scatter and box plots of  $\cos(2\theta)_{Max}$

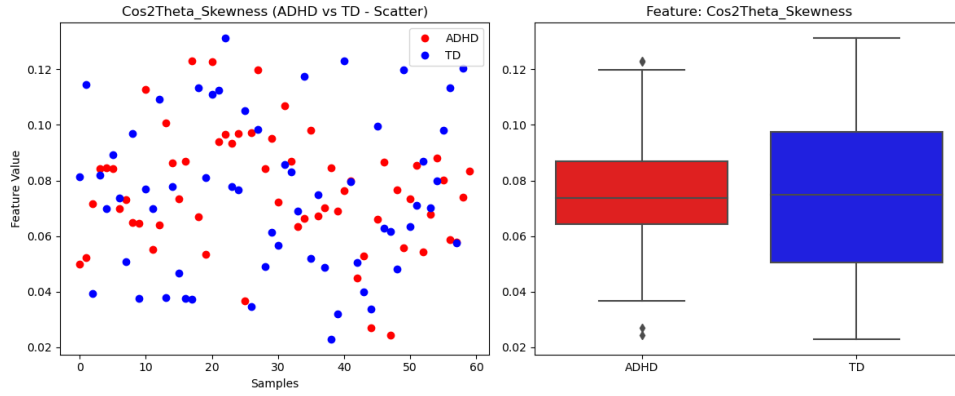
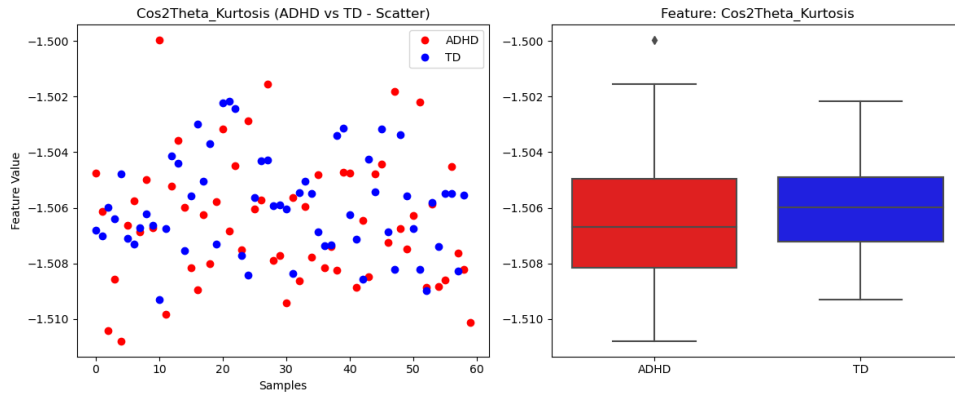
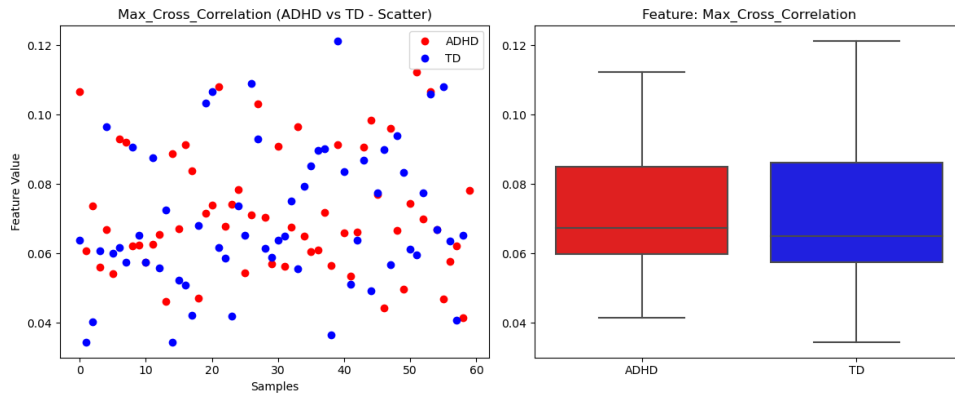
Figure A.54. Scatter and box plots of  $\cos(2\theta)$ \_SkewnessFigure A.55. Scatter and box plots of  $\cos(2\theta)$ \_Kurtosis

Figure A.56. Scatter and box plots of maximum values of cross-correlation between pupil size and velocity

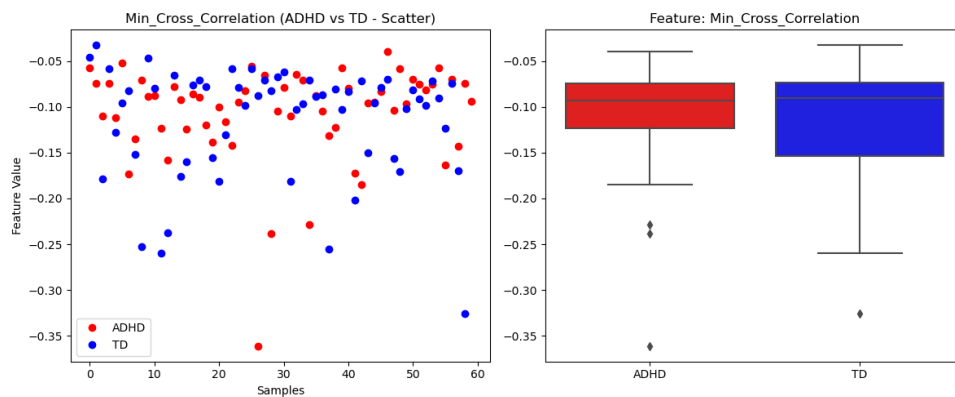


Figure A.57. Scatter and box plots of minimum values of cross-correlation between pupil size and velocity

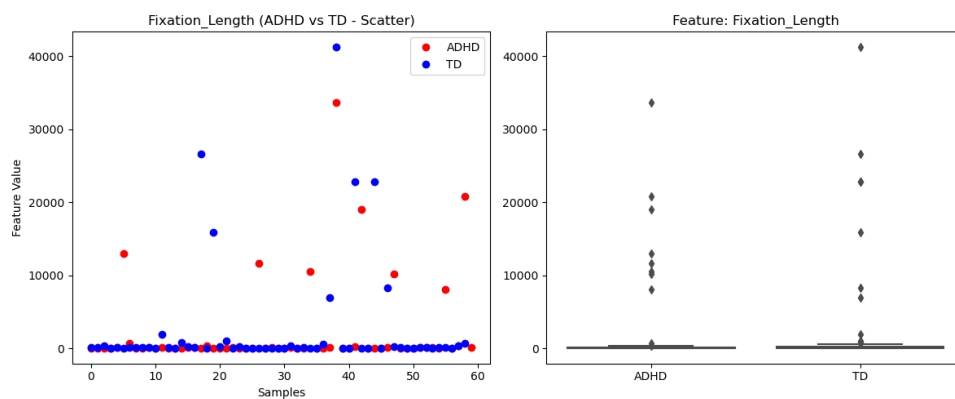


Figure A.58. Scatter and box plots of fixation length

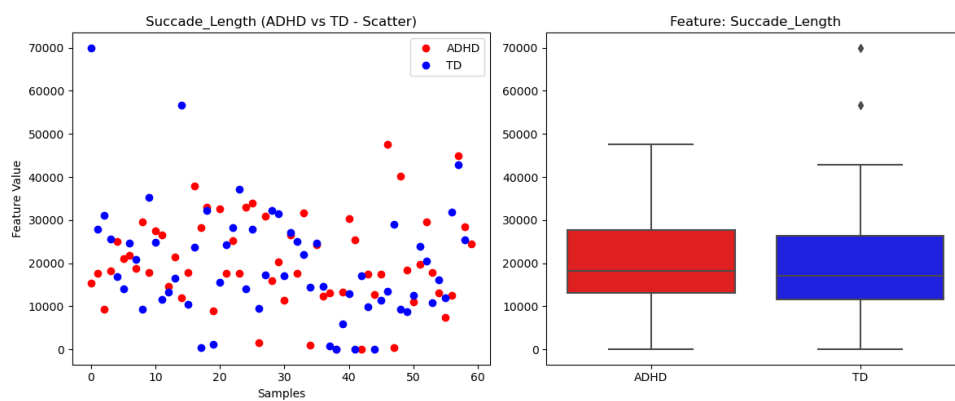


Figure A.59. Scatter and box plots of succade length

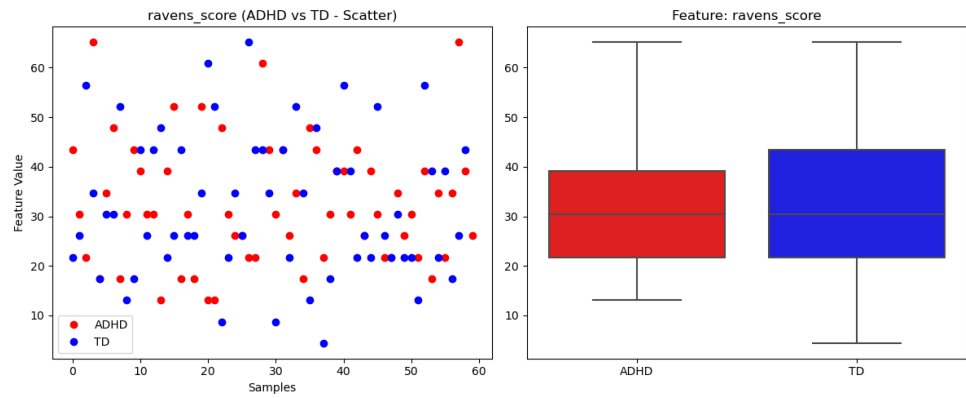


Figure A.60. Scatter and box plots of ravens\_score

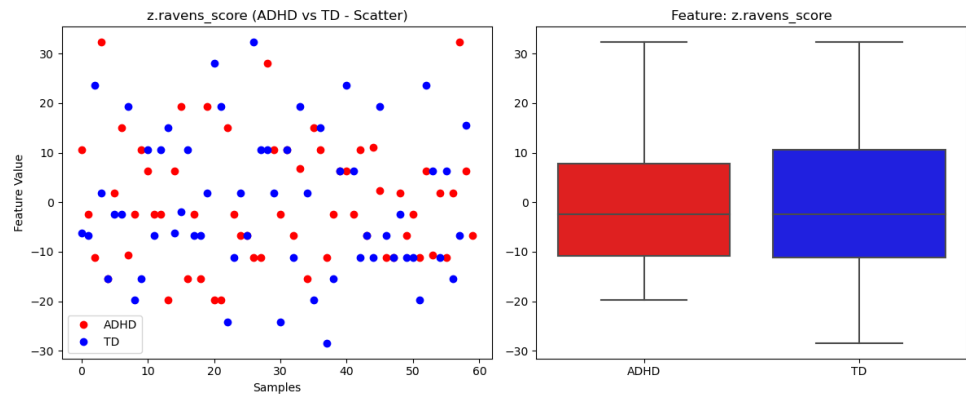


Figure A.61. Scatter and box plots of z.ravens\_score

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